FAA Part 107
Practice Test 2026

567 Realistic Practice Questions with Detailed Explanations

Based on the official FAA Remote Pilot Study Guide (FAA-G-8082-22)
and the Airman Certification Standards (FAA-S-ACS-10B)

Unofficial study material. Not affiliated with or endorsed by the FAA.

Table of Contents

567 questions with answer key and full explanations included

Chapter 1: Aerodynamics & Flight Principles

Q1. On a hot summer afternoon, you need to fly a multirotor from a high-elevation launch site. The density altitude is much higher than the field elevation. How will this affect your drone's performance?

A) It will have increased lift because the air is less dense, causing it to rise faster

B) It will have no effect because unmanned aircraft propellers are not affected by air density

C) It will have reduced lift and less efficient propellers, requiring more power to maintain flight

Answer: C - High density altitude indicates lower air density. This reduces the lift generated by propellers and decreases engine/rotor efficiency. The drone will require more throttle and may struggle to climb or maintain altitude, especially in hot, high-elevation conditions.

Q2. You are operating a small UA in the late afternoon on a hot, humid summer day at a field 5,000 feet above sea level. Which condition will the remote pilot likely experience?

A) Improved lift and longer endurance due to warmer air expanding

B) Reduced lift and less responsive climb because the air is less dense

C) No change, because small UA performance is unaffected by air density

Answer: B - Higher temperature, high humidity, and higher elevation all decrease air density. Less dense air reduces rotor thrust and efficiency, degrading climb performance and overall responsiveness.

Q3. You are planning a flight at sunrise when temperatures are cool, compared to the same location at midday. How will the difference in temperature affect the drone's performance?

A) Cooler air is denser, so the rotors produce more lift for the same RPM

B) Cooler air is less dense, requiring more throttle to maintain hover

C) Temperature has no measurable effect on rotor performance

Answer: A - Density varies inversely with temperature at constant pressure. Cooler air is denser, giving the rotors more air to push and therefore producing more lift for the same rotor speed.

Q4. During a flight, you notice your drone's battery drains faster than normal and it seems sluggish. The weather is hot with high relative humidity. Which factor is the primary cause?

A) High humidity directly increases battery internal resistance and drain

B) Warm, moist air is denser, increasing parasitic drag

C) Warm, moist air is less dense, reducing rotor effectiveness and requiring more throttle

Answer: C - Warm air is less dense, and water vapor is lighter than air, so moist air is even less dense. The rotors move less air per revolution, so the drone needs more RPM and power, draining the battery faster.

Q5. What is the effect of high humidity on density altitude and small UA performance?

A) Moist air is heavier, decreasing density altitude and improving performance

B) Moist air is lighter, increasing density altitude and decreasing performance

C) Humidity does not affect air density or aircraft performance

Answer: B - Water vapor is lighter than dry air, so increasing humidity makes the air less dense. This raises density altitude and reduces lift and thrust, degrading performance.

Q6. You are flying at a location with a density altitude of 7,000 feet. Compared to sea-level conditions, what should you expect?

A) The drone will have less lift and require more throttle to maintain a hover

B) The drone will fly identically, but GPS accuracy will improve at altitude

C) The drone will have more lift because the air is thinner

Answer: A - As altitude increases, pressure drops rapidly and density decreases. With less dense air, the rotors produce less lift, so the pilot must increase throttle to maintain altitude.

Q7. A manufacturer's performance data for a small UA was obtained on a cool, dry day at sea level. To safely apply that data on a hot, humid day at 4,000 feet pressure altitude, the remote pilot should recognize that:

A) Performance will be degraded because density altitude is much higher than pressure altitude

B) Performance will change only if the battery is fully charged

C) The aircraft will outperform the published data

Answer: A - Hot temperature and high humidity increase density altitude above the already elevated pressure altitude. This combination lowers air density, reducing aerodynamic performance compared to the sea-level standard conditions used for the manufacturer's data.

Q8. Which set of conditions will result in the lowest air density and therefore the greatest negative impact on small UA performance?

A) Low temperature, low humidity, low altitude

B) High temperature, high humidity, high altitude

C) High temperature, low humidity, low altitude

Answer: B - High temperature decreases density, high humidity adds lighter water vapor, and high altitude brings lower pressure. All three together produce the lowest air density and the worst performance.

Q9. If a small UA manufacturer does not provide performance data for its aircraft, what is the recommended starting point for determining performance?

A) Seek performance data published by other users of the same make and model

B) Assume performance is identical to all other small UAs

C) Use the performance data printed on the retail packaging

Answer: A - Manufacturer performance data is not standardized. When it is unavailable, the FAA recommends using performance data already determined and published by other users of the same small UA model as a starting point.

Q10. You are a remote pilot flying a small UAS on a hot, humid summer afternoon. Compared to a cool, dry day, what should you expect?

A) No difference because humidity has no effect on aircraft performance

B) Improved climb performance because warm air is lighter

C) Decreased performance because moist air is less dense

Answer: C - The excerpt states that moist air is lighter than dry air, so as water content increases, air becomes less dense, increasing density altitude and decreasing performance.

Q11. A remote pilot checks the weather forecast: temperature is expected to rise significantly between morning and afternoon, but pressure will remain constant. How will this affect air density?

A) Density will remain constant

B) Density will increase as temperature increases

C) Density will decrease as temperature increases

Answer: C - The excerpt states that increasing the temperature of a substance decreases its density at a constant pressure.

Q12. You are operating a small UA from a high-elevation airport in the mountains. As your aircraft gains altitude, what is the general trend in air density?

A) Density increases because temperature decreases

B) Density decreases because pressure drops rapidly with altitude

C) Density stays the same because temperature and pressure offset each other

Answer: B - The excerpt explains that a fairly rapid drop in pressure as altitude increases usually has a dominating effect, so density decreases with altitude.

Q13. Two identical small UAS are flying under the same temperature and pressure, but one is flying in very humid air and the other in dry air. Which will likely experience better performance?

A) The UA flying in dry air because dry air is denser

B) The UA flying in humid air because water vapor adds lift

C) They will perform identically because water vapor is negligible

Answer: A - The excerpt states water vapor is lighter than air, making moist air less dense; denser air improves performance, so dry air is better.

Q14. You just acquired a new small UAS model and the manufacturer has not published performance data. What is an appropriate way to begin understanding its performance?

A) Use performance data published by other users of the same model as a starting point

B) Contact the FAA for the official performance data

C) Assume it performs exactly like all other small UAS

Answer: A - The excerpt recommends seeking out performance data that may have already been determined and published by other users of the same small UA manufacturer model and use that data as a starting point.

Q15. A remote pilot is flying near a Class C airport in the summer. The temperature is high and humidity is high. What does increased density altitude mean for the small UA?

A) The aircraft will perform better because the air is denser

B) The aircraft will be unaffected because temperature and humidity offset each other

C) The aircraft will have decreased performance because the air is less dense

Answer: C - High temperature and humidity make air less dense, increasing density altitude and decreasing performance, as stated in the excerpt.

Q16. Which of the following statements about relative humidity and air density is correct?

A) Warm air holds less water vapor, making it denser

B) Cold air holds more water vapor, making it less dense

C) Warm air holds more water vapor, which makes the air less dense

Answer: C - The excerpt says warm air holds more water vapor, and water vapor is lighter than air, so humid air is less dense.

Q17. You are planning a flight and have two options: a cold, dry morning or a warm, humid afternoon. In which condition would your small UA likely generate more lift?

A) Warm, humid afternoon because the air is lighter

B) Cold, dry morning because the air is denser

C) Both would be equal because lift depends only on airspeed

Answer: B - Denser air increases performance; cold, dry air is denser than warm, humid air, so the aircraft will likely generate more lift.

Q18. Before a flight, you inspect your small UAS and discover a cracked propeller. What is the appropriate action?

A) Proceed with the flight if the crack is small and the aircraft is stable in a hover.

B) Fly at a slower speed to reduce stress on the propeller.

C) Replace the propeller before flight.

Answer: C - A preflight inspection is required to verify that the sUAS is in a condition for safe operation. A cracked propeller is a structural defect that could fail during flight and cause loss of control, so it must be replaced before any flight.

Q19. You are scheduled to conduct a mapping flight at a high-elevation site on a hot afternoon. Compared with flying the same mission on a cool morning, how will the aircraft's performance likely be affected?

A) The propeller will be more efficient because warm air is less viscous, increasing thrust.

B) The air will be less dense, resulting in reduced lift and thrust and poorer climb performance.

C) There will be no change in aircraft performance because density altitude only affects manned aircraft.

Answer: B - Higher temperature and higher altitude both contribute to lower air density. Less dense air reduces the lift generated by rotors and the thrust produced by propellers, so the aircraft will need longer takeoff rolls (or more power) and will have reduced climb performance.

Q20. During your preflight inspection, you discover a 1/2-inch crack in one of the propellers. What is the proper action?

A) File the crack smooth and continue if the prop balances

B) Clean the crack and monitor it during the flight

C) Replace the propeller before flight

Answer: C - A preflight inspection must ensure the sUAS is in a condition for safe operation. A cracked propeller is a structural defect that can lead to failure in flight. The only safe action is to replace the propeller before flying.

Q21. You are a remote pilot planning an afternoon flight in the middle of a hot, humid summer. How will these conditions affect your small UA's performance compared to a cool, dry morning?

A) Performance will increase because warm air is less viscous and flows more easily over the propellers.

B) Performance will decrease because high temperature and humidity both lower air density, increasing density altitude.

C) Performance will remain the same because temperature and humidity only affect manned aircraft, not small unmanned aircraft.

Answer: B - High temperature and high humidity decrease air density. Less dense air means reduced lift and thrust for a given rotor RPM, which degrades a drone's performance and increases density altitude.

Q22. You are flying your drone from a high-altitude location on a very hot day. What should you expect regarding the aircraft's ability to climb and hover?

A) Hot temperatures raise air density, giving the rotors more air to push and improving climb performance.

B) The lower air density requires the rotors to spin faster or the blade pitch to be greater to produce the same lift, so climb performance may be degraded.

C) High altitude and heat have no effect on small UA because they are powered by electric motors.

Answer: B - Both high altitude and high temperature decrease air density. With less dense air, rotors produce less thrust for the same RPM, which can reduce climb performance and payload capability.

Q23. Why does adding humidity to the air reduce the air density, thereby degrading drone performance?

A) Water vapor molecules are heavier than the nitrogen and oxygen molecules they replace, making the air more dense.

B) Water vapor displaces warmer air, which cools the atmosphere and reduces performance.

C) Water vapor is lighter than air, so moist air is less dense than dry air at the same temperature and pressure.

Answer: C - Water vapor (H_2O) has a lower molecular weight than the nitrogen and oxygen it displaces. Therefore, moist air is less dense, and lower density air degrades aerodynamic performance.

Q24. Two identical drones are flying at the same barometric pressure and the same temperature, but one is flying in humid air and the other in dry air. What is true about the drone flying in humid air?

A) It will generate more lift because water vapor increases the mass of the air.

B) It will generate the same lift because temperature and pressure are the same.

C) It will generate less lift for a given rotor speed because the humid air is less dense.

Answer: C - Humid air is less dense than dry air at the same temperature and pressure because water vapor is lighter than air. The lower density reduces the lift generated by the rotors at a given RPM.

Q25. According to the FAA study material, if the manufacturer has not provided performance data for your small unmanned aircraft, what should you do?

A) Assume the aircraft performs identically to all other small UAs on the market.

B) Do not fly the aircraft until the manufacturer publishes the data.

C) Use performance data that other users of the same model have published as a starting point.

Answer: C - The FAA guidance states that if manufacturer performance data is unavailable, it is advisable to seek out performance data published by other users of the same small UA model and use it as a starting point.

Q26. You are flying a small UA near an airport on a summer day. The reported density altitude is significantly higher than the actual altitude. What effect will this have on your drone?

A) The propellers will have less air to move, reducing thrust and requiring more rotor RPM to maintain hover.

B) The propellers will be more efficient because the lower pressure reduces drag on the blades.

C) The drone will gain extra lift because higher density altitude means the air is denser.

Answer: A - A high density altitude indicates less dense air. With fewer air molecules to accelerate, the rotors produce less thrust, so the drone requires a higher RPM to maintain lift, which consumes more battery power.

Q27. Assuming constant pressure, what happens to air density as temperature increases?

A) Air density increases proportionally with temperature.

B) Air density decreases as temperature increases.

C) Air density remains constant because pressure is the controlling factor.

Answer: B - Increasing the temperature of a substance decreases its density when pressure is held constant. Therefore, *warmer air is less dense, reducing aerodynamic performance.*

Q28. You are a remote pilot operating a small UA on a 100°F day with 90% relative humidity. How will these conditions most likely affect the aircraft's battery endurance?

A) Battery endurance will increase because the hot, humid air forces the motors to work more efficiently.

B) Battery endurance will remain unchanged because humidity has no effect on electric motors.

C) Battery endurance will likely decrease because the drone must use more power to maintain lift in the less dense air.

Answer: C - Hot, humid air is less dense. The rotors must spin faster to generate the same thrust, drawing more current from the battery and reducing overall flight time.

Q29. You are reviewing a sectional chart and see a solid blue line surrounding an airport with a small 'P' inside. What does this indicate, and can you fly your drone in this area?

A) A prohibited area; flight is absolutely prohibited at all times unless specifically authorized by the controlling agency.

B) A warning area; you may fly but should exercise caution due to potential hazards.

C) A controlled firing area; you may fly if you coordinate with the range control.

Answer: A - A solid blue line containing a 'P' on a sectional chart denotes a prohibited area (e.g., P-56 near Washington, D.C.). Entry into a prohibited area is absolutely forbidden for any aircraft, including drones, except as authorized by the appropriate agency. No other condition or coordination permits entry.

Q30. You are flying your small UA from a field at 6,000 feet MSL on a hot, dry afternoon. What should you expect regarding density altitude and aircraft performance?

A) Density altitude will be higher than field elevation, reducing aircraft performance.

B) Density altitude will be lower than field elevation, improving aircraft performance.

C) Density altitude will equal field elevation because the air is dry.

Answer: A - High temperatures increase density altitude above the actual field elevation, and the resulting lower air density degrades lift, thrust, and climb performance.

Q31. A weather report shows a relative humidity of 100% when the temperature is 40°F. Why is the effect of this moisture on air density likely to be small?

A) Cold air holds very little water vapor, so the moisture's density-reducing effect is small.

B) Saturated air is denser than dry air, so it cancels out any temperature effect.

C) High humidity always lowers density altitude, improving performance.

Answer: A - Because cold air cannot hold much water vapor, the actual amount of moisture in the air is low, and its effect on density is minimal.

Q32. Two identical drones are flying at the same altitude, one on a dry day and one on a humid day with all other conditions equal. What difference can the remote pilot expect?

A) The drone on the humid day will perform better because the air is thicker.

B) The drone on the humid day may have slightly reduced performance because moist air is less dense.

C) There will be no difference in performance because humidity does not affect air density.

Answer: B - Water vapor is lighter than air, so humid air is less dense than dry air, increasing density altitude and slightly *degrading performance.*

Q33. Your sUAS manufacturer's performance tables show takeoff distance at standard sea level conditions. On a day when the temperature is well above standard at your location, how should you use that information?

A) Use the tables exactly as published because the test conditions are standard.

B) Expect a longer takeoff distance because warm air is less dense and produces less lift.

C) Expect a shorter takeoff distance because warm air increases lift and thrust.

Answer: B - Higher temperature decreases air density, which degrades performance; the pilot should adjust expectations to account for higher density altitude and longer takeoff distance.

Q34. You are flying in a desert where the afternoon temperature has risen from 20°C to 40°C. This change will cause what effect on your drone's performance?

A) Better performance because air pressure rises with temperature.

B) No change in performance because pressure is unchanged.

C) Worse performance because air density decreases, raising density altitude.

Answer: C - Increasing temperature decreases air density, which raises density altitude and reduces the drone's aerodynamic performance.

Q35. Which combination of conditions would likely result in the lowest density altitude and therefore the best small UA performance?

A) High elevation, high temperature, high humidity.

B) Low elevation, low temperature, low humidity.

C) Low elevation, high temperature, low humidity.

Answer: B - Low elevation, cold temperatures, and dry air all increase air density, which lowers density altitude and improves performance.

Q36. On a 30°C day with 80% relative humidity, how does the air density compare to a dry day at the same temperature?

A) It is higher because water vapor is heavier than the other components of air.

B) It is lower because water vapor is lighter than air.

C) It is the same because temperature is the only factor that affects density.

Answer: B - Water vapor has a lower density than dry air, so the more moisture the air contains, the less dense it becomes, which increases density altitude.

Q37. Your drone's owner's manual provides takeoff and climb performance data. How should you use this data when launching from a high-altitude airport on a cold day?

A) Compare the field elevation and temperature to standard conditions and compute density altitude to estimate performance.

B) Assume the manufacturer's data is accurate for all altitudes and temperatures.

C) Multiply all performance figures by 1.5 for any airport above 2,000 feet MSL.

Answer: A - Performance data must be adjusted for density altitude, which accounts for local temperature and pressure; manufacturer data is not standardized and must be interpreted carefully.

Q38. You are preparing to fly during the heat of the day when the ambient temperature is significantly higher than usual. According to the study guide, how does this temperature affect the density of the air?

A) The density of the air increases.

B) The density of the air decreases.

C) The density of the air remains unchanged.

Answer: B - The study guide states that 'increasing the temperature of a substance decreases its density.' This relationship holds true at a constant pressure.

Q39. You are flying in a tropical region where the air is very humid. How does the water vapor content of the air impact the air's density and your drone's performance?

A) Moist air is heavier than dry air, which increases performance.

B) Moist air is lighter than dry air, which increases density altitude and decreases performance.

C) Humidity has no significant effect on density or performance.

Answer: B - The text explains that 'water vapor is lighter than air; consequently, moist air is lighter than dry air.' Therefore, increased water content reduces density, increases density altitude, and decreases performance.

Q40. As you ascend in the atmosphere, how do both temperature and pressure contribute to the density of the air?

A) Both temperature and pressure decrease, causing density to decrease.

B) Pressure decreases, causing density to increase, while temperature decreases, causing density to decrease.

C) Both temperature and pressure decrease, causing density to increase.

Answer: A - The guide notes that 'both temperature and pressure decrease with altitude and have conflicting effects.' However, the 'fairly rapid drop in pressure... usually has a dominating effect,' resulting in a decrease in density.

Q41. When reviewing the manufacturer's operational information for your drone, what is the primary reason you should understand the performance data provided?

A) To ensure you are using the exact same model as other operators for comparison.

B) To make practical use of the aircraft’s capabilities and limitations for safe and efficient operation.

C) To calculate the specific page number where the emergency procedures are located.

Answer: B - The text states it is 'essential to understand the significance of the operational data' to 'make practical use of the aircraft’s capabilities and limitations' such as takeoff, climb, and range.

Q42. You are analyzing a weather report that indicates high temperatures. How does this affect the air's capacity to hold water vapor?

A) High temperature causes the air to hold less water vapor.

B) High temperature allows the air to hold more water vapor.

C) Temperature does not affect the air's ability to hold water vapor.

Answer: B - The study guide states that 'warm air holds more water vapor, while cold air holds less.'

Q43. Comparing a dry environment to a moist environment under the same pressure conditions, which air is less dense?

A) Dry air is less dense.

B) Moist air is less dense.

C) Both are equally dense.

Answer: B - The text clarifies that 'water vapor is lighter than air; consequently, moist air is lighter than dry air.'

Q44. If the relative humidity of the air is 100%, what can you conclude about the water vapor content in the atmosphere?

A) The air is completely dry and contains no water vapor.

B) The air is saturated and cannot hold any more water vapor.

C) The air contains an equal amount of dry air and water vapor.

Answer: B - The study guide defines saturated air as 'which cannot hold any more water vapor' and notes that this state has a relative humidity of 100 percent.

Q45. You are planning a flight during a very humid, rainy day. Based on the relationship between humidity and density, how will this affect your drone's flight characteristics?

A) Performance will increase due to lower density altitude.

B) Performance will decrease due to higher density altitude.

C) Performance will remain exactly the same as on a dry day.

Answer: B - The text explains that 'as the water content of the air increases, the air becomes less dense, increasing density altitude and decreasing performance.'

Q46. You are planning a drone flight on a hot, humid afternoon. According to aerodynamic principles regarding humidity, how will this affect your aircraft compared to a cold, dry morning?

A) Air density will remain unchanged regardless of humidity.

B) Takeoff performance will improve due to increased air density.

C) Takeoff performance will decrease due to decreased air density.

Answer: C - Moist air is lighter than dry air because water vapor is lighter than air. As water content increases, air becomes less dense, increasing density altitude and decreasing performance.

Q47. A remote pilot is analyzing performance data and notes that the ambient temperature is rising while pressure remains constant. What is the effect on air density?

A) Air density decreases.

B) Air density increases.

C) Air density remains constant.

Answer: A - Temperature and density vary inversely at a constant pressure; increasing temperature decreases air density.

Q48. As altitude increases, both temperature and pressure decrease. Which factor has the dominant effect on the decrease of air density at higher altitudes?

A) The increase in water vapor content.

B) The decrease in temperature.

C) The decrease in pressure.

Answer: C - While both temperature and pressure decrease with altitude, a rapid drop in pressure has a dominating effect, causing density to decrease with altitude.

Q49. You are operating a small unmanned aircraft and notice the manufacturer’s specific performance data for climb rate is unavailable. What is the recommended procedure for determining performance capabilities?

A) Assume maximum performance capabilities based on the drone's general category.

B) Do not fly until the specific manufacturer data is obtained.

C) Seek out data determined and published by other users of the same model.

Answer: C - Since manufacturer-published performance data is not standardized, it is advisable to seek data from other users of the same model to use as a starting point.

Q50. Regarding relative humidity, which statement accurately describes the relationship between air temperature and water vapor holding capacity?

A) Cold air holds more water vapor than warm air.

B) Temperature does not affect the amount of water vapor the air can hold.

C) Warm air holds more water vapor than cold air.

Answer: C - Warm air holds more water vapor, while cold air holds less. This relationship is expressed as a percentage of the maximum amount the air can hold.

Q51. When calculating density altitude for flight planning, how is humidity classified in terms of its importance to aircraft performance?

A) It is a contributing factor, though usually not the main one.

B) It has no significant impact on aircraft performance.

C) It is the primary factor for calculating density altitude.

Answer: A - Humidity is a contributing factor in calculating density altitude and aircraft performance, though it is usually not considered the most important factor.

Q52. You are comparing two air samples with identical conditions. Sample A contains water vapor, and Sample B is perfectly dry. Which sample is denser?

A) Sample B (dry air) is denser.

B) Both samples have the same density.

C) Sample A (moist air) is denser.

Answer: A - Water vapor is lighter than air; therefore, moist air is lighter than dry air.

Q53. If you are flying at a high altitude where pressure is dropping rapidly, how does this generally affect air density?

A) Air density will increase.

B) Air density will remain constant.

C) Air density will decrease.

Answer: C - A fairly rapid drop in pressure as altitude increases usually has a dominating effect, causing density to decrease with altitude.

Q54. A remote pilot is conducting a pre-flight inspection on a hot summer day. As the temperature rises, what is the primary effect on the density of the air the drone must fly through?

A) The air density increases.

B) The air density decreases.

C) The air density remains constant.

Answer: B - According to the study material, increasing the temperature of a substance decreases its density. Thus, as temperature rises, air density decreases.

Q55. While flying in a humid environment, a remote pilot observes that the drone is not climbing as efficiently as it does on dry days. Why does this occur?

A) The increased water vapor in the air makes the air heavier, which decreases performance.

B) The increased water vapor in the air makes the air lighter, which decreases performance.

C) The increased water vapor increases air pressure, which helps the drone climb faster.

Answer: B - The study material states that water vapor is lighter than air, making moist air lighter than dry air. This results in increased density altitude and decreased performance.

Q56. A remote pilot is flying at a higher altitude and notices the drone's performance is different than at sea level. What factor is the primary cause of the density decrease at higher altitudes?

A) The rapid drop in pressure.

B) The temperature decrease.

C) The decrease in wind speed.

Answer: A - The text notes that while both temperature and pressure decrease with altitude, the rapid drop in pressure has a dominating effect on the density of the air.

Q57. Comparing warm air to cold air at the same altitude, which condition allows the atmosphere to hold more water vapor?

A) Cold air holds more water vapor than warm air.

B) Warm air holds more water vapor than cold air.

C) Both hold the same amount of water vapor.

Answer: B - The study material states that warm air holds more water vapor, while cold air holds less.

Q58. When monitoring weather conditions, what does relative humidity specifically measure?

A) The total weight of the air molecules.

B) The absolute difference between the current temperature and dew point.

C) The percentage of the maximum amount of water vapor the air can hold.

Answer: C - Relative humidity is defined as the amount of water vapor contained in the atmosphere expressed as a percentage of the maximum amount of water vapor the air can hold.

Q59. If a remote pilot encounters air that is completely saturated with water vapor, what is the relative humidity percentage?

A) 0 percent

B) 50 percent

C) 100 percent

Answer: C - Saturated air, which cannot hold any more water vapor, has a relative humidity of 100 percent.

Q60. A remote pilot needs to determine the range and endurance of their drone but does not have the manufacturer's official specifications. What is the best course of action?

A) Use the manufacturer's data as it is standardized for all drones.

B) Do not use performance data, as it is not standardized and unreliable.

C) Seek out performance data that has been determined and published by other users of the same drone model.

Answer: C - The text states that manufacturer-published performance data is not standardized. If unavailable, it is advisable to seek data from other users of the same model.

Q61. How does the presence of water vapor affect the weight of the air?

A) It makes the air heavier.

B) It makes the air lighter.

C) It has no effect on the weight of the air.

Answer: B - The material states that water vapor is lighter than air; consequently, moist air is lighter than dry air.

Q62. You are planning a flight on a hot summer day. According to the study material, how does increasing the temperature of the air affect the air density and the performance of your small UA?

A) Density decreases, which decreases performance.

B) Density increases, which improves performance.

C) Density remains constant, so performance is unaffected.

Answer: A - The text states that increasing the temperature of a substance decreases its density. This decrease in density results in decreased performance for the aircraft.

Q63. You are flying over water in conditions of high humidity. The study material notes that water vapor is lighter than air. How does this affect the air density and your drone's performance compared to flying in dry air?

A) Moist air is lighter, decreasing density and performance.

B) Moist air is heavier, decreasing density and performance.

C) Moist air is heavier, increasing density and performance.

Answer: A - Since water vapor is lighter than air, moist air is lighter than dry air. As the water content increases, the air becomes less dense, which increases density altitude and decreases performance.

Q64. You are flying at a higher altitude. According to the provided text, which atmospheric factor has a dominating effect on air density as altitude increases?

A) The constant pressure, which maintains density.

B) The decrease in temperature, which increases density.

C) The rapid drop in pressure, which decreases density.

Answer: C - The text states that while both temperature and pressure decrease with altitude, a fairly rapid drop in pressure usually has a dominating effect, causing density to decrease with altitude.

Q65. You are reviewing weather data and see a reading of 85% relative humidity. Based on the study material, what does this percentage represent?

A) The weight of water vapor currently in the air.

B) The rate at which water vapor is evaporating from the surface.

C) The percentage of maximum water vapor the air can hold at that specific temperature.

Answer: C - The text defines relative humidity as the amount of water vapor contained in the atmosphere, expressed as a percentage of the maximum amount of water vapor the air can hold.

Q66. Your drone manufacturer has not published specific performance data for your intended flight profile. According to the text, what is the best course of action to obtain performance data?

A) Use data from a completely different aircraft model.

B) Estimate performance based on a theoretical calculation.

C) Seek out performance data published by other users of the same model.

Answer: C - The text advises that if manufacturer-published data is unavailable, it is best to seek out performance data determined and published by other users of the same small UA manufacturer model.

Q67. When air is saturated with water vapor, it cannot hold any more water. At this point, what is the relative humidity percentage?

A) 0%

B) 50%

C) 100%

Answer: C - The text states that saturated air, which cannot hold any more water vapor, has a relative humidity of 100 percent.

Q68. The study material explains that density varies inversely with temperature at a constant pressure. If the temperature rises while pressure remains constant, what happens to the density of the air?

A) Density decreases.

B) Density remains the same.

C) Density increases.

Answer: A - The text explicitly states that increasing the temperature of a substance decreases its density, and this relationship is true at a constant pressure.

Q69. You are flying in conditions where the air contains the maximum amount of water vapor possible. How does this affect the drone's performance?

A) Performance increases because the air is lightest and least dense.

B) Performance decreases because the air becomes less dense.

C) Performance is unaffected because the density remains constant.

Answer: B - The text states that air is lightest (least dense) when it contains the maximum amount of water vapor. This lower density increases density altitude and decreases performance.

Q70. You are preparing to fly your small unmanned aircraft system (sUAS) on a hot summer day. According to aerodynamic principles, how does increasing the temperature of the air affect its density?

A) Density increases because the air molecules become heavier.

B) Density decreases because the air molecules spread further apart.

C) Density remains unchanged as temperature is a constant in flight operations.

Answer: B - The study material states that increasing the temperature of a substance decreases its density, provided the pressure remains constant. This inverse relationship is a key concept in aerodynamics for Part 107 pilots.

Q71. You are flying in a tropical environment where the humidity is very high. How does this moisture content affect the air density and your aircraft's performance?

A) The air becomes denser, which will improve climb performance and lift.

B) The air becomes less dense, increasing density altitude and decreasing performance.

C) The air density remains the same, but the aircraft will experience less drag.

Answer: B - The text explains that water vapor is lighter than air; therefore, moist air is lighter than dry air. As water content increases, air becomes less dense, which increases density altitude and decreases aircraft performance.

Q72. During a pre-flight briefing, a controller defines the air mass as having 100% relative humidity. What does this mean regarding the air's ability to hold water vapor?

A) The air is completely dry and cannot hold any water vapor.

B) The air is saturated and has reached its maximum capacity for water vapor.

C) The air holds more water vapor than warm air typically holds at sea level.

Answer: B - The study guide defines saturated air as having a relative humidity of 100%, meaning it cannot hold any more water vapor, as opposed to dry air (0% relative humidity).

Q73. As you ascend to higher altitudes in your drone, what is the primary effect of altitude on air density?

A) Air density decreases because the rapid drop in pressure has a dominating effect.

B) Air density increases because the pressure drop is negligible at high altitudes.

C) Air density remains constant regardless of the altitude change.

Answer: A - The study material states that while both temperature and pressure decrease with altitude, the rapid drop in pressure has a dominating effect, causing density to decrease with increasing altitude.

Q74. You are comparing the properties of air in two different weather scenarios. Why is moist air considered lighter or less dense than perfectly dry air?

A) Humidity actually increases the overall weight of the air mass significantly.

B) Water vapor is heavier than the nitrogen and oxygen molecules in the air.

C) Water vapor is lighter than the nitrogen and oxygen molecules in the air.

Answer: C - The provided text explicitly states that 'Water vapor is lighter than air; consequently, moist air is lighter than dry air.'

Q75. When calculating density altitude, which statement accurately reflects the role of temperature and humidity?

A) Temperature and humidity are the primary factors determining density altitude.

B) Temperature is the primary factor, while humidity is a contributing factor.

C) Humidity is the primary factor, while temperature has little to no effect.

Answer: B - The study material notes that temperature and pressure have conflicting effects on density, but temperature is usually the primary factor. Humidity is mentioned as a contributing factor, not the primary one.

Q76. You are planning a flight to a location with very cold temperatures. How does the cold temperature affect the air density compared to a warm day?

A) Cold air is less dense because cold air expands more than warm air.

B) Cold air is more dense because the air molecules are packed closer together.

C) Cold air density is the same as warm air density at the same altitude.

Answer: B - The text states that decreasing the temperature increases the density of air at a constant pressure, meaning cold air is more dense than warm air.

Q77. Which of the following scenarios best describes the relationship between air temperature and the amount of water vapor the air can hold?

A) Cold air holds more water vapor than warm air.

B) Warm air holds more water vapor than cold air.

C) Both warm and cold air hold the same amount of water vapor regardless of temperature.

Answer: B - The study material states that 'Warm air holds more water vapor, while cold air holds less,' explaining how the capacity for water vapor is temperature-dependent.

Q78. You are flying a small unmanned aircraft system (sUAS) on a hot summer afternoon. According to the relationship between temperature and air density, how will the increased temperature affect your aircraft's performance compared to a flight on a cold morning?

A) Performance will improve because hot air is denser.

B) Performance will be degraded because hot air is less dense.

C) Performance will remain unchanged as temperature does not significantly affect density.

Answer: B - The study material states that increasing the temperature of a substance decreases its density. Since hot air is less dense, this leads to a decrease in performance.

Q79. As you ascend to a higher altitude, you observe a drop in air pressure. According to the study guide, which factor usually has the dominating effect on air density as altitude increases?

A) The decrease in temperature has a dominating effect.

B) The decrease in pressure has a dominating effect.

C) The decrease in both temperature and pressure have equal effects.

Answer: B - The text explains that while both temperature and pressure decrease with altitude, a rapid drop in pressure usually has a dominating effect, causing density to decrease with altitude.

Q80. You are analyzing weather data for a flight in a tropical environment with high humidity. How does this moist air compare to perfectly dry air in terms of density?

A) Moist air is denser than dry air.

B) Moist air is lighter than dry air, making it less dense.

C) Moist air has no effect on density because water vapor is invisible.

Answer: B - The study guide notes that water vapor is lighter than air; consequently, moist air is lighter than dry air and becomes less dense as water content increases.

Q81. A remote pilot is reviewing the impact of humidity on aircraft performance. How does high humidity contribute to density altitude and performance?

A) It decreases density altitude and improves climb performance.

B) It increases density altitude but has no effect on performance.

C) It increases density altitude and decreases performance.

Answer: C - The text states that as water content increases, the air becomes less dense, which increases density altitude and decreases performance.

Q82. You have purchased a new small UA but the manufacturer has not provided published performance data. What is the recommended course of action regarding operational performance data?

A) Assume standard performance characteristics based on the drone's size.

B) Do not fly the aircraft until official data is downloaded from the manufacturer.

C) Seek out performance data that has been published by other users of the same model.

Answer: C - The study guide advises that if manufacturer-published data is unavailable, it is advisable to seek out data determined and published by other users of the same model.

Q83. When meteorologists report relative humidity, what are they describing?

A) The total volume of water vapor in the atmosphere.

B) The percentage of water vapor that is heavier than air.

C) The percentage of the maximum amount of water vapor the air can hold.

Answer: C - The study guide defines humidity (relative humidity) as the amount of water vapor contained in the atmosphere, expressed as a percentage of the maximum amount the air can hold.

Q84. According to the study material, how does the capacity of warm air to hold water vapor compare to cold air?

A) Warm air holds less water vapor than cold air.

B) Warm air holds the same amount of water vapor regardless of temperature.

C) Warm air holds more water vapor than cold air.

Answer: C - The text explicitly states that 'Warm air holds more water vapor, while cold air holds less.'

Q85. You are preparing for a flight and notice the weather report indicates a 100% relative humidity. What does this tell you about the air?

A) The air is perfectly dry.

B) The air cannot hold any more water vapor.

C) The air is heavier than standard air.

Answer: B - The study guide defines saturated air (100% relative humidity) as air that cannot hold any more water vapor.

Q86. You are planning a flight on a hot summer day. How does the increase in air temperature affect the air density and your drone's performance?

A) It increases air density, allowing for greater lift and endurance.

B) It decreases air density, which decreases lift and performance.

C) It has no effect on air density because pressure remains constant.

Answer: B - According to the study material, increasing the temperature of air decreases its density, which in turn decreases performance.

Q87. You are flying in a tropical region with high humidity. How does the presence of water vapor in the air affect the air density and your drone's capabilities?

A) The air becomes denser, increasing lift and battery efficiency.

B) The air becomes less dense, increasing density altitude and decreasing performance.

C) The air becomes heavier, requiring the drone to use more power to maintain altitude.

Answer: B - The study material states that water vapor is lighter than air, so moist air is lighter than dry air, leading to increased density altitude and decreased performance.

Q88. While flying at a higher altitude, you notice the air density is changing. Why is the air density expected to decrease with altitude despite the temperature also changing?

A) Because temperature decreases with altitude, which increases density.

B) Because pressure decreases rapidly with altitude and has a dominating effect on density.

C) Because the air pressure remains constant regardless of altitude.

Answer: B - The material explains that while both temperature and pressure decrease with altitude, the rapid drop in pressure usually has a dominating effect, causing density to decrease.

Q89. You are attempting to fly but realize the manufacturer's performance data for your drone is unavailable. What is the recommended course of action regarding operational data?

A) Ignore the performance data entirely and fly based on visual estimation.

B) Do not fly the drone at all as there is no reliable data available.

C) Seek out performance data that has been determined and published by other users of the same model.

Answer: C - The study guide states that if manufacturer data is unavailable, it is advisable to seek out data published by other users of the same model.

Q90. If you are flying in an area with saturated air (100% relative humidity), how does this condition affect the air compared to dry air under the same conditions?

A) The air is heavier and denser than dry air.

B) The air is lighter and less dense than dry air.

C) The air has no density difference from dry air.

Answer: B - The study material notes that air is lightest or least dense when it contains the maximum amount of water vapor *(saturated air).*

Q91. Which of the following operational performance data categories is explicitly listed as information provided by the manufacturer?

A) Takeoff, climb, range, endurance, descent, and landing data.

B) Federal airspace restrictions and airport frequency lists.

C) Cargo capacity and pilot weight limits.

Answer: A - The study guide lists operational performance data specifically pertaining to takeoff, climb, range, endurance, descent, and landing.

Q92. You are conducting a flight in a dry environment versus a humid environment. Which statement correctly describes the relationship between water vapor and air density?

A) Water vapor is heavier than air, making moist air denser.

B) Water vapor is lighter than air, making moist air less dense.

C) Water vapor and air density are not related to one another.

Answer: B - The material explicitly states that water vapor is lighter than air; consequently, moist air is lighter than dry air.

Q93. You are flying a small unmanned aircraft and need to understand the aircraft's capabilities. Which type of data provides the practical operational performance for safe flight operations?

A) Weather reports from local airports.

B) The pilot's personal logbook entries from previous flights.

C) Manufacturer-published operational and performance information.

Answer: C - The study guide emphasizes that using manufacturer-published performance data is essential for making practical use of the aircraft's capabilities and limitations.

Q94. A remote pilot is preparing to take off from a high-altitude airport during a summer day when temperatures are unusually high. According to aerodynamic principles regarding temperature and density, how will the high temperature affect the air density and the drone's performance?

A) Increase air density, resulting in better performance.

B) Decrease air density, resulting in reduced performance.

C) Increase air density, resulting in reduced performance.

Answer: B - The study material states that increasing the temperature of a substance decreases its density, which will reduce the drone's performance.

Q95. A remote pilot is operating a small UA in a tropical region with high humidity. Compared to flying in a dry climate, how does the moisture in the air affect the aircraft's performance capabilities?

A) It increases air density, improving performance.

B) It decreases air density, decreasing performance.

C) It has no effect on performance.

Answer: B - The study material notes that water vapor is lighter than air; therefore, moist air is lighter and less dense, which decreases performance.

Q96. A remote pilot is flying at a high altitude where pressure decreases rapidly. What is the general effect of increasing altitude on air density and drone

performance?

A) Air density increases, leading to improved performance.

B) Air density decreases, leading to reduced performance.

C) Air density remains constant.

Answer: B - While temperature decreases with altitude, the rapid drop in pressure usually has a dominating effect, causing the density to decrease with altitude.

Q97. A remote pilot observes that the air feels very warm on a summer day but is also humid. Why does warm air hold more water vapor than cold air?

A) Warm air molecules are more attracted to water vapor than cold air molecules.

B) Warm air is less dense, allowing more water vapor to exist.

C) Warm air has more kinetic energy and can hold a greater amount of water vapor.

Answer: C - The study material explicitly states that 'Warm air holds more water vapor, while cold air holds less.'

Q98. When a remote pilot checks the weather report and sees that the relative humidity is 100%, what does this indicate about the amount of water vapor in the air?

A) The air is saturated with water vapor.

B) The air is completely dry.

C) The air contains more water vapor than dry air.

Answer: A - The study guide defines saturated air as the condition where the air cannot hold any more water vapor.

Q99. A remote pilot purchases a used small UA but finds the operational manual is missing. To determine safe flight parameters, what should the pilot prioritize?

A) Seek out published performance data from other users of the same model.

B) Ignore performance data as it is not standardized.

C) Use estimates based solely on the drone's weight.

Answer: A - The study material advises that if manufacturer-published performance data is unavailable, it is advisable to seek out data determined and published by other users of the same model.

Q100. A remote pilot is calculating density altitude for a flight. Why is humidity often considered a minor factor compared to temperature and pressure?

A) Humidity actually increases air density, counteracting the effects of temperature.

B) Humidity is not a contributing factor to performance calculations.

C) While humidity is a contributing factor, it is usually not considered an important factor for calculating density altitude compared to temperature and pressure.

Answer: C - The text clarifies that 'Humidity alone is usually not considered an important factor in calculating density altitude and aircraft performance, but it is a contributing factor.'

Q101. A remote pilot compares flight performance between a cold morning and a hot afternoon at the same altitude and pressure. Which statement best describes the relationship between temperature and density?

A) As temperature increases, density increases.

B) As temperature increases, density decreases.

C) Density remains constant regardless of temperature changes.

Answer: B - The study material states that increasing the temperature of a substance decreases its density (at constant pressure).

Chapter 2: Airspace & Charts

Q102. What is the primary characteristic of controlled airspace?

A) Airspace that is restricted due to specific hazards or military operations.

B) Airspace below 1,200 feet AGL where ATC has the right to control traffic.

C) Airspace where ATC has the right to control air traffic.

Answer: C - Controlled airspace is defined in the study material as airspace in which ATC has the right to control air traffic.

Q103. What is Special Use Airspace (SUA)?

A) Airspace designated for VFR flight only.

B) Areas containing specific hazards or restricted activities.

C) Airspace that has no specific hazards to aircraft operations.

Answer: B - The study material defines Special Use Airspace as areas containing specific hazards or restricted activities.

Q104. What is an Alert Area?

A) An area with a high volume of flight training or unusual aerial activity.

B) An area restricted to military operations only.

C) An area with a specific hazard to aircraft, such as aerial spraying.

Answer: A - An Alert Area is defined in the 'Other Airspace Areas' section as an area with a high volume of flight training or unusual aerial activity.

Q105. What is the primary function of Air Traffic Control (ATC) in the National Airspace System?

A) Providing separation and control of aircraft.

B) Issuing weather forecasts and aviation charts.

C) Selling airline tickets and managing airport infrastructure.

Answer: A - The 'Air Traffic Control and the National Airspace System' section indicates ATC's role is providing separation and control.

Q106. What is the primary characteristic of Visual Flight Rules (VFR)?

A) Flying only in clear weather conditions.

B) Navigation with respect to the surface.

C) Following instrument approach procedures.

Answer: B - The 'Visual Flight Rules (VFR) Terms & Symbols' section defines VFR as navigation with respect to the surface.

Q107. What is a Notice to Airmen (NOTAM)?

A) A notice containing information concerning the establishment, condition, or change in any NAS facility, service, procedure, or hazard.

B) A notice about airport runway construction schedules.

C) A notice regarding the issuance of new pilot certificates.

Answer: A - The 'Notices to Airmen (NOTAMs)' section defines a NOTAM as a notice containing information concerning the establishment, condition, or change in any National Airspace System (NAS) facility, service, procedure, or hazard.

Q108. What does the chapter 'Airspace Classification, Operating Requirements, and Flight Restrictions' primarily cover?

A) The history of aviation and aircraft construction.

B) Classification, operating requirements, and flight restrictions.

C) Only the specific regulations for helicopters.

Answer: B - This is directly stated in the chapter title provided in the study material excerpt.

Q109. What is Uncontrolled Airspace?

A) Airspace below 500 feet AGL where ATC has no authority.

B) Airspace where ATC does not have the right to control air traffic.

C) Airspace that is permanently restricted due to environmental hazards.

Answer: B - The 'Uncontrolled Airspace' section refers to airspace where ATC does not have the right to control air traffic.

Q110. What is a Military Operation Area (MOA)?

A) An area with a high volume of flight training or unusual aerial activity.

B) An area established for military training purposes.

C) An area where pilots must remain in visual contact with the ground at all times.

Answer: B - The 'Other Airspace Areas' section lists Military Operation Areas as areas established for military training purposes.

Q111. You are approaching an airport that has an operating control tower and a control zone designated as Class C airspace. What is the primary requirement for operating a small UA in this airspace?

A) You must fly at or above 2,500 feet AGL to avoid the control zone.

B) You must contact ATC immediately and receive clearance to enter.

C) You must maintain visual separation from other aircraft and follow VFR visibility and cloud clearance rules.

Answer: C - In Class C airspace, a remote pilot operating under Visual Flight Rules (VFR) must maintain visual separation from other aircraft and adhere to VFR visibility and cloud clearance requirements. While radio contact is recommended, the primary requirement is maintaining visual separation and VFR flight rules.

Q112. You are planning a flight over an area where there is a high volume of flight training and unusual aerial activity. This area is likely designated as what type of airspace?

A) Prohibited Area

B) Restricted Area

C) Alert Area

Answer: C - Alert Areas indicate areas where there is a high volume of flight training or unusual aerial activity. The purpose is to inform non-participating pilots of these activities, not to restrict their flight.

Q113. You are checking for any potential hazards or changes to the National Airspace System before a flight. Which publication provides this information?

A) Notices to Airmen (NOTAMs)

B) The National Weather Service (NWS) forecasts

C) The Federal Aviation Regulations (FARs)

Answer: A - Notices to Airmen (NOTAMs) contain information about temporary hazards or changes to facilities, services, and procedures that could affect the safety of flight. Pilots must check for current NOTAMs before every flight.

Q114. While flying over the coast, you enter a Warning Area. What does this designation primarily indicate?

A) The area is permanently closed to all civilian aviation.

B) It is a designated area of US airspace extending from 3 nautical miles to 12 nautical miles off the coast.

C) It is an area where US military operations pose a hazard to non-participating aircraft.

Answer: C - Warning Areas indicate areas where hazardous military operations occur, extending from 3 to 12 nautical miles off the coast. They alert non-participating pilots to the activity and potential hazards.

Q115. You are flying over a rural area with no control tower and no designated airspace boundaries. You are in Uncontrolled Airspace (Class G). What is the minimum altitude requirement?

A) You may fly at any altitude as long as you stay 500 feet away from people and property.

B) You must maintain 500 feet above the ground and comply with VFR visibility and cloud clearance rules.

C) You must remain at least 1,000 feet above the ground.

Answer: B - In uncontrolled airspace, remote pilots must still adhere to the general VFR minimums: flying at or above 500 feet above the ground, maintaining 500 feet clearance from people/property, and complying with visibility and cloud clearance rules.

Q116. You are reading a sectional chart and see a shaded area designated as a Military Operations Area (MOA). What does this indicate?

A) No aircraft are allowed to fly in this area at any time.

B) Military training exercises occur frequently in this area.

C) Commercial air traffic is restricted during peak hours.

Answer: B - MOAs are established areas where military activities are conducted. They are used to segregate these activities from civilian traffic, though civilian aircraft are permitted to fly through them, but pilots should be aware of potential activity.

Q117. You are flying under VFR near a major airport with Class B airspace. The floor of the Class B airspace is 1,000 feet above ground level (AGL). What is the minimum altitude requirement to enter this airspace?

A) You must be below 1,000 feet AGL to avoid the airspace.

B) You must maintain 1,000 feet AGL to enter the airspace.

C) You must remain clear of other aircraft and follow VFR rules.

Answer: C - When operating in Class B airspace, a remote pilot under VFR must maintain visual separation from other aircraft and adhere to all applicable VFR rules, including visibility and cloud clearance requirements.

Q118. You see a Restricted Area on your chart. How does a Restricted Area differ from a Prohibited Area?

A) Restricted areas are permanently closed to all aircraft, while prohibited areas can be entered with permission.

B) Restricted areas may be authorized for access by persons or agencies under certain conditions, while prohibited areas are never open to civil aircraft.

C) Prohibited areas are only active during the day, while restricted areas are active 24 hours a day.

Answer: B - The key difference is that access to a Restricted Area is permitted to persons or agencies authorized by the controlling agency under certain conditions. Access to a Prohibited Area is strictly prohibited to civil aircraft.

Q119. You are flying over a rural area at 1,200 feet AGL. The chart indicates that Class E airspace begins at this altitude. What is the primary operating requirement for this flight?

A) You must maintain visual separation from other aircraft.

B) You must have an instrument flight rules (IFR) plan.

C) You must contact ATC immediately to receive a clearance.

Answer: A - Class E airspace is controlled airspace, but when operating under VFR, the pilot must maintain visual separation from other aircraft and adhere to VFR visibility and cloud clearance rules.

Q120. You are planning to fly your sUAS within the lateral boundaries of a Class C airport's airspace. What must you do before entering that airspace?

A) Establish two-way radio communication with ATC

B) Equip the aircraft with ADS-B Out

C) Post a notice to airmen (NOTAM) with the airport tower

Answer: A - Class C airspace requires the remote pilot to establish two-way radio communication with ATC before entering. ADS-B Out is not required for sUAS operations in Class C airspace, and NOTAMs do not provide authorization.

Q121. You need to operate your sUAS within the controlled airspace of a Class D airport. How can you obtain the required authorization?

A) Call the control tower and receive a verbal clearance before launching

B) Use an FAA-approved UAS service supplier, such as LAANC, to request authorization

C) File a flight plan with the nearest Flight Service Station

Answer: B - For operations in controlled airspace, the remote pilot must obtain an airspace authorization from the FAA, commonly done through LAANC (Low Altitude Authorization and Notification Capability). Verbal approval from a tower or a filing a flight plan does not satisfy the requirement.

Q122. You plan to conduct a Part 107 drone flight in the vicinity of a Class C airport. What must you do before operating?

A) Obtain authorization from ATC before operating in the Class C airspace

B) Fly only at or below 400 feet and no special authorization is needed

C) Submit a NOTAM to the airport traffic control tower

Answer: A - Under Part 107, operations in controlled airspace (Classes B, C, D, and E) require prior ATC authorization. Simply staying below 400 feet does not allow operations in Class C airspace without approval.

Q123. You are planning a commercial sUAS flight over undeveloped rural land. The sectional chart shows no airspace markings around the area (no magenta or blue shading, no dashed lines). You intend to fly at 100 ft AGL. What type of airspace are you operating in, and do you need ATC authorization?

A) Class E controlled airspace; authorization is required

B) Class B airspace; authorization is required

C) Class G uncontrolled airspace; no ATC authorization is required

Answer: C - Class G airspace is uncontrolled and exists anywhere not designated as controlled (e.g., Class A, B, C, D, or E). In this scenario, with no charted controlled airspace, the surface to 700 or 1200 ft AGL is Class G, so Part 107 operations may proceed without ATC authorization.

Q124. You are hired to inspect a cell tower located inside the outer ring (shelf) of Class C airspace, but not within the inner core. The weather is clear. According to Part 107, what must you do before flying?

A) Obtain an ATC authorization (e.g., LAANC) for the flight

B) No action is needed because the outer ring is not controlled

C) File a VFR flight plan and squawk a transponder code

Answer: A - Class C airspace is controlled airspace from the surface to its designated top, including the outer shelf. Part 107 requires remote pilots to obtain an airspace authorization from ATC before operating in Class B, C, D, or surface-based Class E airspace.

Q125. A temporary flight restriction (TFR) has been issued for a major sporting event. Your proposed flight is 3 miles away from the stadium, but the TFR covers that area. What is the correct course of action?

A) Proceed because the TFR only applies to manned aircraft

B) Contact the event organizer to get permission

C) Do not fly in the TFR area until it expires or the restriction is removed

Answer: C - TFRs are issued to restrict all aircraft, including UAS, from operating in a specified area. Part 107 pilots are responsible for checking NOTAMs and must not operate in an area subject to a TFR unless they are the approving agency or have explicit authorization.

Q126. You see a small airport on the sectional chart with a magenta dashed circle drawn around it. This indicates Class E airspace starting at the surface. You intend to fly at 150 ft AGL near that airport. What is required before you begin operations?

A) Hire a visual observer because the flight is near an airport

B) Ensure you remain clear of the traffic pattern and no authorization is needed

C) Obtain ATC authorization because this is controlled airspace

Answer: C - A magenta dashed circle denotes Class E airspace that begins at the surface. This is controlled airspace, so Part 107 remote pilots must obtain an ATC authorization (often via LAANC) before flying in that area, regardless of altitude.

Q127. You are checking NOTAMs before a construction site survey. A NOTAM is issued for a temporary crane within your planned flight area. What should you do?

A) Review the NOTAM and adjust your flight plan to avoid the hazard if necessary

B) Cancel the flight because any NOTAM prohibits all UAS operations

C) Ignore the NOTAM because it only applies to manned aircraft operations

Answer: A - NOTAMs alert pilots to hazards such as cranes, parachute jumps, or TFRs. Part 107 remote pilots are required to check NOTAMs before flight and must ensure they do not operate in areas where a NOTAM indicates unsafe conditions, adjusting their plan as needed.

Q128. You are planning a Part 107 flight to inspect a cell tower located within the inner ring of a Class C airport's airspace, below 4,000 feet AGL. What action must you take before launching your drone?

A) File a flight plan with the local Flight Service Station.

B) Obtain an airspace authorization from Air Traffic Control, typically via LAANC or the FAA DroneZone.

C) Contact the airport tower on the CTAF frequency and announce your position.

Answer: B - Class C airspace is controlled airspace. Under Part 107 (§107.41), you must obtain ATC authorization before operating in controlled airspace. LAANC or DroneZone are standard methods for obtaining this authorization.

Q129. While reviewing preflight information, you discover a Temporary Flight Restriction (TFR) has been issued for a VIP movement near your planned flight area. The TFR is active during your planned flight time. What must you do?

A) You may fly as long as you remain below 400 feet AGL and keep the drone within visual line of sight.

B) You may enter the TFR if you operate the drone from an FAA registered aircraft.

C) You must not operate the drone within the TFR unless you have specific authorization from the issuing agency.

Answer: C - TFRs are airspace restrictions issued to protect people, property, or operations. Operating contrary to an active TFR is a violation. You may only enter if authorized by the issuing agency (e.g., ATC, Secret Service, military).

Q130. You are flying your drone in a remote rural location. The airspace is Class G from the surface up to 1,200 feet AGL, with Class E airspace above. You are flying at 200 feet AGL. Do you need an ATC authorization for this operation?

A) Yes, because the flight is under controlled airspace.

B) No, because Class G is uncontrolled airspace and ATC authorization is not required.

C) Yes, because you are above 100 feet AGL.

Answer: B - Class G is uncontrolled airspace. Under Part 107, ATC authorization is only required for controlled airspace (Classes B, C, D, or E). Since you are below the Class E floor, you are in Class G and no authorization is needed.

Q131. You are planning a flight along a route where the Sectional Chart shows Class E airspace with a floor of 1,200 feet AGL. You intend to fly at 350 feet AGL. Which airspace class are you operating in?

A) Class D airspace.

B) Class E airspace.

C) Class G airspace.

Answer: C - The floor of Class E is 1,200 feet AGL in this area, so any altitude below that floor is Class G uncontrolled airspace. At 350 feet AGL you are in Class G.

Q132. As part of your preflight planning for a Part 107 flight near a rural airport, which is the most appropriate source for checking airspace NOTAMs that could affect your drone operation?

A) The airport's automated weather observing system (AWOS).

B) The FAA's NOTAM search tool or Flight Service (e.g., 1800wxbrief.com).

C) The local newspaper.

Answer: B - Remote pilots must be aware of NOTAMs before flight. The FAA publishes NOTAMs through flight planning tools, mobile apps, and the official NOTAM search, so you can identify temporary hazards or restrictions.

Q133. Before operating a small UAS within any portion of Class B airspace, what must the remote pilot in command obtain?

A) A waiver from the FAA Administrator for all Part 107 operations.

B) An automatic altitude readout displayed in the ground station.

C) A specific ATC authorization for the planned altitude and location, usually obtained through LAANC or DroneZone.

Answer: C - Part 107 §107.41 requires an ATC authorization to operate in Class A, B, C, D, or E airspace. Class B surrounds the busiest airports, so the authorization is mandatory for any drone flight within its boundaries.

Q134. Which of the following airspace classes is considered uncontrolled airspace?

A) Class B

B) Class C

C) Class G

Answer: C - Controlled airspace includes Class A, B, C, D, and E. Class G is uncontrolled airspace, meaning there is no ATC authority to regulate IFR or VFR flights; however, Part 107 rules still apply.

Q135. You are checking a Sectional Chart and notice a Restricted Area (e.g., R-2508) near your proposed flight path. What does this mean for your drone operation?

A) You may fly through the Restricted Area if you stay below 400 feet AGL.

B) The area may contain hazardous activities; flight is prohibited unless the area is not active or you have authorization from the controlling agency.

C) You can fly through as long as you maintain ADS-B out.

Answer: B - Restricted Areas are special use airspace where flight may be hazardous and is generally subject to restriction. You should check the active status/schedule and obtain authorization if required, or avoid the area entirely.

Q136. You plan to fly a small UAS within 5 miles of a Class D airport. What must you do before starting the flight?

A) Operate only below 400 feet AGL to avoid controlled airspace.

B) Obtain an airspace authorization from Air Traffic Control (ATC).

C) Call the airport manager and request permission.

Answer: B - Class D airspace is controlled airspace from the surface to a certain altitude. Operations in Class D require prior authorization from ATC through the FAA's approved process. Flying below 400 feet does not remove the need for authorization inside controlled airspace.

Q137. You are hired to photograph a real estate property that lies within the lateral boundaries of a Class C airport's surface area. You are launching your sUAS from the property before coordinating with air traffic control. What must you do before beginning the flight?

A) Obtain authorization from ATC prior to operating in Class C airspace

B) Only call the airport operator to notify them

C) Avoid any aircraft by staying below 200 feet AGL

Answer: A - Class C airspace is controlled airspace. Under Part 107, operations in controlled airspace require prior ATC authorization. Notifying the airport operator is not sufficient, and there is no blanket altitude exemption.

Q138. You plan to fly a quadcopter through a valley in Class G airspace for agricultural monitoring. The nearest weather reporting station indicates visibility is 2.5 statute miles. What should you do?

A) Fly lower than 100 feet to avoid the visibility requirement

B) Proceed because Class G airspace has no visibility requirement for sUAS

C) Cease operations because visibility is below the required 3 statute miles

Answer: C - Part 107 requires the remote PIC to maintain at least 3 statute miles of flight visibility. With weather reporting showing 2.5 miles visibility, you cannot legally operate under Part 107, regardless of airspace class.

Q139. You plan to fly your small UAS within the lateral boundary of the Class C airspace surrounding a busy airport, at an altitude of 300 feet AGL. The airport has an operating control tower. What action must you take before beginning the operation?

A) No action required because operations below 400 feet AGL are automatically permitted

B) Call the control tower on the common traffic advisory frequency and receive verbal permission

C) Obtain an airspace authorization through an approved FAA Low Altitude Authorization and Notification Capability (LAANC) service provider

Answer: C - Class C airspace is controlled airspace, so Part 107 operations require FAA authorization. LAANC provides near-real-time authorization for operations in many controlled airspace areas. Verbal permission from the tower is not sufficient.

Q140. You are operating in an area where no controlled airspace exists below 1,200 feet AGL (Class G). The visibility is 2.5 statute miles. You plan to fly at 300 feet AGL. Can you legally operate?

A) Yes, because there are no weather minimums for sUAS in Class G airspace

B) No, because visibility is less than 3 statute miles

C) Yes, because 2.5 SM is acceptable for drones

Answer: B - Part 107 requires that small UAS operations maintain at least 3 statute miles visibility, regardless of airspace class. With only 2.5 SM visibility, you cannot operate.

Q141. A temporary flight restriction (TFR) has been issued for the President's motorcade near your planned flight route. The TFR extends from the surface to 3,000 feet AGL. You want to fly at 400 feet AGL. What should you do?

A) Wait until the TFR has expired before operating in that airspace

B) Fly above the TFR ceiling at 3,100 feet AGL

C) Fly under the TFR ceiling at 400 feet AGL

Answer: A - A published TFR is a flight restriction, and entering it is not permitted. Altitude restrictions in the NOTAM must be respected; you cannot fly below or above the TFR ceiling; you must avoid the area entirely until lifted.

Q142. You are planning a flight through a Military Operations Area (MOA) that is not currently active. There are no other airspace restrictions in the area. Is it legal to operate your small UAS there?

A) No, MOAs are permanent restricted airspace

B) Yes, but you should exercise caution and, if possible, avoid the MOA even when inactive

C) Only with approval from the controlling military base

Answer: B - An MOA is not prohibited airspace for small UAS, but a high volume of military operations may occur. Pilots should avoid MOAs when active and maintain caution at all times.

Q143. The weather minimums are: ceiling 800 feet overcast, visibility 4 SM. You plan to fly at 200 feet AGL. How far will your drone be from the cloud base, and is this legal under Part 107?

A) It will be 600 feet below the clouds, which satisfies the minimum of 500 feet below clouds

B) It will be 200 feet below the clouds, which is illegal

C) It will be 1,000 feet below the clouds, which is legal

Answer: A - Below 3,000 feet AGL, Part 107 requires the UAS to remain at least 500 feet below a cloud and 2,000 feet horizontally. At 200 feet AGL with an 800-foot ceiling, you are 600 feet below the clouds, so the operation is legal.

Q144. Your planned route takes you inside the surface area of Class D airspace, which is surrounded by uncontrolled (Class G) airspace. Which statement is true about operating your drone in this area?

A) You may fly in either Class D or Class G without authorization, as long as you stay below 400 feet AGL

B) You must have an FAA airspace authorization to fly inside the Class D surface area, but you do not need one for the adjacent Class G airspace

C) You need approval from the local police department before flying in Class D airspace

Answer: B - Class D is controlled airspace from the surface to a specified altitude. Part 107 operations inside Class D require prior FAA authorization, while operations in Class G do not. Local police approval is not an FAA authorization.

Q145. You are flying at 500 feet AGL in a rural location where the floor of the overlying Class E airspace begins at 700 feet AGL. What type of airspace is your drone currently in?

A) Class G (uncontrolled) airspace

B) Class E (controlled) airspace

C) Class B (controlled) airspace

Answer: A - Class E airspace often has a floor at 700 feet AGL over certain areas. If you are below that floor, you are in Class G airspace, which is uncontrolled and does not require an airspace authorization.

Q146. A NOTAM has been issued for a restricted area that will be used for live-fire military exercises. You see the area on a chart as a purple hatched shape. What is your obligation as a remote pilot?

A) You may fly over the restricted area at 400 feet AGL if the exercise is not in your altitude band

B) You must avoid the restricted area when the NOTAM says it is active; you cannot enter it without permission from the controlling agency

C) You can enter the restricted area as long as you remain clear of clouds and maintain visual line of sight

Answer: B - Restricted airspace is special use airspace where flight is subject to restrictions, typically when active. The NOTAM identifies the activity. Safe and legal operation requires avoiding the area when active unless you have permission from the using agency.

Q147. You plan to fly your sUAS within 3 nautical miles of a Class C airport. What must you have before beginning the flight?

A) A visual observer for the entire flight

B) Air traffic control authorization obtained through proper procedures

C) Written permission from the airport manager

Answer: B - Operating in Class B, C, or D airspace requires ATC authorization. Flight within 3 NM of a Class C airport is within Class C airspace (usually), so you must obtain ATC authorization prior to the operation.

Q148. You are operating in Class G airspace where the ground elevation is 1,200 ft MSL. What is the maximum altitude you may fly your small UAS?

A) 1,700 ft MSL

B) 1,500 ft MSL

C) 1,600 ft MSL

Answer: C - Part 107.51 limits small UAS operations to 400 feet above ground level (AGL). With terrain at 1,200 ft MSL, 1,600 ft MSL is exactly 400 ft AGL and is the maximum allowed altitude.

Q149. You request ATC authorization through LAANC to operate your small UAS in Class B airspace near a busy airport. What does receiving this authorization allow you to do?

A) Operate in that airspace for the requested time and location with ATC approval

B) Operate beyond visual line of sight within the authorized area

C) Exceed the 400-foot AGL limit if you use a visual observer

Answer: A - Part 107.41 requires ATC authorization before operating in Class B, C, D, or surface-based Class E airspace. LAANC provides near-real-time authorization for a specific time and location, but it does not waive visual line of sight or altitude requirements.

Q150. You plan to fly a small UAS inside the Class D airspace surrounding a towered airport to inspect a cell tower. Under Part 107, what must you obtain before beginning the operation?

A) ATC authorization from the controlling facility

B) A certificate of waiver from the FAA regional office

C) Permission from the airport manager and a filed flight plan

Answer: A - Part 107.41 prohibits operating in controlled airspace, including Class D, without authorization from the appropriate ATC facility. LAANC or another FAA-approved method is used to request this authorization, not a waiver.

Q151. You are planning a drone flight that will take you inside the lateral boundary of a Class C airport's surface area. What must you do before entering this airspace?

A) Obtain an ATC authorization (e.g., LAANC) and comply with any ATC instructions.

B) Do nothing, since Class C airspace is uncontrolled below 400 feet AGL.

C) Equip your drone with a Mode C transponder and squawk 1200.

Answer: A - Class C airspace is controlled airspace. Under Part 107, you must obtain an ATC authorization (often through LAANC) before operating in Class B, C, D, or surface-level Class E airspace. Simply remaining under 400 feet AGL does not exempt you from this requirement.

Q152. You are flying a drone in an area you have confirmed is Class G airspace. What is the primary difference from flying in controlled airspace?

A) You do not need ATC authorization, but you must still maintain visual line of sight and remain clear of clouds.

B) You must file a VFR flight plan with ATC before launching.

C) You are allowed to fly higher than 400 feet AGL without a waiver.

Answer: A - Class G is uncontrolled airspace. No ATC clearance or authorization is required, but Part 107 rules continue to apply, including the 400-foot AGL ceiling (or 400 feet above a structure), ≥3 statute miles visibility, and cloud clearance requirements.

Q153. While reviewing a sectional chart, you see a large MOA (Military Operations Area) along your planned drone route. When can you operate your drone inside the MOA?

A) Unless flying above 400 feet AGL, there are no restrictions.

B) You may fly at any time because MOAs are not restricted airspace.

C) You should avoid the MOA when it is active and check NOTAMs or Flight Service for activity times.

Answer: C - A MOA denotes airspace where military training activities occur. It is not a 'prohibited' area, but drone pilots should avoid it when active. Always check NOTAMs and other sources for whether the MOA is hot before flying near or through it.

Q154. Before taking off for a real-estate photo shoot, you review NOTAMs and find a Temporary Flight Restriction (TFR) that applies to your intended area. What is your correct course of action?

A) Proceed with the flight as long as you stay below 400 feet AGL.

B) Contact Flight Service to receive a standard remote pilot exemption.

C) Refrain from operating in the TFR unless you have authorization from the issuing agency.

Answer: C - A TFR restricts drone and manned aircraft operations in a specified area for safety or security reasons. Unless you have specific authorization from the issuing agency (e.g., FAA or the entity that requested the TFR), you must not enter the airspace covered by the TFR.

Q155. You intend to launch your drone just outside a Class D airport's surface boundary. The control tower is active. What is required before you can operate in the Class D airspace?

A) You may fly outside the boundary without ATC coordination, but you cannot penetrate the airspace without authorization.

B) You must establish two-way radio communication with ATC and receive explicit clearance before each entry.

C) No authorization is required because Class D airspace is uncontrolled at night.

Answer: A - Class D airspace is controlled. If you remain entirely outside its lateral boundary, you may fly without ATC authorization in the surrounding uncontrolled airspace (subject to other Part 107 rules). To enter the Class D airspace, you *must obtain ATC authorization, typically via LAANC or a prior ATC approval.*

Q156. On a VFR sectional chart, a solid blue line forms a large circular pattern around a major city. Which airspace class does this represent?

A) Class B airspace.

B) Class C airspace.

C) Class D airspace.

Answer: A - Class B airspace is depicted by a solid blue line on sectional charts. It typically surrounds the busiest airports and extends from the surface to 10,000 feet MSL. Flying a drone in Class B requires specific ATC authorization.

Q157. On a sectional chart, a dashed magenta vignette surrounds an airport. What does this symbol indicate?

A) Class E airspace begins at the surface.

B) Class G airspace begins at the surface.

C) Class D airspace begins at the surface.

Answer: A - The dashed magenta line (vignette) on a VFR sectional indicates Class E airspace that starts at the surface. This is controlled airspace, so Part 107 remote pilots must have ATC authorization to operate within it.

Q158. Under Part 107, in which of the following airspace classes can a remote pilot fly without ATC authorization, assuming the flight stays in uncontrolled airspace and complies with all other Part 107 requirements?

A) Class G airspace.

B) Class B airspace.

C) Class C airspace.

Answer: A - Class G is the only uncontrolled airspace class listed. It normally exists beneath a controlled airspace ceiling or in areas with no designated controlled airspace. Drone operations in Class G do not require an ATC authorization, but must still comply with Part 107 rules such as VLOS and altitude limits.

Q159. You are planning to fly your small UAS within 4 miles of a Class C airport but just outside its controlled airspace. What is required before you begin the flight?

A) Obtain ATC authorization from the control tower

B) Call the airport manager and file a flight plan

C) No action is required because you are outside the Class C airspace

Answer: C - ATC authorization is required when operating in Class B, C, or D airspace or within the lateral boundaries of Class E surface areas designated for an airport (14 CFR §107.41). If you are outside that controlled airspace, no ATC authorization is needed.

Q160. You plan to operate a small UAS within the 5-mile radius of a Class C airport, below the floor of the Class C airspace, but inside the outer ring shown on the sectional chart. What must you do before flying?

A) No ATC authorization is required because you are below the Class C airspace floor.

B) File a flight plan and obtain a waiver from the FAA.

C) Obtain ATC authorization from the Class C airspace facility before operating.

Answer: C - Class C airspace typically has a surface area extending from the surface to a certain altitude, and an outer area with a floor usually at 1,200 or 2,000 ft AGL. If you are within the lateral boundary of the Class C airspace or its outer ring, you must have ATC authorization from the controlling facility before operating a UAS in controlled airspace, even if below the floor of the inner core, depending on the specific chart. For Part 107, any operation in controlled airspace (Class B, C, D, or E with surface area) requires prior ATC authorization.

Q161. You are preflight planning and see a dashed blue circle on your sectional chart surrounding a small airport. What does this symbol indicate, and what is the operational impact for your drone flight?

A) Class D airspace; you must obtain ATC authorization before flying within it.

B) Class B airspace; you must obtain a waiver from the FAA to enter.

C) Class G airspace; no ATC authorization is needed, but you must remain clear of clouds.

Answer: A - A dashed blue circle on a sectional chart depicts Class D airspace. Class D airspace is controlled airspace, and under Part 107 you must receive ATC authorization from the controlling tower before operating a small UAS within its lateral boundaries.

Q162. While checking local NOTAMs before a flight, you see a temporary flight restriction (TFR) has been issued for a major sporting event near your planned route. What is the appropriate course of action?

A) Cancel or reroute your flight to stay clear of the TFR airspace.

B) Fly at a higher altitude to avoid the TFR, or descend below the TFR floor.

C) Proceed as planned since the TFR only applies to manned aircraft.

Answer: A - Temporary flight restrictions (TFRs) apply to all aircraft, including UAS, within the defined area. You must not operate within a TFR unless you have specific authorization from the issuing authority. The correct action is to avoid the restricted area by canceling or rerouting your flight.

Q163. You are flying a drone in an area that appears on a sectional chart as a magenta-shaded region (with a solid magenta boundary) around an airport. The floor of this airspace is noted as 700 ft AGL. At your planned altitude of 400 ft AGL, what is the airspace classification, and what are the authorization requirements?

A) This is Class G airspace; you may fly without ATC authorization.

B) This is Class E airspace; ATC authorization is required only if the area is marked as a surface area.

C) This is Class C airspace; you must contact the tower before flying.

Answer: B - A magenta shaded area adjacent to an airport indicates Class E airspace with a floor of 700 ft AGL. Since you are operating at 400 ft AGL, you remain below the floor of Class E airspace, so the airspace is uncontrolled (Class G) at your altitude. However, you should be aware that if you were inside the lateral boundary of a Class E surface area, ATC authorization would be required.

Q164. You are planning a flight in a rural area that is not within any controlled airspace. The nearest airport has no tower, and the weather is clear. At an altitude of 400 ft AGL, which airspace classification are you in, and what is required to operate?

A) Class G airspace; no ATC authorization is needed, but you must comply with all Part 107 rules.

B) Class E airspace; you must receive ATC authorization before flying.

C) Class D airspace; you must establish two-way radio communication with a tower.

Answer: A - If the area is not designated as controlled airspace from the surface, it is Class G (uncontrolled) airspace. At 400 ft AGL, you are still within Class G if the floor of Class E is 700 ft or 1,200 ft AGL. Part 107 does not require ATC authorization in Class G airspace, but you must still follow all other applicable rules.

Q165. During a preflight briefing, you see a restricted area (shown as a blue hatched area) on the sectional chart. The NOTAMs indicate it is 'hot' (active). What are the operational limitations for your drone?

A) You may enter the restricted area if you remain below 400 ft AGL and yield to manned aircraft.

B) You may enter the restricted area only after getting permission from the controlling agency.

C) You must avoid the restricted area entirely whenever it is active, regardless of your altitude.

Answer: C - Restricted areas are special use airspace where flight is subject to restrictions for safety or national security. When a restricted area is active, unauthorized aircraft (including UAS) must not enter. Permission from the controlling agency is required for entry, so the safest and correct action is to avoid the area entirely.

Q166. You are inspecting a remote solar farm that lies under a Military Operations Area (MOA) that is depicted as a magenta hatched area. The MOA is active, but no NOTAMs indicate current military activity. Which statement is true regarding your planned flight?

A) You cannot fly in a MOA because it is restricted airspace.

B) You may fly in the MOA, but you should remain vigilant for military aircraft and avoid any active operations.

C) You must obtain a special ATC authorization before flying in the MOA.

Answer: B - A Military Operations Area (MOA) is not a restricted area; it is designated to separate military training activities from IFR traffic. VFR flight (including UAS operations) is allowed in a MOA, but pilots should exercise extreme caution and be alert for high-speed military aircraft. No ATC authorization is required for UAS operations in a MOA.

Q167. You are operating a drone near a Class B airport and are within the airspace boundaries. What equipment requirements apply to your operation?

A) A transponder is required.

B) ADS-B In is required.

C) Both a transponder and ADS-B In are required.

Answer: C - FAA Part 107.33 requires the use of both a transponder with Automatic Dependent Surveillance-Broadcast (ADS-B) Out capability and an ADS-B In receiver in Class B and Class C airspace.

Q168. You are approaching a Class C airport and wish to operate your drone in the airspace around the airport. What is the minimum requirement to do so?

A) You must have a transponder.

B) You must have ATC clearance.

C) You must have a waiver from the FAA.

Answer: B - FAA Part 107.33 requires that you receive ATC clearance before operating a small UA in Class B, Class C, or Class D airspace.

Q169. You are flying in an Alert Area that is known for high-density flight activity, such as near an airshow. What does this designation mean for you?

A) The area is closed to all aircraft operations.

B) There are no special restrictions, but you should exercise increased vigilance.

C) You are prohibited from entering if you are a recreational pilot.

Answer: B - Alert Areas are established to inform pilots of areas that contain a high volume of flight activity or unusual types of aeronautical activity, but they do not impose special restrictions.

Q170. You are flying in uncontrolled airspace (Class G) over a rural area during daylight VFR conditions. What is the minimum altitude you must maintain?

A) 500 feet above ground level.

B) 700 feet above ground level.

C) 1,200 feet above ground level.

Answer: B - FAA Part 107.35 states that VFR minimums in Class G airspace are 700 feet above ground level over rural areas and 1,200 feet above ground level over urban areas.

Q171. You are flying near the White House and see a Prohibited Area on your sectional chart. What does this imply regarding drone operations?

A) You may enter if you obtain special permission from the airport manager.

B) You are completely prohibited from entering the airspace.

C) You must maintain a higher altitude than other aircraft to avoid the area.

Answer: B - Prohibited Areas are areas where the public has a special interest in the use of the airspace, and entry is generally prohibited for all aircraft, including drones.

Q172. You are flying over international waters off the coast of the United States. You enter a Warning Area. What is the primary characteristic of this airspace?

A) It involves foreign military activities.

B) It is permanently restricted to civilian aircraft.

C) It is closed to all aviation due to wildlife migration.

Answer: A - Warning Areas extend beyond U.S. jurisdiction and involve activities that may be hazardous to nonparticipating aircraft, often involving U.S. or foreign military activities.

Q173. A Temporary Flight Restriction (TFR) has been issued for a large sporting event downtown. What is the most critical step you must take to ensure compliance?

A) You may fly in the area if you stay under 400 feet.

B) You can fly in the area if you have a recreational certificate.

C) You must check the Notices to Airmen (NOTAMs) for the current status and details of the restriction.

Answer: C - TFRs are temporary restrictions, and pilots must consult Notices to Airmen (NOTAMs) to determine the validity, location, and duration of the restriction before flying.

Q174. You are operating under Visual Flight Rules (VFR) and plan to fly at 500 feet above ground level (AGL) near a small airport with an operating control tower. You maintain two-way radio communication with the tower and remain clear of all aircraft. Which airspace classification are you operating in?

A) Class C

B) Class B

C) Class D

Answer: C - Class D airspace is established around airports to provide IFR and VFR services to aircraft operating in the vicinity of the airport surface. For VFR flight, you must maintain two-way radio communication with the tower. Class C requires both radio communication and an altitude report, while Class B requires the most restrictive requirements and equipment.

Q175. You are planning a flight path that requires crossing a Restricted Area (R-5210) that is currently active. The area is managed by a military facility and is depicted on your sectional chart. What is your requirement regarding this area?

A) You must obtain authorization from the controlling agency (A/FAC) before entering the area.

B) You may fly over the area without permission as long as you stay below 400 feet AGL.

C) You must land immediately and wait until the area becomes inactive.

Answer: A - Restricted areas indicate the presence of identified objects or aerial areas which may be dangerous to unprotected aircraft. Entry into restricted areas requires specific authorization from the controlling agency (usually the military or FAA facility managing the area), typically via the Airspace Facility (A/FAC).

Q176. While flying in uncontrolled airspace (Class G) at 500 feet AGL during daylight hours, you encounter a haze that reduces visibility to 2 miles. According to standard

operating requirements for VFR flight in Class G, can you continue your flight?

A) Yes, as long as you remain clear of clouds.

B) Yes, provided you have a transponder.

C) No, you must land immediately due to insufficient visibility.

Answer: C - The standard Visual Flight Rule (VFR) minimums for Class G airspace at or below 1,200 feet AGL during the day require 3 miles of visibility and to remain clear of clouds. 2 miles of visibility is below the minimum required for VFR flight.

Q177. You are flying a drone toward a Prohibited Area (P-56) designated around a sensitive government facility. What is your course of action?

A) You must avoid the area entirely.

B) You must request permission from the controlling agency to enter the area.

C) You may enter the area as long as you stay below 400 feet AGL.

Answer: A - Prohibited areas are established for reasons of national security or safety and explicitly prohibit all aircraft, regardless of permission, from entering the airspace. Unlike restricted areas, there is no authorization process to enter a prohibited area.

Q178. You are preparing for a flight and review the Notices to Airmen (NOTAMs). What specific type of information does a NOTAM provide to a remote pilot?

A) Information about hazards to air navigation that may affect safe operation of aircraft.

B) Long-term forecasts for weather conditions up to 30 days in advance.

C) Information about the fuel prices at local airports.

Answer: A - NOTAMs contain information about temporary hazards to air navigation, such as runway closures, construction activities, or disabled navigational aids. They are critical for ensuring the safety of flight operations.

Q179. You are flying in Class E airspace at 1,500 feet above ground level (AGL) during daylight hours. What visibility and cloud clearance requirements apply to your flight?

A) 1 mile of visibility and remain clear of clouds.

B) 3 miles of visibility and remain clear of clouds.

C) 5 miles of visibility and 1,000 feet above/below clouds.

Answer: B - In Class E airspace, the VFR weather minimums for daylight flight at or above 1,200 feet AGL generally require 3 miles of visibility and to remain clear of clouds. This is the standard baseline for VFR flight at higher altitudes in uncontrolled airspace.

Q180. You are flying in an Alert Area (A-730) known for intense glider and general aviation activity. You maintain your 400-foot altitude limit and scan the sky frequently. What does this classification indicate about your flight?

A) The area is prohibited to drone operations.

B) The area is subject to high levels of aircraft activity, and the pilot should remain alert for other aircraft.

C) You must land immediately if you see any other aircraft.

Answer: B - Alert areas indicate areas with a high volume of flight training or unusual aerial activity. While entry is not restricted, pilots are responsible for seeing and avoiding other aircraft in these busy areas.

Q181. You are operating near a major airport located in Class B airspace. You do not have a transponder or an ADS-B Out capability installed on your drone. Can you legally fly in this airspace?

A) Yes, as long as you maintain visual contact with the tower.

B) No, you must have a transponder and ADS-B Out capability to operate in Class B airspace.

C) Yes, if you stay below 1,000 feet AGL.

Answer: B - For VFR operations in Class B airspace, a transponder and ADS-B Out capability are required. This equipment allows air traffic control to identify and track aircraft accurately within the complex airspace system.

Q182. You are planning a flight near an airport that has an operational control tower. To ensure you are not violating airspace regulations, you check the charts for the vertical limits of Class D airspace. What is the maximum altitude for Class D airspace?

A) 1,200 feet above ground level (AGL)

B) 4,000 feet above the airport elevation (AGL)

C) 2,500 feet above the airport elevation (AGL)

Answer: C - Class D airspace generally extends from the surface to 2,500 feet above the airport elevation.

Q183. You are flying your drone near a major airport and the controller informs you that you are operating in Class C airspace. Based on the standard configuration, what is the vertical range of this airspace?

A) From the surface to 10,000 feet MSL

B) From 1,200 feet AGL to 4,000 feet above the airport elevation

C) From 14,500 feet MSL to 18,000 feet MSL

Answer: B - Class C airspace is generally a 10 NM radius that extends from 1,200 feet above ground level to 4,000 feet above the airport elevation.

Q184. You are flying over an area where the sectional chart does not depict a Class E airspace base. At what altitude does Class E airspace begin in this specific situation?

A) 700 feet above ground level (AGL)

B) 1,200 feet above ground level (AGL)

C) 14,500 feet above mean sea level (MSL)

Answer: C - In areas where charts do not depict a class E base, class E begins at 14,500 feet MSL.

Q185. You are operating at an altitude of 15,000 feet MSL. You check your chart and see that the lower limit of Class A airspace is 18,000 feet MSL. Which airspace classification are you currently in?

A) Class G

B) Class E

C) Class D

Answer: B - Class E airspace typically extends up to, but not including, 18,000 feet MSL (the lower limit of Class A airspace).

Q186. You are flying over a rural area and notice that the Class E airspace base is at 1,200 feet AGL. You want to fly lower than this to capture ground-level footage. At what altitude are you legally operating in uncontrolled airspace?

A) 1,199 feet AGL

B) 1,100 feet AGL

C) 500 feet AGL

Answer: C - Class E airspace typically begins at 1,200 feet AGL (in most cases). Flying below this altitude places you in Class G (uncontrolled) airspace.

Q187. You are navigating a route and see a blue line on your sectional chart labeled as a Federal Airways. What is the minimum altitude for these airways?

A) 700 feet above ground level (AGL)

B) 10,000 feet above mean sea level (MSL)

C) 1,200 feet above ground level (AGL)

Answer: C - Federal Airways are shown as blue lines on a sectional chart and usually start at 1,200 feet AGL.

Q188. You are flying over a restricted area depicted on the instrument charts where military activities are confined. You do not have permission from the controlling agency to enter this area. What type of Special Use Airspace are you flying over?

A) Prohibited Area

B) Restricted Area

C) Warning Area

Answer: B - Restricted areas are a type of Special Use Airspace where certain activities must be confined or limitations imposed on aircraft operations.

Q189. You are flying in an area where the Class E airspace base is depicted on the chart at an MSL altitude rather than an AGL altitude. You want to fly below this base. At what altitude are you in uncontrolled airspace?

A) 1,200 feet above ground level (AGL)

B) 700 feet above ground level (AGL)

C) 500 feet above ground level (AGL)

Answer: C - Class G airspace extends from the surface to the base of the overlying Class E airspace. Flying below the Class E base places you in Class G, regardless of whether the base is depicted at an MSL or AGL altitude.

Q190. You are planning a flight near an area designated as 'Restricted' airspace. What is the primary requirement for operating your drone in this zone?

A) You must obtain specific authorization from the controlling agency prior to entering.

B) You are not allowed to fly in restricted airspace, regardless of altitude.

C) You only need to contact Air Traffic Control (ATC) if you are below 1,000 feet.

Answer: A - Restricted airspace is a type of Special Use Airspace (SUA) where activities must be confined because of their nature, or because a hazard exists to nonparticipating aircraft. Authorization is required for entry.

Q191. The study material distinguishes between Controlled and Uncontrolled Airspace. Which statement best describes Uncontrolled Airspace?

A) It is airspace where ATC does not have the authority to provide air traffic control services.

B) It is airspace where aircraft must fly strictly under Instrument Flight Rules (IFR).

C) It is airspace that is restricted to military aircraft only.

Answer: A - Uncontrolled airspace, such as Class G, is generally not subject to ATC control, meaning pilots are responsible for their own separation from other aircraft unless in conflict.

Q192. You are operating a small UA and notice a new, very bright light source near an airport. Where should you look to determine if this light requires you to deviate from your flight path?

A) Notices to Airmen (NOTAMs)

B) The Sectional Aeronautical Chart

C) The airport's landing approach lights

Answer: A - The material lists 'Notices to Airmen (NOTAMs)' as a key topic. Temporary flight restrictions or changes to airport *facilities, like new bright lights, are typically announced via NOTAMs.*

Q193. Regarding 'Prohibited' airspace, what is the general status of the airspace?

A) It is closed to all aircraft, including civil aviation, without explicit authorization.

B) It is open to all civil aviation operations during daylight hours.

C) It is restricted to military operations only.

Answer: A - Prohibited airspace is a type of Special Use Airspace where the airspace is generally forbidden for entry. It is typically designated to protect high-value assets or sensitive areas.

Q194. You are flying near a Military Operations Area (MOA). What is the primary purpose of this airspace?

A) To restrict civilian aircraft from entering a specific height band.

B) To provide airspace for military training and testing activities.

C) To separate civil and military air traffic at a specific airport.

Answer: B - Military Operations Areas (MOAs) are designated areas where military activities are conducted. They are established to separate military training from civil air traffic.

Q195. The chapter discusses 'Visual Flight Rules (VFR) Terms & Symbols.' Why is this section relevant for airspace classification?

A) It provides the visual cues and symbols used on charts to determine airspace restrictions.

B) It lists the exact radio frequencies required for each airspace class.

C) It helps pilots identify the specific altitude bands for different classes of airspace.

Answer: A - VFR terms and symbols are standard elements used on aeronautical charts to represent airspace boundaries, restrictions, and other critical information visually.

Q196. In the context of 'Other Airspace Areas,' what is the primary restriction for operating a small UA in National Parks?

A) You must adhere to specific restrictions regarding height and distance from people or objects.

B) You are allowed to fly freely as long as you stay within the park boundaries.

C) You must obtain a waiver from the National Park Service before taking off.

Answer: A - While the text references 'National Parks' under Other Airspace Areas, the operational reality for remote pilots involves adhering to specific restrictions, such as height limits and distance from people, to ensure safety.

Q197. The section on 'Air Traffic Control and the National Airspace System' explains the broader framework of airspace management. What is the primary function of the National Airspace System (NAS)?

A) To track every individual drone registered in the United States.

B) To provide a grid of airports for private pilots to land.

C) To organize the airspace into a system of airspace classes and air traffic services.

Answer: C - The National Airspace System is the network of airspace, ground facilities, electronic and communication systems, and personnel or procedures necessary for the safe and efficient use of the airspace.

Q198. You are operating your drone under Part 107 and approach an airport with a control tower. You have established two-way radio communication with the air traffic controller. What is the next step required to enter Class C airspace?

A) Obtain visual confirmation of the airport traffic by sight

B) Make a visual inspection of the aircraft's position lights

C) Maintain a two-way radio communication frequency and have your transponder on (if required)

Answer: C - Part 107 regulations require a remote pilot to establish two-way communication with ATC and have the transponder turned on when operating in Class C airspace.

Q199. You are planning a flight route over an area marked as 'R-4401' on your sectional chart. This area is a Restricted Area. What action must you take before entering?

A) Obtain authorization from the controlling agency

B) Obtain authorization from the airport manager

C) Obtain authorization from the local police department

Answer: A - Restricted airspace indicates restricted flying activities and authorized persons may only enter it with authorization from the controlling agency. Drones are generally prohibited without this authorization.

Q200. You are operating a small unmanned aircraft in uncontrolled airspace (Class G) during daylight hours. You are operating within the visual line of sight (VLOS). What are the minimum visibility requirements for VFR operations?

A) 1 mile

B) 3 miles

C) 5 miles

Answer: A - During daylight operations in uncontrolled airspace, the minimum visibility requirement for VFR is 1 statute mile.

Q201. You are flying over an area designated as a Military Operation Area (MOA). You notice military aircraft are actively using the area. What is the best practice for your drone flight?

A) Avoid the MOA completely

B) Continue flight but stay clear of the aircraft

C) Contact the controlling agency to request permission to enter

Answer: C - MOAs are established for the purpose of separating military training from IFR and VFR traffic. It is standard practice to avoid entering an active MOA or to obtain authorization from the controlling agency if entry is necessary.

Q202. Under Visual Flight Rules (VFR), what is the minimum distance a small unmanned aircraft must maintain from clouds?

A) 500 feet horizontally

B) 500 feet vertically and 2,000 feet horizontally

C) 1,000 feet vertically and 1,000 feet horizontally

Answer: C - VFR operations require the remote pilot to maintain at least 1,000 feet of horizontal distance from clouds and remain clear of tops of clouds.

Q203. Before your Part 107 flight, you check the Notices to Airmen (NOTAMs) and see that a temporary flight restriction (TFR) has been issued for a nearby stadium due to a high-profile event. What should you do?

A) Avoid the airspace immediately

B) Contact the FAA to request an exemption

C) Proceed with flight as planned

Answer: A - Remote pilots must avoid operating within the boundaries of a TFR. TFRs are established for national security, disaster relief, or other emergency situations where air operations are restricted.

Q204. You are operating a small unmanned aircraft and approach an airport with a Class B airspace surface area. You have established two-way radio communication

with ATC. What additional equipment requirement applies to enter Class B airspace?

A) A 4096-code transponder with Mode C

B) A 1090ES transponder

C) A 4096-code transponder with Mode A

Answer: A - Part 107 requires a transponder capable of displaying a 4096-code and providing altitude information (Mode C) to operate in Class B airspace.

Q205. You are flying over an area designated as an 'Alert Area' (e.g., J-707). This area has high air traffic but no restricted access. What is the primary concern for your operation?

A) The weather is unpredictable

B) You are prohibited from entering

C) There is a high volume of flight activity and you must remain vigilant

Answer: C - Alert Areas indicate a high volume of flight activity for military or other training. The primary concern is the increased likelihood of encountering aircraft, requiring the remote pilot to exercise extra caution.

Q206. You are flying a drone at 3,000 feet above ground level near a major airport. The sectional chart depicts a circle with a ten-mile radius around the airport. Based on the provided study material, which airspace class are you operating in, and what authorization is required?

A) Class C airspace; you must receive ATC authorization before operating.

B) Class D airspace; you must receive ATC authorization before operating.

C) Class E airspace; no authorization is required.

Answer: A - Class C airspace is generally airspace from 1,200 feet to 4,000 feet above the airport elevation, often depicted as a circle with a ten-mile radius. According to the study material, a remote pilot must receive authorization before operating in Class C airspace.

Q207. You are approaching a towered airport and your drone is at an altitude of 1,500 feet above the airport elevation. You do not have a transponder or radio, but you are maintaining visual line of sight. What is the airspace classification?

A) Class D airspace; you must receive ATC authorization before operating.

B) Class C airspace; you must receive ATC authorization before operating.

C) Class G airspace; no authorization is required.

Answer: A - Class D airspace is generally airspace from the surface to 2,500 feet above the airport elevation surrounding airports with an operational control tower. The study material states a remote pilot must receive ATC authorization before operating in Class D airspace.

Q208. You are reviewing a sectional chart and notice a blue line extending across the area. Based on the provided text, what does this typically represent in terms of airspace and altitude?

A) Federal Airways, usually found within Class E airspace from 1,200 feet AGL to but not including 18,000 feet MSL.

B) Class C airspace, generally from 1,200 feet to 4,000 feet above the airport elevation.

C) Special Use Airspace, which is generally above FL 600.

Answer: A - Federal Airways are shown as blue lines on a sectional chart. They are usually found within Class E airspace, starting at 1,200 feet AGL and going up to, but not including, 18,000 feet MSL.

Q209. You are flying a drone at an altitude of 17,500 feet MSL. You are below the lower limit of Class A airspace (which begins at 18,000 feet MSL). What is the classification of the airspace you are in?

A) Class B airspace.

B) Class A airspace.

C) Class E airspace.

Answer: C - Class E airspace typically extends up to, but not including, 18,000 feet MSL (the lower limit of Class A airspace). At 17,500 feet MSL, you are within Class E airspace.

Q210. You are flying over an area where the sectional chart does not depict a Class E base, and your drone is below 14,500 feet MSL. Where does the Class E airspace base begin for your location?

A) At the surface.

B) At 1,200 feet above ground level.

C) At 14,500 feet MSL.

Answer: C - In areas where charts do not depict a Class E base (and the altitude is below 14,500 feet MSL), Class E begins at 14,500 feet MSL.

Q211. You are flying over an uncontrolled airport where there is no control tower and no charted airspace base. What is the airspace classification?

A) Class G airspace, which extends from the surface to the base of the overlying Class E airspace.

B) Class D airspace, requiring ATC authorization.

C) Class E airspace, which begins at 1,200 feet AGL.

Answer: A - Class G airspace is uncontrolled airspace that extends from the surface to the base of the overlying Class E airspace.

Q212. You want to fly near a military operating area that poses a hazard to non-participating aircraft but is not necessarily a restricted area. This type of airspace is best described as:

A) A prohibited area.

B) A warning area.

C) Class E airspace.

Answer: B - Special use airspace includes warning areas, which are located over the high seas or other areas where activity may be hazardous to non-participating aircraft.

Q213. You are flying an unmanned aircraft system at an altitude of 20,000 feet MSL. What is the airspace classification?

A) Class E airspace.

B) Class A airspace.

C) Class G airspace.

Answer: A - The study material states that all airspace above FL 600 (60,000 feet) is Class E airspace.

Q214. You are operating within the lateral and horizontal dimensions of Class C airspace. What is the most restrictive requirement regarding ATC authorization for your drone?

A) ATC permission is required at all times while operating in this airspace.

B) ATC permission is required only when operating at night.

C) ATC permission is required only when operating within 5 miles of the airport.

Answer: A - According to Part 107 regulations, if you are operating a UAS within the lateral and horizontal dimensions of Class C airspace, you must receive ATC authorization.

Q215. While conducting pre-flight planning, you find a NOTAM for your proposed route. What specific type of information does a NOTAM provide to a remote pilot?

A) The current altitude restrictions for all aircraft operating in the area.

B) Current weather forecasts and wind speeds.

C) Information about hazards or conditions that affect the safe operation of aircraft.

Answer: C - NOTAMs (Notices to Airmen) provide information on hazards or conditions that affect the safety of flight, such as temporary flight restrictions or malfunctioning equipment.

Q216. You are flying in Class G airspace during the day. What are the specific VFR weather minimums you must maintain to ensure you remain in visual meteorological conditions?

A) Stay clear of clouds and maintain 3 miles visibility.

B) Stay clear of clouds and maintain 1 mile visibility.

C) Remain within 500 feet of the ground and maintain 3 miles visibility.

Answer: A - For VFR operations during the day, a remote pilot must maintain 3 miles visibility and remain clear of cloud surfaces to operate safely in visual meteorological conditions.

Q217. You are reviewing a sectional chart and notice a symbol indicating an area is 'VFR Only.' What does this symbol signify regarding flight operations in that specific area?

A) Visual Flight Rules are the only flight rules permitted in that area.

B) Instrument Flight Rules are prohibited in that area.

C) Visual Flight Rules are the only flight rules permitted in that area.

Answer: C - The 'VFR Only' symbol on a chart indicates that the airspace is designated for visual flight rules only, often used to restrict IFR operations in certain zones.

Q218. You are operating in the vicinity of a towered airport with a Class D airspace designation. What communication requirement is imposed on your drone operation?

A) You must establish two-way radio communication with the tower.

B) You must contact ATC only when you are within 2 miles of the airport.

C) You must contact ATC only after entering the airspace.

Answer: A - When operating a UAS within Class D airspace, the remote pilot must establish two-way radio communication with the air traffic controller before entering the airspace.

Q219. You are planning a flight route that would take you over the grounds of a federal facility. What type of airspace usually restricts access to unauthorized aircraft?

A) Prohibited airspace.

B) Restricted airspace.

C) Controlled airspace.

Answer: A - Prohibited airspace is designated specifically to prohibit all aircraft entry, including drones, into areas such as federal facilities for national security reasons.

Q220. You notice an area on your chart designated as a BETA (Basic Experimental Test Area). What does this designation typically indicate regarding airspace usage?

A) It is a permanently closed area for military training.

B) It is a temporary restriction for military testing and development.

C) It is a high-density commercial air traffic zone.

Answer: B - BETA (Basic Experimental Test Area) is a type of Special Use Airspace used for military testing and development activities, often imposing temporary restrictions.

Q221. You are operating under Visual Flight Rules (VFR). What is the minimum visibility requirement you must maintain to ensure you are operating in visual meteorological conditions?

A) 5 miles.

B) 1 mile.

C) 3 miles.

Answer: C - Under Part 107, to operate under VFR, a remote pilot must generally maintain 3 miles of visibility and remain clear of clouds.

Q222. You are planning a flight near a Class C airport and must receive authorization from ATC before entering the airspace. What is the vertical range of Class C airspace?

A) From 1,200 feet to 3,000 feet above ground level

B) From 1,200 feet to 4,000 feet above the airport elevation

C) From 1,500 feet to 4,000 feet above ground level

Answer: B - Class C airspace is depicted as a solid magenta line on sectional charts and extends from 1,200 feet to 4,000 feet above the airport elevation. A remote pilot must receive ATC authorization before operating in this airspace.

Q223. You are flying over an airport that has an operational control tower and you are below 2,500 feet above the airport elevation. What class of airspace are you operating in?

A) Class E Airspace

B) Class B Airspace

C) Class D Airspace

Answer: C - Class D airspace generally extends from the surface to 2,500 feet above the airport elevation for airports with an operational control tower. A remote pilot must receive ATC authorization to operate in Class D airspace.

Q224. On a sectional chart, you notice a blue line indicating a Federal Airways. What is the standard altitude range for these airways?

A) From 1,200 feet above ground level up to 18,000 feet MSL

B) From 1,500 feet above ground level up to 14,500 feet MSL

C) From 2,500 feet above ground level up to 18,000 feet MSL

Answer: A - Federal Airways, shown as blue lines on a sectional chart, typically start at 1,200 feet above ground level and extend up to, but not including, 18,000 feet MSL.

Q225. You are operating an unmanned aircraft in uncontrolled airspace below 14,500 feet MSL. What is the designation of this airspace?

A) Class C Airspace

B) Class E Airspace

C) Class G Airspace

Answer: C - Uncontrolled airspace is designated as Class G. It extends from the surface to the base of the overlying Class E airspace. A remote pilot will not need ATC authorization to operate in Class G airspace.

Q226. You are flying at an altitude of 15,000 feet MSL over the continental United States. What is the class of airspace you are in?

A) Class A Airspace

B) Class E Airspace

C) Class G Airspace

Answer: B - Class E airspace typically extends up to, but not including, 18,000 feet MSL (the lower limit of Class A airspace). All airspace above FL 600 is also Class E.

Q227. You are preparing to fly near an area where military activities are occurring, and limitations may be imposed on aircraft operations. This area is likely a Warning area. Which of the following is also a type of Special Use Airspace?

A) Prohibited Area

B) Class C Airspace

C) Class G Airspace

Answer: A - Special Use Airspace includes Prohibited areas, Restricted areas, Warning areas, and Military Operation Areas (MOA). These areas restrict or limit operations for safety or national security reasons.

Q228. You are flying over an area where you see a solid blue line on your chart, which is a Class D airspace area. You are operating under VFR. What is required to operate in this airspace?

A) No ATC authorization is needed since you are flying below 1,000 feet AGL

B) ATC authorization is required before operating in the airspace

C) ATC authorization is required only if you are flying under 500 feet AGL

Answer: B - Class D airspace surrounds airports with an operational control tower. A remote pilot must receive ATC authorization before operating in Class D airspace, regardless of altitude.

Q229. In most areas of the United States, if a Class E airspace base is not depicted on your chart, what is the base altitude?

A) 1,200 feet above ground level

B) 14,500 feet above sea level

C) 700 feet above ground level

Answer: B - In areas where charts do not depict a Class E base, Class E begins at 14,500 feet MSL. In most other areas, the base is 1,200 feet AGL or the surface.

Chapter 3: Weather & Weather Sources

Q230. What is the primary function of an Aviation Weather Report?

A) To predict future weather conditions.

B) To report current weather conditions.

C) To issue warnings for convective storms.

Answer: B - Aviation weather reports provide current information on existing weather conditions.

Q231. What does the introduction to Weather Sources discuss?

A) The availability of weather information.

B) The history of aviation weather reporting.

C) The physics of atmospheric pressure.

Answer: A - The introduction to Weather Sources discusses the general availability of weather information.

Q232. What is a SPECI weather report?

A) A report issued only at night.

B) A report issued every hour.

C) A report issued for significant weather changes.

Answer: C - A SPECI is an unscheduled or special aviation weather report issued when significant changes occur.

Q233. What are Surface Aviation Weather Observations used for?

A) To forecast weather 24 hours in advance.

B) To observe weather conditions at the surface.

C) To monitor aircraft fuel levels.

Answer: B - Surface Aviation Weather Observations provide data on conditions observed at the surface.

Q234. What is an Aviation Forecast?

A) A record of past weather events.

B) A prediction of future weather conditions.

C) A warning of immediate danger.

Answer: B - Aviation Forecasts are predictions of weather conditions for a future time period.

Q235. You are planning a cross-country flight for tomorrow afternoon and need to know if a thunderstorm is expected in the area. Which type of weather product should you consult?

A) Aviation Forecast

B) METAR

C) Convective Significant Meteorological Information

Answer: A - Aviation Forecasts provide predictions of weather conditions for a future time period. METARs provide current conditions, and Convective SIGMETs provide specific alerts for existing weather.

Q236. While preparing for a flight, you check the automated weather observation at the departure airport to see the current wind speed and visibility. Which weather observation service is providing this data?

A) ASOS

B) TAF

C) SPECI

Answer: A - ASOS stands for Automated Surface Observing System, which is a type of Surface Aviation Weather Observation that provides automated current weather conditions.

Q237. You are reviewing the weather reports for your flight route and notice a report that indicates a significant change in the current weather conditions occurred recently. What type of report is this?

A) Aviation Forecast

B) SPECI

C) Convective Significant Meteorological Information

Answer: B - A SPECI is an unscheduled aviation weather report that is issued when conditions differ significantly from the expected forecast or when significant weather changes occur.

Q238. You are flying over a region known for severe thunderstorms. Which weather product specifically provides information about convective weather, such as thunderstorms, tornadoes, and ice storms?

A) METAR

B) Aviation Forecast

C) Convective Significant Meteorological Information

Answer: C - Convective Significant Meteorological Information (WST) is a specific product providing information on convective weather.

Q239. You need to determine the current wind speed and visibility at your destination airport. Which weather report provides current conditions at a specific location?

A) Aviation Forecast

B) Convective SIGMET

C) Aviation Weather Report

Answer: C - Aviation Weather Reports provide current weather conditions (like wind, visibility, and cloud ceilings) observed at a specific location.

Q240. You are planning a flight for the next 24 hours and need to know the expected wind conditions and weather trends. Which document should you review?

A) ASOS

B) METAR

C) Aviation Forecast

Answer: C - Aviation Forecasts provide predicted weather conditions for a future time period, unlike ASOS and METAR which provide current conditions.

Q241. You are checking the weather at an airport and see a report labeled 'WST.' What does this code indicate?

A) Wind Speed and Temperature

B) Aviation Forecast

C) Convective Significant Meteorological Information

Answer: C - The code WST stands for Convective Significant Meteorological Information.

Q242. You are at an airport and see a sign indicating 'ASOS.' What does this system provide?

A) Current surface weather observations

B) Future weather forecasts

C) Pilot reports from other pilots

Answer: A - ASOS (Automated Surface Observing System) provides current surface weather observations.

Q243. The study material outlines various categories of weather sources. If you are looking for a report that provides current conditions at a specific location, you are looking for which category?

A) Aviation Forecasts

B) Surface Aviation Weather Observations

C) Pilot Reports

Answer: B - Surface Aviation Weather Observations provide current conditions at a specific location.

Q244. You launch your drone on a clear day. The nearest weather report says visibility is 2 statute miles, and there are scattered clouds at 1,200 ft AGL. Your planned flight is at 100 ft AGL. Can you operate under Part 107?

A) No, because visibility is less than the required 3 statute miles

B) Yes, because no clouds affect your altitude

C) Yes, because you are 1,100 ft below the clouds

Answer: A - Part 107 requires at least 3 statute miles of flight visibility for all sUAS operations. Even though cloud clearance is met, the visibility below 3 miles makes the flight illegal.

Q245. The sky is covered by a broken cloud layer at 700 ft AGL. You plan to fly at 400 ft AGL in Class G airspace. Are you compliant with Part 107 cloud clearance requirements?

A) Yes, because you are not within 500 ft of the clouds

B) No, because you need to maintain at least 500 ft below clouds, and you are only 300 ft below the layer

C) Yes, because you can fly directly beneath clouds without a minimum distance

D) D) Only if you request a waiver from the FAA

Answer: B - Part 107 requires that sUAS remain at least 500 ft below any cloud. From 400 ft AGL to the cloud base at 700 ft AGL is only 300 ft, so this flight would not meet the minimum cloud clearance. Correct option is B.

Q246. You are checking the latest METAR before a mapping flight near your home airport. The report reads: 'KJFK 131455Z 24015G22KT 10SM FEW040 28/18 A2992'. What does '24015G22KT' indicate for your preflight planning?

A) Wind is from 240° true at 15 knots with gusts to 22 knots

B) Wind is from 240° magnetic at 15 knots with gusts to 22 knots

C) Wind direction varies from 240° at 15 knots to 220 knots

Answer: A - In METAR, wind direction is referenced to true north, and the first number is the direction the wind is coming from. '24015G22KT' means wind direction 240° true, speed 15 knots, with gusts up to 22 knots.

Q247. A convective SIGMET (WST) has been issued for the area where you intend to fly your small unmanned aircraft. Which hazardous weather conditions are you specifically being warned about?

A) Widespread dust storms and volcanic ash

B) Thunderstorms capable of producing severe turbulence, destructive winds, and icing

C) Dense fog and low ceilings that will rapidly reduce visibility

Answer: B - Convective SIGMETs (WST) are issued for severe thunderstorms that may produce destructive winds, large hail, severe turbulence, and icing. These conditions can be extremely hazardous to small unmanned aircraft.

Q248. Your planned flight is at 18:00 UTC, and the 12:00 UTC TAF for the area contains the line: 'TEMPO 1818/1822 2SM TSRA'. How should you interpret this forecast?

A) Between 18:00 and 22:00 UTC, expect temporary conditions of 2-mile visibility with thunderstorm rain

B) At 18:00 UTC, visibility will be 2 miles but thunderstorms are only a low probability

C) Between 18:00 and 22:00 UTC, the visibility will be reduced to 2 miles due to rain showers that may last all day

Answer: A - A TEMPO group in a TAF means temporary fluctuations from the prevailing conditions. '1818/1822' indicates the time period 18:00 to 22:00 UTC; '2SM TSRA' means visibility of 2 statute miles with thunderstorms and rain, which is hazardous and below Part 107 minima.

Q249. While flying in a steady breeze, your drone suddenly experiences a rapid loss of altitude close to the ground after a gust shifts from a headwind to a tailwind. What is the most likely weather phenomenon causing this?

A) Wind shear, which reduces the relative airflow over the rotors and decreases lift

B) Convective turbulence caused by solar heating, which pushes the aircraft upward

C) Mountain wave turbulence, which creates a strong updraft at low altitude

Answer: A - Wind shear is a sudden change in wind speed or direction. When a headwind becomes a tailwind, the relative airflow over the aircraft can be reduced, causing a sudden loss of lift and an unexpected descent—especially dangerous during takeoff and landing.

Q250. A METAR at your departure airport reports 'OVC010'. You plan to inspect a structure at 600 feet AGL while remaining clear of clouds. Does this altitude comply with Part 107 cloud clearance requirements?

A) No, because the cloud ceiling is 1,000 feet and you must stay at least 500 feet below, which puts the maximum altitude at 500 feet

B) No, because the cloud ceiling is 1,000 feet and you must stay at least 600 feet below, which puts the maximum altitude at 400 feet

C) Yes, because 600 feet AGL is still below the cloud deck at 1,000 feet

Answer: A - OVC010 means an overcast ceiling at 1,000 feet above ground level. Part 107 requires no closer than 500 feet below a cloud. At 600 feet AGL, you would be 400 feet below the 1,000-foot ceiling, which violates the requirement. You must remain at or below 500 feet AGL in this case.

Q251. Your local automated surface observation station reports 'TSRA' in the weather group. You are logged in to your ground control station ready to fly. What does this observation mean, and what should you do?

A) Thunderstorm with rain; postpone the flight due to gusty winds, lightning, and potential updrafts/downdrafts

B) Light rain with mist; you may fly if you use the drone's rain-resistant rating

C) Temperature and rain; fly at low altitude to avoid the rain

Answer: A - TSRA is an aviation weather code for 'thunderstorm with rain.' Thunderstorms can produce lightning, severe turbulence, microbursts, and heavy precipitation, all of which are dangerous to small drones. The safest action is to delay the *flight until conditions improve.*

Q252. You are preflighting your small UA near an airport. The local METAR reads: 'KJFK 151800Z 17015G25KT 2SM BR BKN010 OVC020 20/18 A2992'. Which of the following is the correct interpretation for your flight planning?

A) Wind from 170° at 15 knots gusting to 25 knots, visibility 2 statute miles in mist, clouds broken at 1,000 ft and overcast at 2,000 ft

B) Wind from 170° at 25 knots gusting to 15 knots, visibility 2 nautical miles in mist, ceiling 2,000 ft

C) Wind from 170° at 15 knots steady, visibility 2 kilometers in mist, ceiling 1,000 ft

Answer: A - In METAR, wind direction is given in degrees true, speed in knots, and gusts are appended after 'G'. Visibility is in statute miles and cloud layers are in hundreds of feet AGL. Misreading these values could lead to launching in conditions that exceed aircraft limitations or violate visual line of sight.

Q253. A TAF issued for your flight area says 'TEMPO 1908G18KT 1/2SM TSRA'. How should you plan your sUAS flight during that period?

A) Expect temporary wind from 190° at 8 knots gusting to 18 knots with visibility 1/2 statute mile in thunderstorms and rain; avoid flying

B) Expect a permanent wind shift, but the thunderstorms are forecast only at the airport and will not affect your flight

C) Expect temperature 19°C, dew point 8°C, and light rain; continue flying with caution

Answer: A - TEMPO in a TAF indicates temporary changes lasting less than an hour at a time. The mention of thunderstorms (TSRA) and low visibility means hazardous conditions for small UAS operations, so the flight should be postponed.

Q254. While monitoring aviation weather broadcasts, you hear a Convective SIGMET (WST) issued for your region. What does a WST specifically alert you to?

A) Thunderstorms, severe turbulence, and icing conditions

B) Widespread sandstorms or dust storms reducing visibility to below 1 mile

C) Volcanic ash clouds drifting into the flight area

Answer: A - Convective SIGMETs are issued for hazardous convective weather such as embedded thunderstorms, severe turbulence, and icing. These conditions can be violently dangerous to a small unmanned aircraft.

Q255. You plan to fly near an airport that has an Automated Surface Observing System (ASOS). Which type of weather information does ASOS provide?

A) Real-time observations of wind, visibility, and cloud height

B) Forecasts for the next 24 hours including probability of precipitation

C) Regional radar interpretation of thunderstorm movement

Answer: A - ASOS and AWOS are automated weather observation systems that provide current surface conditions. They do not give forecasts—for those you would use a TAF or other forecast products.

Q256. You are launching on a cold morning with an ambient temperature of -5°C. How does this temperature affect the lithium-ion batteries in your small unmanned aircraft?

A) Battery performance is reduced, causing a more rapid voltage drop and shorter flight time

B) Battery performance improves because cold air is denser and provides more lift

C) Battery voltage increases, extending the available flight endurance

Answer: A - Cold temperatures cause chemical reactions in lithium-ion batteries to slow down, reducing their ability to deliver current. This results in decreased battery voltage and reduced flight time, so you should plan shorter flights and watch battery levels closely.

Q257. While flying near the coast, you see a line of clouds and a sudden shift in the wind direction. A sea breeze front is passing through. What is the primary hazard for your small UA?

A) A sudden change in wind direction and speed, creating turbulence and control difficulty

B) A rapid drop in temperature that will cause ice to form on the propellers

C) Heavy precipitation that immediately reduces visibility to near zero

Answer: A - Sea breeze fronts are boundaries between cool marine air and warmer land air. They cause abrupt shifts in wind direction, increased gustiness, and local turbulence, which can overwhelm a small UAS's control system.

Q258. A weather forecast says the air mass over your flight area is unstable with cumulus clouds developing. What flight conditions should you expect while flying your small UA?

A) Gusty winds and moderate to severe turbulence beneath and near cumulus clouds

B) Smooth air with excellent visibility and light winds

C) Widespread low ceilings and steady precipitation extending many miles

Answer: A - Unstable air promotes vertical air currents and cumuliform clouds. These convective currents generate turbulence and gusty winds, so a small UAS should be flown with extreme caution, especially near or below cumulus clouds.

Q259. A METAR reports visibility as 10 statute miles and no significant clouds, but the wind is 45 knots gusting to 60 knots. What is the primary concern for your sUAS flight?

A) The wind may exceed the aircraft's control authority, causing loss of control or inability to fly back

B) The high wind will cause the aircraft's battery to overheat and fail

C) The visibility is too low for maintaining visual line of sight

Answer: A - Although visibility is good, wind speeds and gusts far above typical sUAS operating limits make the flight unsafe. Strong winds can cause fly-aways, excessive power draw, or inability to maintain position heading into the wind.

Q260. You are preflighting a drone near a Class D airport. The latest METAR reads in part: '24008KT 10SM FEW040'. Which wind condition is reported?

A) Wind from 240° true at 8 knots

B) Wind from 240° magnetic at 8 knots

C) Wind variable at 8 knots

Answer: A - In METARs, wind direction is reported relative to true north. The format is 'direction (in tens of degrees) + speed' so 24008KT means wind from 240° true at 8 knots. This distinction is critical for determining crosswind exposure during drone operations.

Q261. A remote pilot checks the TAF for the flight area and sees 'VCTS FM1800'. What is the safest course of action for a planned drone flight that will continue past 1800 local time?

A) Proceed as planned because the thunderstorms are only forecast in the vicinity, not overhead

B) Plan to stop drone operations before the thunderstorms are expected to arrive

C) Fly only at a higher altitude to stay above the weather

Answer: B - VCTS (vicinity thunderstorms) indicates thunderstorms forecast within 5–10 miles of the airport. These storms can generate gusty, turbulent winds and lightning, all of which are hazards to small unmanned aircraft. The safe practice is to avoid flying during the period of expected thunderstorms.

Q262. While reviewing a surface aviation weather observation, you see a visibility value of 'M1/4SM' at the airport. Which operational decision is most appropriate?

A) Cancel the flight because the visibility is less than 1/4 statute mile, making visual line of sight impossible

B) Proceed because the drone can fly in low visibility as long as it stays below 400 feet

C) Continue but rely on the drone's camera to maintain separation

Answer: A - The 'M' in a METAR means 'less than' (less than 1/4 statute mile). Under Part 107, the remote pilot must maintain visual line of sight of the small unmanned aircraft at all times. Extremely low visibility prevents this, so the correct action is to cancel the flight.

Q263. A METAR for a coastal airport includes 'TSRA' and 'WND 18015G30KT'. You plan to launch a small drone. What is the most important threat to control?

A) The drone's GPS could be disabled by the thunderstorm

B) Rain will increase the drone's maximum flight time

C) Gusty winds of 30 knots and convective turbulence from the thunderstorm will make stable flight difficult

Answer: C - TSRA indicates a thunderstorm with rain, and the gusty winds to 30 knots represent sudden changes in airspeed and direction. Even if the drone is rated for wind, convective activity causes turbulence that degrades flight control. Safe practice is to avoid flying in or near thunderstorms.

Q264. You see a Convective SIGMET (WST) issued for the county where you plan to fly. What should this prompt you to do?

A) Continue flying because SIGMETs apply only to manned aircraft

B) Postpone the flight until the WST is no longer in effect

C) Move the flight closer to the center of the SIGMET area where conditions are more stable

Answer: B - A Convective SIGMET is issued for severe thunderstorms and related hazards such as tornadoes, hail, and severe turbulence. These conditions pose significant risks to small UAS. The prudent action is to postpone the flight until the SIGMET expires and conditions are safe.

Q265. Which product would give a drone pilot the most accurate, observed current conditions at an airport, including wind and visibility?

A) Terminal Aerodrome Forecast (TAF)

B) Convective SIGMET (WST)

C) Surface Aviation Weather Observation (METAR)

Answer: C - METARs are routine aviation weather reports based on surface observations. They provide actual, current conditions such as wind, visibility, weather phenomena, clouds, and temperature. TAFs are forecasts, not observations, and WSTs are severe-weather alerts for convective activity.

Q266. A remote pilot is planning an outdoor session and the TAF includes 'PROB30 2SM TSRA'. How should the pilot interpret this and act?

A) There is a 30% chance of thunderstorms, so the pilot should have a backup plan and monitor weather updates

B) The TAF is invalid because PROB30 is not a recognized forecast term

C) Thunderstorms are guaranteed within the next 30 minutes

Answer: A - PROB30 indicates a 30% probability of the described conditions occurring. In this case, there is a 30% chance of thunderstorms with 2 statute mile visibility. This is a significant risk to flight safety, so having a backup plan and continuously monitoring the weather before and during operations is essential.

Q267. During a preflight weather check, a METAR reports 'RA' (moderate rain). Which of the following hazards is most directly relevant to a small drone?

A) Rain can infiltrate the drone's body and cause short circuits or corrosion

B) Rain increases air density, making the drone easier to control

C) Rain only affects the drone’s camera, not the propulsion system

Answer: A - Moderate rain can seep into unprotected electronics, connectors, and motors, causing shorts or long-term corrosion. Additionally, rain degrades visibility and can affect sensor performance. The safest choice is to avoid flying in precipitation unless the drone is specifically designed for wet conditions.

Q268. You are flying a sUAS in uncontrolled airspace. The weather report shows scattered clouds at 1,500 ft AGL and visibility is 4 miles. What is the maximum altitude you can fly while complying with Part 107 cloud clearance requirements?

A) 1,000 ft AGL

B) 400 ft AGL

C) 1,500 ft AGL

Answer: B - Although Part 107 requires maintaining at least 500 ft below clouds (which would allow 1,000 ft AGL), the regulation also generally limits sUAS operations to a maximum of 400 ft AGL unless a taller structure applies. Therefore, 400 ft is the maximum altitude for this flight.

Q269. You are reviewing a METAR for the airport near your planned drone operation and see '18015KT'. Which of the following correctly interprets this wind information?

A) Wind is from 180° true at 15 knots

B) Wind is from 180° magnetic at 15 knots

C) Wind is blowing toward 180° at 15 knots

Answer: A - In METAR reports, wind direction is given in degrees true and is always stated as the direction FROM which the wind is blowing, and speed is in knots.

Q270. You are planning a flight later in the day and read a TAF that includes 'FM1800 28020G35KT'. What does 'G35' signify for your operations?

A) The wind is expected to gust to 35 knots

B) The wind will shift from 280° to 35°

C) The ground speed will be 35 knots

Answer: A - In TAFs, 'G' denotes gusts, so '28020G35KT' indicates a wind from 280° at 20 knots with gusts up to 35 knots.

Q271. Your preflight weather briefing shows a convective SIGMET active over your planned operating area. What is the most appropriate course of action?

A) Proceed with the flight but stay below 100 feet AGL to avoid wind shear

B) Cancel or postpone the flight because convective SIGMETs indicate severe thunderstorms and associated hazards

C) Continue as planned but stay clear of any visible clouds

Answer: B - A convective SIGMET (WST) is issued for severe convective weather such as severe thunderstorms, which are unacceptable hazards for small UAS operations.

Q272. A METAR at your local airport reports visibility as '1/2SM BR' with clear skies. Are you legal to operate your small unmanned aircraft under Part 107 in this area?

A) Yes, because this restriction only applies to flights in controlled airspace

B) Yes, if you remain at least 500 feet below any clouds

C) No, because Part 107 requires a minimum visibility of at least 3 statute miles

Answer: C - Part 107 operating limitations require a minimum flight visibility of 3 statute miles regardless of airspace class.

Q273. It is a hot day at a high-altitude airport (e.g., 35°C at 7,000 ft MSL). How will this affect your small unmanned aircraft's performance?

A) The low air density will decrease propeller thrust and control response

B) The high temperature will increase lift and allow a safer payload margin

C) High altitude and temperature have no measurable effect on small UAS

Answer: A - Higher temperature and altitude increase density altitude and lower air density, which reduces propeller efficiency and thrust.

Q274. You are flying your drone at 400 feet AGL while a cloud layer is present at 800 feet AGL. Is this compliant with Part 107 cloud clearance requirements?

A) Yes, because you are staying below the cloud layer

B) No, because you must be at least 500 feet below any cloud, and you are only 400 feet below

C) Yes, as long as you maintain visual line of sight

Answer: B - Part 107 requires no less than 500 feet below clouds. With the cloud base at 800 feet and you at 400 feet, you are only 400 feet below, so it is not compliant.

Q275. While reviewing a METAR, you see 'P6SM' in the visibility field. What does this tell you about visibility?

A) Visibility is exactly 6 statute miles

B) Visibility is less than 6 statute miles

C) Visibility is greater than 6 statute miles

Answer: C - 'P6SM' in a METAR means visibility is greater than 6 statute miles and therefore well above Part 107 minimums.

Q276. You are planning a flight to begin in two hours and need the expected wind direction, visibility, and cloud conditions at your intended location. Which weather product is designed to provide this future information?

A) A current Surface Aviation Weather Observation (METAR)

B) A Convective SIGMET

C) A Terminal Aerodrome Forecast (TAF)

Answer: C - A TAF is a weather product that provides forecast wind, visibility, and cloud conditions for a specific airport over a defined period, while METARs are current observations and convective SIGMETs address severe convective hazards.

Q277. You are a remote pilot inspecting a cell tower. The METAR at a nearby airport reports 'SCT025'. What does this indicate about the cloud cover over your flight area?

A) Scattered clouds at 2,500 feet AGL

B) Sky clear with 2,500 feet visibility

C) Broken clouds at 250 feet AGL

Answer: A - In a METAR, SCT025 is a cloud layer code meaning scattered clouds (3/8 to 4/8 coverage) at 2,500 feet above ground level (AGL).

Q278. While reviewing a TAF for your flight site, you see 'TEMPO 1818/1822 1/2SM -RA BR'. How should you interpret this forecast?

A) Temporary change between 1800Z and 2200Z with 1/2-mile visibility, light rain and mist

B) Winds from 180 degrees at 18 knots, visibility 1/2 mile, rain

C) Periods of heavy rain between 1800Z and 2200Z with 2,500-foot ceiling

Answer: A - TEMPO indicates a temporary fluctuation in conditions during the listed time period. Here, 1/2SM visibility, light rain (-RA), and mist (BR) are expected between 1800Z and 2200Z.

Q279. A convective SIGMET (WST) has been issued for your flight area. What does this mean for your small UA operation?

A) Operations should be suspended because WST implies severe thunderstorms with hazards such as strong winds, hail, and tornadoes

B) Continue operations but maintain visual line of sight since WST is only a forecast for moderate turbulence

C) The SIGMET is only for manned aircraft, so small UA are unaffected

Answer: A - Convective SIGMETs are issued for severe thunderstorms that pose hazards like damaging winds, large hail, and tornadoes. These conditions are dangerous to small UA and operations should be suspended until they pass.

Q280. While checking the METAR for your launch site, you notice 'VCTS' in the remarks section. What does this signify?

A) Thunderstorm with rain at the station

B) Thunderstorm in the vicinity of the airport, within 10 miles

C) Visibility variable with ceiling at 10,000 feet

Answer: B - VCTS is an abbreviation for 'thunderstorm in the vicinity,' meaning a thunderstorm is occurring within about 10 nautical miles of the observation point but not directly over the airport.

Q281. The TAF for your proposed flight time includes 'FZRA'. As a drone pilot, what specific hazard should you anticipate?

A) Freezing rain causing ice to accumulate on your sUAS surfaces and affecting performance

B) Flaming rain due to volcanic activity

C) During rain, visibility is always greater than 3 miles

Answer: A - FZRA designates freezing rain, which can rapidly form ice on an airframe, including small UAS, degrading lift and control. This is a serious hazard for drone operations.

Q282. You are planning early-morning operations near an uncontrolled airfield. Which weather product provides the most current observed surface conditions, such as wind, visibility, and cloud cover?

A) METAR

B) TAF

C) Convective SIGMET

Answer: A - Surface Aviation Weather Observations, or METARs, are hourly (or specially updated) observed conditions at a location, including wind, visibility, weather phenomena, and cloud cover. TAFs are forecasts, and convective SIGMETs are warnings for severe thunderstorms.

Q283. A Part 107 operation requires at least 3 statute miles of visibility. Where would you find this information from a weather report?

A) In the 'wind' group of a METAR as the prevailing visibility in statute miles

B) In the 'visibility' group of a METAR, which is reported in statute miles

C) In the cloud base group of a TAF

Answer: B - METAR visibility is reported in the visibility group in statute miles (e.g., 10SM), so you can directly read it to confirm compliance with the Part 107 3-statute-mile minimum.

Q284. A METAR for your flight location reports 'OVC008'. You plan to operate your drone at 400 feet AGL. How should you interpret this ceiling?

A) The flight is below the cloud base, so you may operate as long as you remain clear of clouds

B) The flight is above the cloud base, which is prohibited

C) The ceiling has no impact on small UA operations

Answer: A - OVC008 means an overcast ceiling at 800 feet AGL. Flying at 400 feet AGL puts you below the cloud base, so operations are allowed as long as you remain in clear air and maintain visual line of sight, avoiding the clouds.

Q285. You are flying a drone for commercial photography near a coastal airport and check the current weather conditions via a METAR. The report indicates 'VRB 10KT G20KT'. What does this tell you about the wind?

A) The wind is blowing from the north at 10 knots, with gusts up to 20 knots.

B) The wind is variable in direction at 10 knots, with gusts up to 20 knots.

C) The wind is blowing from the south at 20 knots.

Answer: B - In aviation weather reports (METAR), 'VRB' stands for variable, indicating the wind direction is changing. The number before 'KT' is the sustained wind speed, and the number after 'G' is the gust speed.

Q286. You are planning a delivery flight for the afternoon and review the TAF (Terminal Aerodrome Forecast) for your route. The forecast states 'BECMG 1200 18012KT'. What does this indicate?

A) The wind will be calm at 12:00 UTC.

B) Wind direction will change to 180 degrees at 12:00 UTC.

C) Wind direction will change gradually at 12:00 UTC.

Answer: C - In TAFs, 'BECMG' stands for 'becoming,' indicating a gradual change in weather conditions. The time following it is the time the change is expected to begin.

Q287. While monitoring a Convective Significant Meteorological Information (WST) bulletin during a flight over a mountainous region, you see the code 'TSRA'. What weather phenomenon is this reporting?

A) Thunderstorm with Rain and associated turbulence.

B) Thunderstorm with Severe turbulence only.

C) Heavy rain and fog.

Answer: A - The WST code 'TSRA' indicates a Thunderstorm with Rain, which implies the presence of turbulence and other hazardous convective activity.

Q288. You are conducting a coastal survey and check the current surface aviation weather observations. The report lists '4SM' in the visibility section. What is the current visibility?

A) 4 nautical miles

B) 4 kilometers

C) 4 statute miles

Answer: C - Surface aviation weather observations (METAR) report visibility in statute miles.

Q289. Your drone is tasked with inspecting a transmission line. The TAF for the area forecasts 'TEMPO 1500 3SM'. What does this indicate regarding the weather conditions?

A) Conditions will occur continuously for 15 hours.

B) Conditions will cease at 15:00 UTC.

C) Conditions will occur intermittently for a short period.

Answer: C - In TAFs, 'TEMPO' indicates temporary weather conditions that may occur for short periods of time, not continuously.

Q290. You are analyzing a WST alert and see the code 'TSE'. This code specifically identifies which type of hazard associated with thunderstorms?

A) Thunderstorm with hail

B) Thunderstorm with embedded severe turbulence

C) Thunderstorm with severe icing

Answer: B - The WST code 'TSE' stands for Thunderstorm with embedded severe turbulence, a critical hazard for small unmanned aircraft.

Q291. You check the aviation weather reports to prepare for a flight. The report indicates a wind shift to the north with a decrease in wind speed. What is the best course of action to determine the safety of your flight path?

A) Check the Doppler weather radar to confirm the wind shift.

B) Ignore the report and rely on visual flight conditions.

C) Land immediately as wind shear is unavoidable.

Answer: A - Doppler weather radar is a weather source that allows pilots to identify wind shear and rotation within storms, helping to assess safety.

Q292. You are reviewing a TAF and notice the notation 'TN 1600 12005KT'. What does this indicate?

A) The forecast time is 16:00 UTC.

B) The wind direction will change at 16:00 UTC.

C) The specified weather conditions will end at 16:00 UTC.

Answer: C - In TAFs, 'TN' stands for 'until,' indicating the end time of a specific weather condition.

Q293. You are planning a drone delivery flight over a city tomorrow. You check the Aviation Routine Weather Report (METAR) for your departure airport and the Terminal Aerodrome Forecast (TAF) for your destination. Which forecast provides the most accurate prediction of wind, visibility, and weather conditions at the destination airport for a future time?

A) Surface Aviation Weather Observations (ASOS)

B) Aviation Routine Weather Reports (METAR)

C) Terminal Aerodrome Forecasts (TAF)

Answer: C - Terminal Aerodrome Forecasts (TAF) are designed to predict weather conditions (wind, visibility, weather, clouds) at airports for a future time frame (up to 30 hours out). ASOS and METAR provide current conditions.

Q294. You are flying a small unmanned aircraft near a small airport. You want to know the current wind direction and speed. Which source provides the most reliable, current wind data?

A) Aviation Routine Weather Report (METAR)

B) Terminal Aerodrome Forecast (TAF)

C) Surface Aviation Weather Observations (ASOS)

Answer: C - Surface Aviation Weather Observations (ASOS) are automated systems that provide current conditions, including wind direction, speed, and other surface weather data.

Q295. You are flying over a region known for thunderstorms. A Convective Significant Meteorological Information (WST) or Convective SIGMET is issued. What is the primary purpose of this product?

A) To forecast the exact time a thunderstorm will pass your location in 24 hours.

B) To provide significant weather information affecting IFR and VFR aircraft, specifically concerning convective activity.

C) To list the current temperature and dew point readings at all airports in the region.

Answer: B - Convective SIGMETs (WST) are issued to alert pilots to significant weather that can affect aircraft, including thunderstorms, tornadoes, ice storms, and dust storms.

Q296. You are monitoring the weather for a drone delivery flight. You notice the Terminal Aerodrome Forecast (TAF) has a 'TEMPO' group. What does this indicate in the forecast?

A) The specified weather conditions are expected to occur for a short period, interspersed with periods of normal weather.

B) The weather is expected to be the same as the current METAR reading at all times.

C) The weather will be deteriorating continuously over the next 24 hours.

Answer: A - TEMPO indicates temporary conditions expected for a short period (less than 1 hour) within the specified period, interspersed with periods when the conditions are not occurring.

Q297. While flying, you check the Aviation Routine Weather Report (METAR) to assess the current visibility at your location. What does the visibility report in a METAR represent?

A) The distance an aircraft can fly in a straight line assuming no obstacles.

B) The greatest distance one can see and identify a prominent unlighted object.

C) The distance required for an aircraft to land safely under current conditions.

Answer: B - Visibility in a METAR refers to the prevailing visibility, defined as the greatest distance one can see and identify a prominent unlighted object or a high contrast lighted object.

Q298. You are preparing to launch your drone. You consult the Surface Aviation Weather Observations (ASOS) and see the wind is reported as '27010KT.' How do you interpret this?

A) Wind blowing at 10 knots from the west.

B) Wind blowing at 270 knots from the east.

C) Wind blowing at 10 knots from the west (270 degrees).

Answer: C - In aviation weather reports (METAR/ASOS), the first number is the direction the wind is coming from (in degrees), and the second number is the speed in knots. 270 is West.

Q299. You are looking at a Surface Aviation Weather Observation (ASOS) and see the cloud layer is reported as 'BKN045.' What does this tell you about the weather conditions?

A) The sky is clear, and visibility is unrestricted.

B) The sky is broken with a base altitude of 4,500 feet.

C) The sky is overcast, and the ceiling is below 1,000 feet.

Answer: B - BKN indicates broken clouds. The number (045) represents the altitude in hundreds of feet, so 4,500 feet.

Q300. You are a remote pilot needing to know the current altimeter setting to ensure your drone's GPS altitude is accurate relative to local pressure. Which source provides this information?

A) Convective SIGMET (WST)

B) Aviation Routine Weather Report (METAR)

C) Terminal Aerodrome Forecast (TAF)

Answer: B - METARs include the altimeter setting (QNH), which is critical for pressure altitude calculations. TAFs are forecasts, and SIGMETs are for significant convective weather.

Q301. You are reviewing the weather at your departure airport and notice a METAR report followed by a SPECI report. What does the SPECI report indicate?

A) A significant change in weather conditions has occurred.

B) The weather forecast has been updated for the next 24 hours.

C) A routine weather observation has occurred.

Answer: A - A SPECI (Special) report is an unscheduled aviation weather report issued when significant changes have occurred or are expected to occur in the weather conditions since the last regular report.

Q302. You are planning a drone flight that will take 4 hours to complete. You consult the Terminal Aerodrome Forecast (TAF) for your area. What is the maximum validity period of a TAF?

A) 6 hours

B) 12 hours

C) 24 hours

Answer: C - Aviation Forecasts, such as the TAF, are valid for periods up to 24 hours and are updated every 6 hours.

Q303. You are monitoring weather sources and see a Convective SIGMET (WST) alert. What type of weather information does this provide?

A) A forecast of general weather conditions for the next 24 hours.

B) A warning of existing or developing thunderstorms, icing, and turbulence.

C) A routine daily weather observation at the nearest airport.

Answer: B - Convective SIGMETs (WST) provide information about convective weather, including thunderstorms, hail, and associated icing or turbulence.

Q304. You need to know the current temperature and dew point at a specific location to assess fog potential before takeoff. Which section of Weather Sources provides these surface observations?

A) Aviation Forecasts

B) Surface Aviation Weather Observations

C) Aviation Weather Reports

Answer: B - Surface Aviation Weather Observations provide current weather conditions at a specific location, including temperature, dew point, and visibility.

Q305. You are reviewing the Aviation Forecasts section to determine if winds will be favorable for a landing in 2 hours. Which document specifically provides these predictions?

A) TAF

B) SPECI

C) METAR

Answer: A - The Terminal Aerodrome Forecast (TAF) is a specific type of Aviation Forecast that predicts weather conditions, including wind, at a specific airport for a specified time period.

Q306. You are checking the Aviation Weather Reports to determine current wind speed and direction at your location. Which report provides this current snapshot?

A) METAR

B) TAF

C) WST

Answer: A - METAR reports provide current weather conditions, including wind speed and direction, temperature, and visibility.

Q307. You are reviewing the list of Weather Sources and want to identify the document that warns of existing or developing thunderstorms, icing, and turbulence. What is this document called?

A) Surface Aviation Weather Observations

B) Aviation Weather Reports

C) Convective SIGMET (WST)

Answer: C - Convective SIGMETs (WST) are issued to warn pilots of weather conditions that are potentially hazardous to aircraft, specifically regarding thunderstorms and associated hazards.

Q308. You are checking the Surface Aviation Weather Observations. What does this section primarily provide to the remote pilot?

A) A forecast of future weather conditions.

B) A history of past weather events.

C) Current weather conditions at a specific location.

Answer: C - Surface Aviation Weather Observations provide the current state of the weather at a specific point in time.

Q309. You are preparing for a flight and need to know the current wind speed and direction at the airport. Which weather source provides this real-time data?

A) TAF

B) METAR

C) AIRMET

Answer: B - METAR (Meteorological Aerodrome Report) is a surface aviation weather observation that provides current conditions, including wind speed and direction.

Q310. You are planning a cross-country flight tomorrow and need to know the expected wind conditions at your destination airport. Which weather report should you consult?

A) METAR

B) Convective SIGMET

C) TAF

Answer: C - TAF (Terminal Aerodrome Forecast) is an aviation forecast that provides projected weather conditions for a future period, including expected wind.

Q311. While reviewing weather reports, you see a Convective SIGMET (WST) indicating severe turbulence and embedded thunderstorms. What does this information signify regarding aviation safety?

A) It is a routine weather report with no safety implications.

B) It indicates significant convective weather significant to the safety of flight.

C) It is a forecast valid only for the immediate area and next hour.

Answer: B - Convective SIGMETs provide information on significant convective weather, such as severe thunderstorms and turbulence, which is significant to the safety of flight.

Q312. You need to determine if weather conditions are expected to change or improve during your upcoming flight window. Which weather product is specifically designed to provide these projected conditions?

A) METAR

B) TAF

C) Convective SIGMET

Answer: B - TAFs are aviation forecasts that provide projected weather conditions for a future period, allowing pilots to anticipate changes.

Q313. You receive a Convective SIGMET (WST) regarding tornadoes and cumulonimbus clouds. What is the primary purpose of this product?

A) To alert pilots to significant convective weather affecting the safety of flight.

B) To forecast the exact time the sun will set.

C) To provide a list of nearby airports.

Answer: A - Convective SIGMETs are issued to alert pilots to significant convective weather that poses a hazard to aviation operations.

Q314. Which surface aviation weather observation provides the current wind direction, speed, visibility, and weather conditions?

A) METAR

B) TAF

C) AIRMET

Answer: A - METARs are the standard format for surface aviation weather observations, reporting current wind direction, speed, visibility, and weather.

Q315. A Convective SIGMET is issued when convective weather is significant to the safety of flight. Which of the following conditions is typically NOT the reason a Convective SIGMET is issued?

A) Light rain and drizzle.

B) Severe turbulence.

C) Embedded thunderstorms.

Answer: A - Convective SIGMETs are specifically for significant convective weather, such as severe turbulence and embedded thunderstorms, not light weather like rain or drizzle.

Q316. A remote pilot needs to check the current weather conditions at a specific airport to determine wind speed and visibility. Which aviation weather report provides these routine observations?

A) SPECI

B) METAR

C) TAF

Answer: B - A METAR (Meteorological Aerodrome Report) is a routine hourly report of weather observations at an airport. SPECI is a special report for significant changes, and TAF is a forecast for future conditions.

Q317. A remote pilot observes that a weather condition at an airport has changed significantly since the last routine observation. Which aviation weather report indicates this significant change?

A) TAF

B) METAR

C) SPECI

Answer: C - A SPECI is a significant weather report issued to provide current weather information when it differs significantly from the routine METAR.

Q318. A remote pilot is planning a cross-country flight for the next day and needs to know the expected conditions at the destination airport. Which source provides these predictions?

A) Aviation Forecasts

B) NOTAMs

C) METAR

Answer: A - Aviation Forecasts provide predictions for future weather conditions, such as wind, visibility, and weather phenomena, which are essential for flight planning.

Q319. A remote pilot is scanning a weather map and notices a region of developing thunderstorms. Which product specifically highlights hazardous convective weather to help the pilot avoid it?

A) METAR

B) WST

C) PIREP

Answer: B - WST (Significant Meteorological Information) is a chart designed to alert pilots to potentially hazardous convective weather.

Q320. A remote pilot is flying over mountainous terrain and experiences strong, turbulent air. This type of turbulence is most likely caused by:

A) Wind shear

B) Thermal turbulence

C) Mountain wave turbulence

Answer: C - Mountain wave turbulence is caused by airflow over mountains and can create violent updrafts and downdrafts, even without visible clouds.

Q321. A remote pilot is flying through visible moisture while the air temperature is below freezing. This condition poses a risk of:

A) Structural icing

B) Carburetor icing

C) Dew point fog

Answer: A - Structural icing occurs when supercooled water droplets freeze on the aircraft's surfaces, which is a danger when flying in visible moisture below freezing temperatures.

Q322. A remote pilot is taking off into a strong headwind. How does this headwind affect the takeoff distance compared to a no-wind condition?

A) Increases takeoff distance

B) Has no effect on takeoff distance

C) Decreases takeoff distance

Answer: C - A headwind assists the aircraft by providing additional lift and groundspeed, resulting in a shorter takeoff distance compared to calm wind conditions.

Q323. A remote pilot is flying and cannot distinguish the ground or other aircraft clearly due to reduced visibility. This is known as:

A) Low visibility

B) High winds

C) Icing

Answer: A - Low visibility conditions, such as those caused by fog or heavy precipitation, reduce the pilot's ability to see and are a critical weather hazard.

Q324. You are preparing for a flight and need to know the exact temperature, wind speed, and visibility at the airport right now. Which report should you consult?

A) TAF

B) Area Forecast

C) METAR

Answer: C - METARs are Surface Aviation Weather Observations that provide current weather conditions, including temperature, wind, and visibility.

Q325. You are planning a cross-country flight that will take place several hours from now. Which forecast report is specifically designed to provide the expected weather conditions for your destination?

A) SIGMET

B) AIRMET

C) TAF

Answer: C - TAFs (Terminal Aerodrome Forecasts) are forecasts issued for specific airports, providing expected weather conditions for a future time period.

Q326. You receive a notification that a Convective Significant Meteorological Information (WST) alert has been issued for your flight path. What specific weather hazard is being reported?

A) Low-level wind shear

B) Convective activity (thunderstorms)

C) Ceiling and visibility restrictions

Answer: B - WST stands for Convective Significant Meteorological Information and is specifically used to warn of significant convective weather activity, such as thunderstorms.

Q327. You are reviewing significant weather advisories to determine if you should divert your flight due to hazardous conditions. Which category of reports is best suited to identify severe weather hazards?

A) Aviation Weather Reports

B) Area Forecasts

C) Surface Weather Observations

Answer: A - Aviation Weather Reports include significant weather advisories such as SIGMETs, AIRMETs, and WSTs, which highlight hazardous conditions.

Q328. You are flying over a large region and need to understand the general weather trends for the area, rather than just a single airport. Which forecast type is most appropriate?

A) Winds and Temperature Aloft

B) TAF

C) Area Forecast

Answer: C - Area Forecasts are provided for large geographic areas and describe general weather trends, cloud cover, and weather conditions for a period of time.

Q329. Before taking off, you check the weather and see a 'RMK' remark in the METAR report. What does this abbreviation indicate?

A) Runway Maintenance

B) Report Modified

C) Remarks

Answer: C - RMK stands for Remarks in a Surface Aviation Weather Observation, providing additional information or details not included in the standard data fields.

Q330. You are concerned about specific weather phenomena like moderate icing or mountain obscuration that may affect your small unmanned aircraft. Which report provides these significant advisories?

A) WST

B) SIGMET

C) AIRMET

Answer: C - AIRMETs provide information on weather phenomena that are of interest to all aircraft but usually less hazardous than SIGMETs, such as icing or mountain obscurations.

Q331. You need to know the wind direction and speed at a specific altitude, such as 3,000 feet above ground level, for your flight planning. Which forecast component should you check?

A) TAF

B) METAR

C) Winds and Temperature Aloft

Answer: C - Winds and Temperature Aloft forecasts provide information on wind direction and speed at various altitudes, which is crucial for planning the altitude of flight.

Q332. You are reviewing current weather conditions at an airport, which includes wind direction, visibility, and current weather phenomena. Later, you receive a new report for the same airport. What is the most likely reason for the second report?

A) It is a Convective Significant Meteorological Information (WST), which updates you on thunderstorms.

B) It is a METAR, which provides a routine report every hour.

C) It is a SPECI, which is an unscheduled report for significant changes.

Answer: C - A SPECI (Special Weather Report) is an unscheduled aviation weather report issued when significant changes have occurred or are expected to occur to existing significant weather conditions.

Q333. While planning a cross-country flight, you need information about expected conditions at your destination airport for the next 24 hours. Which weather source provides this type of information?

A) A surface aviation weather observation (METAR).

B) An aviation weather forecast (TAF).

C) Convective Significant Meteorological Information (WST).

Answer: B - Aviation Forecasts, such as a Terminal Aerodrome Forecast (TAF), provide expected weather conditions at a specific location for a future period, typically up to 30 hours.

Q334. You are flying in an area known for potential severe weather and want to check for warnings regarding convective activity. Which report type would provide this information?

A) A standard Aviation Weather Report (SPECI).

B) A Surface Aviation Weather Observation (METAR).

C) A Convective Significant Meteorological Information (WST).

Answer: C - Convective Significant Meteorological Information (WST) is specifically designed to alert pilots to the presence and movement of convective storms.

Q335. Before takeoff, you check the current conditions at the airport. The report indicates wind direction, visibility, ceiling, and current weather phenomena. What type of report are you reading?

A) An aviation weather forecast (TAF).

B) A surface aviation weather observation (METAR).

C) Convective Significant Meteorological Information (WST).

Answer: B - Surface Aviation Weather Observations (METAR) provide the current meteorological conditions at an airport at a specific time.

Q336. You are reviewing the provided study material to find the section that categorizes surface observations, aviation weather reports, aviation forecasts, and convective SIGMETs together. Which section title contains these items?

A) Weather Sources (Introduction)

B) Chapter 3b: Effects of Weather on Small Unmanned Aircraft

C) Weather Sources: Introduction

Answer: A - The provided text lists the categories of weather sources, including surface aviation weather observations, aviation weather reports, aviation forecasts, and convective Significant Meteorological Information (WST) under the main heading 'Weather Sources'.

Q337. A sudden change in weather conditions at the airport occurs, such as a drop in visibility or the onset of a weather phenomenon that was not present in the previous routine report. Which report should you check to get the latest information on this change?

A) A routine aviation weather report (METAR).

B) An unscheduled aviation weather report (SPECI).

C) A convective Significant Meteorological Information (WST).

Answer: B - SPECI reports are issued unscheduled when significant changes have occurred to existing significant weather conditions, filling in gaps between routine METARs.

Q338. You are planning a flight over a region prone to severe thunderstorms. Which type of weather information is specifically designed to alert pilots to the presence and movement of convective storms?

A) A routine surface observation (METAR).

B) A routine aviation weather forecast (TAF).

C) Convective Significant Meteorological Information (WST).

Answer: C - The text specifies that Convective Significant Meteorological Information (WST) is used to alert pilots to convective weather.

Q339. You need to know the current weather at a specific airport and the expected weather for the next 24 hours at that same airport. According to the provided study material, which two specific weather products would you consult?

A) METAR and SPECI

B) TAF and WST

C) METAR and TAF

Answer: C - A METAR provides current surface weather observations, while a TAF provides a forecast of future weather conditions. SPECI is an update to a METAR, and WST is a specific alert for convective weather.

Q340. You are planning a flight for tomorrow morning and need a prediction of wind speed, direction, and visibility for your specific airport for the next 24 hours. Which product provides this information?

A) METAR

B) TAF

C) SPECI

Answer: B - TAFs (Terminal Aerodrome Forecasts) provide forecasts for specific airports, typically valid for 24 hours.

Q341. You are on the flight line and need to know the current wind speed, wind direction, and visibility at the airport immediately. Which report do you check?

A) TAF

B) SPECI

C) METAR

Answer: C - METARs (Meteorological Aerodrome Reports) provide current weather observations, including wind and visibility.

Q342. You are monitoring weather reports and receive a WST product. What specific type of weather hazard is this designed to alert you to?

A) Non-convective severe weather like severe icing or severe turbulence

B) Low-level wind shear conditions near the airport

C) Thunderstorms and other convective weather

Answer: C - WST stands for Convective Significant Meteorological Information and is specifically used to alert pilots to thunderstorms and other significant convective weather.

Q343. You checked the METAR an hour ago, but suddenly the weather at the airport is deteriorating rapidly. Which report provides an updated observation of the current weather conditions?

A) TAF

B) FDC

C) SPECI

Answer: C - SPECI reports are updated METARs issued when significant changes occur in weather conditions, such as sudden wind shifts or visibility drops.

Q344. A TAF forecast states '26018KT 2408/2408'. What does the '2408' indicate?

A) Visibility is 2408 feet

B) Visibility is 8 miles with a ceiling at 2400 feet

C) Visibility is 8 miles

Answer: B - In a TAF, the first number (8) is visibility in miles, and the second number (2400) is the ceiling height in hundreds of feet (2400). Thus, 2408 means a ceiling of 2400 feet with visibility of 8 miles.

Q345. You need to find general aviation weather forecasts for a large area that encompasses several airports. Which report should you consult?

A) TAF

B) SPECI

C) FDC

Answer: C - FDC forecasts (Flight Director Chart) provide area forecasts for a large region, whereas TAFs are specific to individual airports.

Q346. When using a smartphone app to check weather for a drone flight, where is the most reliable source of current weather data located?

A) On the home screen widget

B) In the 'Aviation Weather Reports' section

C) In the 'Download Locations' menu

Answer: B - Reliable aviation weather sources (like the Aviation Weather Center website or apps) categorize data into sections such as Reports, Forecasts, and Warnings.

Q347. You see a report indicating 'SIGMET'. How does this differ from a 'WST' report?

A) WST is for convective weather, SIGMET is for icing

B) Both report the same type of weather

C) WST is for convective weather, SIGMET is for non-convective severe weather

Answer: C - SIGMETs alert to non-convective severe weather (like severe icing or severe turbulence), while WSTs (Convective SIGMETs) alert specifically to thunderstorms and other convective weather.

Chapter 4: Loading & Performance

Q348. What is a load factor defined as in the loading section?

A) The ratio of lift to weight

B) The ratio of thrust to drag

C) The ratio of weight to lift

Answer: A - The text defines a load factor as the ratio of lift to weight.

Q349. What does density altitude represent in the Performance section?

A) The altitude above sea level

B) The altitude in the standard atmosphere corresponding to a particular density

C) The altitude above the highest obstacle

Answer: B - Density altitude is defined as the altitude in the standard atmosphere corresponding to a particular density.

Q350. How does the text describe the effect of obstructions on wind?

A) Wind speed decreases and airflow becomes smooth

B) Wind speed increases and turbulence occurs

C) Wind speed remains constant regardless of the obstruction

Answer: B - The section on the effect of obstructions on wind states that wind speed increases and turbulence occurs.

Q351. What is low-level wind shear?

A) A decrease in wind speed due to high humidity

B) A steady increase in wind speed at high altitudes

C) A sudden change in wind speed and/or direction over a short distance

Answer: C - Low-level wind shear is defined as a sudden change in wind speed and/or direction over a short distance.

Q352. What does the difference between temperature and dew point indicate?

A) Wind direction

B) Atmospheric pressure

C) Humidity

Answer: C - The text explains that the difference between temperature and dew point indicates humidity.

Q353. What is the total weight of a small unmanned aircraft?

A) The empty weight only

B) The payload only

C) The sum of the empty weight and the payload

Answer: C - The text defines total weight as the sum of the empty weight and the payload.

Q354. In the context of weight and balance, what does the center of gravity (CG) refer to?

A) The center of the battery

B) A reference point for weight distribution

C) The center of the propeller blades

Answer: B - The text indicates that the center of gravity (CG) is a reference point for weight distribution.

Q355. A remote pilot is operating a multirotor small UAS on a hot summer afternoon from a high-elevation airstrip. The aircraft seems slow to climb and feels less stable even at full throttle. Which of the following best explains this performance loss?

A) The density altitude is higher, which reduces rotor efficiency and motor performance.

B) The pressure altitude is lower, which increases air density and rotor lift.

C) The battery is overcharged, producing too much voltage for the motors.

Answer: A - High temperature and high elevation increase density altitude, meaning the air is thinner. Thinner air reduces rotor lift and propeller efficiency, degrading climb performance and responsiveness. Lower pressure altitude would mean denser air and better performance, not worse.

Q356. A remote pilot must deliver a fragile payload to a survey site. The manufacturer's loading chart shows the center of gravity (CG) will fall outside the acceptable range with the payload attached to the rear mounting point. What should the pilot do before departing?

A) Proceed because the total weight is still below the 55 lb limit.

B) Redistribute or reduce the payload so the CG falls within the approved range.

C) Increase the throttle trim to compensate for the rearward CG during flight.

Answer: B - Operating outside the manufacturer's CG envelope is unsafe, regardless of total weight. The pilot must adjust the load placement or reduce weight to keep the CG within the approved range. Throttle trim cannot fix an out-of-limit CG condition.

Q357. A small UAS has an empty weight of 40 lbs. The pilot installs two batteries each weighing 5 lbs. How much additional payload can be legally and safely carried if the maximum takeoff weight is 55 lbs?

A) 5 lbs

B) 10 lbs

C) 15 lbs

Answer: A - Empty weight (40 lbs) plus two batteries (10 lbs) equals 50 lbs. Subtracting from the 55 lb limit leaves 5 lbs for payload. Exceeding this would violate the maximum takeoff weight limit.

Q358. While preparing for a flight, a remote pilot calculates the moment for a camera payload weighing 2 lbs that is installed 12 inches forward of the datum. What is the moment contribution in pound-inches?

A) 6 lb-in

B) 24 lb-in

C) 14 lb-in

Answer: B - Moment is calculated by multiplying weight by arm: 2 lbs × 12 in = 24 lb-in. This value is used in weight and balance computations to ensure the total CG remains within limits.

Q359. A remote pilot checks the forecast and observes winds aloft at 22 knots with gusts to 30 knots. The UAS manufacturer lists a maximum allowable wind of 15 knots. What is the appropriate decision?

A) Reduce the payload by half to make the aircraft less susceptible to wind.

B) Fly only if the pilot has completed advanced training.

C) Cancel the flight because expected winds exceed the aircraft's operational limit.

Answer: C - Exceeding the manufacturer's wind limit puts the aircraft at risk of loss of control and unsafe operation. Pilots must operate within the UAS's approved operating envelope; reducing payload does not change the wind limit.

Q360. A remote pilot is flying from two different airfields: one is at sea level on a cool morning, and the other is at 6,000 feet MSL on a hot afternoon. Which combination of conditions is most likely to produce the highest density altitude and the worst aircraft performance?

A) High elevation and high temperature

B) Low elevation and low temperature

C) Low elevation and high humidity

Answer: A - Density altitude increases with both altitude and temperature. High elevation and high temperature result in lower air density, reducing lift and propeller efficiency. Low elevation and low temperature produce denser air, while humidity has a small effect compared to altitude and temperature.

Q361. After adding a new camera and gimbal to the front of a fixed-wing UAS, the remote pilot notes that the CG is now near the forward limit. What primary handling characteristic should the pilot expect?

A) The aircraft will be less stable in pitch and hard to control.

B) The aircraft will require more elevator authority to maintain level flight and may stall at higher speeds.

C) The aircraft will be more maneuverable but may become nose-heavy on landing.

Answer: B - A forward CG increases pitch stability but requires more elevator force to raise the nose, especially during takeoff and landing. The wing must produce more lift to compensate, leading to higher stall speeds and reduced climb performance.

Q362. A manufacturer's performance chart states that the maximum takeoff weight is reduced by 5% for every 1,000 feet of elevation above 1,000 feet MSL. The UAS has a rated max takeoff weight of 50 lbs. What is the maximum allowable weight when operating from a field at 5,000 feet MSL?

A) 50 lbs

B) 45 lbs

C) 40 lbs

Answer: C - At 5,000 feet MSL, the aircraft is 4,000 feet above the 1,000-foot reference. 4 × 5% = 20% reduction. 50 lbs × 0.80 = 40 lbs. Exceeding this weight would degrade climb and control performance at high density altitude.

Q363. You are flying a small fixed-wing UAS on a hot summer afternoon from a high-elevation airport. How does the resulting high density altitude affect the aircraft's performance?

A) It has no effect on small unmanned aircraft.

B) It increases lift and propeller efficiency.

C) It decreases lift and propeller efficiency.

Answer: C - High density altitude means the air is less dense. This reduces lift generation and propeller efficiency, which degrades takeoff, climb, and overall performance.

Q364. Your small multirotor has a maximum gross takeoff weight of 4.5 kg. The aircraft with battery weighs 3.2 kg, your camera payload weighs 0.8 kg, and you also want to attach an additional sensor weighing 0.7 kg. What is the total weight and is the flight within limits?

A) 4.0 kg, within limits

B) 4.7 kg, exceeds the maximum gross takeoff weight

C) 3.2 kg, within limits

Answer: B - The total weight is 3.2 + 0.8 + 0.7 = 4.7 kg, which exceeds the 4.5 kg maximum. Operating above the maximum *gross takeoff weight can degrade performance and compromise safety.*

Q365. You are planning a UAS flight during a hot day with high humidity. What effect will these conditions have on the density altitude and your aircraft's performance?

A) Density altitude is low, increasing lift and thrust.

B) Density altitude is high, reducing lift and thrust.

C) Humidity has no effect on performance.

Answer: B - High temperature and high humidity both lower air density, raising density altitude. Lower air density degrades aerodynamic performance and propeller efficiency.

Q366. You are choosing a battery for a mapping mission. A heavier battery provides longer flight time but adds weight, while a lighter battery improves agility but reduces endurance. What is the most appropriate selection consideration?

A) Always use the lightest battery to maximize payload capacity.

B) Use the battery that best balances weight and required flight time while keeping the aircraft within weight and CG limits.

C) Always use the highest-capacity battery, regardless of weight.

Answer: B - The remote pilot must ensure the aircraft remains within specified weight and center of gravity limits. Battery selection requires balancing endurance needs with the aircraft's approved loading envelope.

Q367. A remote pilot observes that their small UAS has a noticeably shorter flight time during cold-weather operations. What is the most likely cause?

A) Cold temperatures reduce battery performance and available capacity.

B) Cold weather has no effect on UAS batteries.

C) Cold air is denser, so the aircraft requires more power to generate lift.

Answer: A - Low temperatures slow the chemical reactions inside batteries, reducing their effective capacity. This results in shorter flight times even though the battery may appear fully charged.

Q368. You need to attach an underslung payload to your multirotor for an aerial survey. How will hanging the payload far below the aircraft affect stability?

A) It lowers the center of gravity but can create a pendulum effect that reduces stability.

B) It always improves stability and controllability.

C) It does not affect stability.

Answer: A - An underslung load moves the center of gravity downward, which can increase the likelihood of a pendulum effect. This can make the aircraft less stable and require extra control inputs.

Q369. You are flying a fixed-wing UAS from a mountainous airstrip on a hot afternoon. Which set of conditions produces the highest density altitude?

A) Low air temperature and low elevation

B) High air temperature and high elevation

C) High air temperature and low elevation

Answer: B - Density altitude increases with higher air temperature and higher elevation because both reduce atmospheric pressure and air density. High density altitude degrades aircraft performance.

Q370. Before a flight, you add a custom camera payload to the front of your small UAS. How should you verify that the aircraft remains safe to operate?

A) Confirm both the total weight and the center of gravity are within the approved limits.

B) Confirm the payload is securely attached and ignore weight and CG.

C) Confirm the total weight is below the maximum takeoff weight.

Answer: A - Loading changes affect both total weight and center of gravity. To maintain controllability and prevent structural failure, the remote pilot must ensure the aircraft remains within its approved weight and CG limits.

Q371. While flying, you notice the battery charge is much lower than expected and your UAS is a long distance downrange. What is the best course of action?

A) Immediately begin a controlled return to the launch point and land before the battery is exhausted.

B) Increase your airspeed to return to the launch point quickly, even if it uses more battery.

C) Continue the mission and land the UAS as soon as it returns to the launch point, even if the battery is depleted.

Answer: A - Managing the battery is a critical operational duty. If the battery level is lower than planned, the safest action is to abort the mission and execute a controlled return to a safe landing area before the battery runs out. Increasing speed may drain the battery faster and increase risk.

Q372. You are preparing for a flight on a hot, humid afternoon at an airport with an elevation of 5,000 feet MSL. The sUAS requires more throttle to hover, and the battery drains faster than expected. What condition is primarily degrading the aircraft's performance?

A) High density altitude

B) Wind shear

C) Magnetic interference

Answer: A - High temperature, high humidity, and high elevation all increase density altitude, which reduces air density. This lowers lift and propeller efficiency, requiring higher throttle settings and causing faster battery drain.

Q373. During a preflight inspection, you calculate that your sUAS with a new camera payload is within the maximum takeoff weight, but the center of gravity is near the aft limit. What is the most likely effect on pitch control?

A) The aircraft will be more stable and easier to control

B) The aircraft will tend to pitch nose-down and require more aft stick

C) It may be difficult to bring the nose down, reducing pitch authority and increasing the risk of loss of control

Answer: C - An aft center of gravity reduces longitudinal stability and makes it harder to pitch the nose down, which can lead to a stall or loss of control. The aircraft must be operated within the manufacturer's CG envelope.

Q374. You need to inspect a rooftop using your quadcopter. The aircraft's empty weight is 1.1 kg, the battery weighs 0.3 kg, and the inspection camera weighs 0.25 kg. The manufacturer's maximum takeoff weight is 1.5 kg. What should you do?

A) Fly with the camera mounted near the center to stay within the CG limits

B) Remove the battery to reduce weight and fly on auxiliary power

C) Do not fly; the total takeoff weight exceeds the manufacturer's limit

Answer: C - The total takeoff weight is 1.1 + 0.3 + 0.25 = 1.65 kg, which exceeds the 1.5 kg maximum takeoff weight. Operating above the manufacturer's structural limit is unsafe and violates the required operating limitations.

Q375. During an aerial survey, your sUAS enters a steady 60-degree banked turn. The pilot observes that the aircraft requires additional angle of attack to maintain level flight. Why does this occur?

A) The banked turn increases the load factor, which raises the stall speed

B) The banked turn decreases the load factor, making the aircraft lighter

C) The banked turn improves propeller efficiency, but reduces battery output

Answer: A - In a coordinated 60-degree bank, the load factor is 2 G. This effectively doubles the wing loading, requiring a higher angle of attack and increasing the stall speed. Pilots must account for this when maneuvering.

Q376. You are planning a flight in cold winter conditions. How does the cold temperature typically affect the sUAS battery's performance?

A) Battery capacity and voltage are diminished, reducing available flight time and power output

B) Battery capacity increases because lower temperatures reduce internal resistance

C) Cold temperatures have no effect on battery performance

Answer: A - Cold temperatures slow the chemical reactions inside batteries, reducing their capacity and voltage. A remote pilot should plan for shorter flight times and keep batteries warm until just before launch.

Q377. You are investigating a reported aircraft performance issue. Which source provides the most authoritative loading and performance information for the specific sUAS you are operating?

A) FAA Part 107 regulations

B) The manufacturer's Flight Manual or operating manual

C) Sectional charts

Answer: B - The manufacturer's operating manual contains the specific limitations, weight and balance data, and performance tables for the exact sUAS model. The remote pilot must comply with those limitations as required by §107.51 and related rules.

Q378. You are flying a small UAS over a private property inspection site. The current flight's total weight exceeds the manufacturer's maximum takeoff weight, but the center of gravity remains within limits. Are you authorized to conduct the flight?

A) Yes, as long as the flight is over private property

B) Yes, if the visual observer approves the extra weight

C) No, because the remote pilot must comply with all operating limitations specified by the manufacturer

Answer: C - Part 107 requires compliance with the sUAS's operating limitations. Exceeding the manufacturer's maximum takeoff weight is not allowed regardless of location or observer approval, and it can compromise structural integrity and controllability.

Q379. You are flying at 3,000 feet MSL on a hot, dry day. The density altitude is calculated to be much higher than the actual altitude. What performance effect should you expect?

A) Reduced propeller efficiency and decreased ability to support weight

B) Increased battery endurance due to lower air resistance

C) Improved climb performance and increased payload capability

Answer: A - High density altitude means less dense air, which reduces propeller thrust and rotor efficiency. This degrades climb performance and reduces the sUAS's ability to carry a given load, so pilots must adjust their performance expectations.

Q380. You are flying a small UAS near the coast on a warm, very humid day. How does the high humidity affect the air density and your aircraft's performance?

A) Water vapor is lighter than air, so moist air is less dense, which increases density altitude and reduces performance.

B) Water vapor is heavier than air, so moist air is denser, which decreases density altitude and improves performance.

C) Humidity has no effect on density altitude because it is not a component of the atmosphere.

Answer: A - Water vapor is lighter than dry air. When humidity increases, the air becomes less dense, raising the density altitude and degrading aircraft performance. Humidity is a contributing factor to density altitude, though temperature and pressure typically dominate.

Q381. While planning a flight at an airport located at a high field elevation, you note the temperature is above standard. What should you expect regarding your small UA's performance?

A) The aircraft will perform better because higher temperatures increase air density.

B) The aircraft will experience less efficient lift and thrust due to a higher density altitude.

C) The aircraft will require less battery power because thinner air reduces drag.

Answer: B - High elevation already reduces air pressure, and high temperature further decreases air density. The combined effect creates a high density altitude, meaning the air is less dense. This reduces lift, thrust, and overall aircraft performance.

Q382. You are operating your sUAS in the outer ring of Class C airspace at 2,500 feet AGL. The airport elevation is 2,000 feet MSL. What action must you take?

A) No authorization is needed because you are below 400 feet AGL.

B) You must receive ATC authorization before operating in this airspace.

C) You may operate without authorization if you stay at least 5 miles from the airport.

Answer: B - The outer circle of Class C airspace extends from 1,200 feet to 4,000 feet above the airport elevation. Since you are at 2,500 feet AGL, you are within Class C airspace, and remote pilots must receive ATC authorization to operate there.

Q383. You plan to fly near an airport that has an operational control tower. The airspace is tailored to the airport and extends from the surface to 2,500 feet above the airport elevation. What is this airspace classification?

A) Class C airspace

B) Class D airspace

C) Class E airspace

Answer: B - Class D airspace is generally surface to 2,500 feet above the airport elevation and surrounds airports with an operational control tower. Remote pilots must receive ATC authorization before operating in Class D airspace.

Q384. You are flying at 600 feet AGL in an area where the sectional chart does not depict any Class E airspace base. What class of airspace are you operating in?

A) Class E airspace

B) Class D airspace

C) Class G airspace

Answer: C - In areas where charts do not depict a Class E base, Class E begins at 14,500 feet MSL. Since you are at 600 feet AGL, the airspace below that base is Class G (uncontrolled), and no ATC authorization is required.

Q385. You are flying at 1,500 feet AGL in an area where the Class E airspace base is 700 feet AGL. Do you need ATC authorization to operate there?

A) Yes, because you are in Class E airspace.

B) No, because Class E airspace generally does not require ATC authorization.

C) No, because you are still in Class G airspace.

Answer: B - You are above 700 feet AGL, so you are in Class E airspace. In most cases, a remote pilot does not need ATC authorization to operate in Class E airspace.

Q386. You are following a Federal airway, shown as a blue line on the sectional chart, at 1,500 feet AGL. What is the airspace classification for this operation?

A) Class E airspace

B) Class G airspace

C) Class D airspace

Answer: A - Federal Airways are usually within Class E airspace. They start at 1,200 feet AGL and extend up to, but not including, 18,000 feet MSL. You are at 1,500 feet AGL, so you are in Class E airspace.

Q387. According to the FAA study guide, which of the following is NOT a listed type of special use airspace?

A) Prohibited areas

B) Restricted areas

C) Parachute jump areas

Answer: C - The excerpt lists Prohibited, Restricted, Warning, and Military areas as examples of special use airspace. Parachute jump areas are not mentioned as a type of special use airspace in the provided material.

Q388. A remote pilot must receive ATC authorization before operating in which of the following airspace classes?

A) Class G airspace

B) Class E airspace

C) Class D airspace

Answer: C - The study guide states that a remote pilot must receive ATC authorization before operating in Class D airspace. Class G and Class E airspace do not require such authorization.

Q389. Above what altitude is all airspace classified as Class E airspace per the FAA study guide?

A) Flight Level 180

B) 14,500 feet MSL

C) Flight Level 600

Answer: C - The excerpt states that all airspace above FL 600 is Class E airspace. FL 600 is 60,000 feet MSL. Class E typically extends up to, but not including, 18,000 feet MSL, while Class A begins at 18,000 feet MSL.

Q390. You are a remote pilot planning an inspection of a tower located within Class C airspace. What must you do before operating?

A) No authorization is required if the drone weighs less than 250 grams.

B) Receive ATC authorization to operate in Class C airspace.

C) File a flight plan with the nearest Automated Flight Service Station.

Answer: B - The excerpt states that a remote pilot must receive authorization before operating in Class C airspace. ATC authorization is required regardless of drone weight.

Q391. You are flying near an airport with an operational control tower, and the airspace is Class D. What is the typical vertical extent of this airspace above the airport elevation?

A) From the surface up to 2,500 feet above the airport elevation (MSL).

B) From 1,200 feet AGL up to 4,000 feet MSL.

C) From the surface up to 4,000 feet above the airport elevation.

Answer: A - Class D airspace is generally airspace from the surface to 2,500 feet above the airport elevation (charted in MSL).

Q392. A sectional chart shows magenta shading indicating the floor of Class E airspace is 700 feet AGL. You plan to fly your small UAS at 500 feet AGL in that area.

What airspace are you operating in?

A) Class G airspace.

B) Class E airspace.

C) Class D airspace.

Answer: A - Class G airspace extends from the surface to the base of the overlying Class E airspace. Since the Class E base is 700 feet AGL, the airspace below 700 feet AGL is Class G.

Q393. While reviewing a sectional chart for a remote flight, you notice a blue line that represents a Federal Airway. At what altitude does this Federal Airway typically begin?

A) 1,200 feet AGL.

B) 700 feet AGL.

C) 14,500 feet MSL.

Answer: A - Federal Airways, shown as blue lines on a sectional chart, start at 1,200 feet AGL and go up to, but not including, 18,000 feet MSL.

Q394. Which of the following is correctly described as special use airspace?

A) Air Defense Identification Zone (ADIZ).

B) Class E airspace.

C) Restricted area.

Answer: C - Special use airspace usually consists of prohibited areas, restricted areas, warning areas, and military areas. Restricted areas are a listed example.

Q395. You are operating a small UAS in Class G airspace near a non-towered airport. What is true regarding ATC authorization?

A) No ATC authorization is required.

B) You must contact the nearest tower before flight.

C) ATC authorization is always required.

Answer: A - Class G is uncontrolled airspace, and a remote pilot will not need ATC authorization to operate in Class G airspace.

Q396. In which of the following airspace classes does a remote pilot generally NOT need to obtain ATC authorization?

A) Class B airspace.

B) Class D airspace.

C) Class E airspace.

Answer: C - In most cases, a remote pilot will not need ATC authorization to operate in Class E airspace. Class B, C, and D airspace all require authorization.

Q397. All airspace above FL 600 (60,000 feet MSL) is classified as which class of airspace?

A) Class A.

B) Class B.

C) Class E.

Answer: C - According to the excerpt, all airspace above FL 600 is Class E airspace.

Q398. You are planning to fly a drone at 3,500 feet MSL over a city that has an operational control tower. The sectional chart indicates you are within 10 nautical miles of the airport. What must you do?

A) You must receive ATC authorization before operating in this airspace.

B) You must maintain visual line of sight (VLOS) only.

C) You may operate freely as long as you are below 4,000 feet MSL.

Answer: A - According to the provided material, Class C airspace is generally a 10 NM radius area extending from 1,200 feet to 4,000 feet above the airport elevation, and a remote pilot must receive authorization before operating in it.

Q399. You are looking at a sectional chart and notice a blue line representing a Federal Airways, such as V123. You are flying at 10,000 feet MSL. Which airspace classification applies to this flight?

A) Class G airspace.

B) Class A airspace.

C) Class E airspace.

Answer: C - Federal Airways are usually found within Class E airspace, which typically extends up to but not including 18,000 feet MSL.

Q400. You are flying over an airport that has an operational control tower. The airport elevation is 500 feet MSL, and you are maintaining 2,000 feet MSL. What is the airspace classification and requirement?

A) Class E airspace; no ATC authorization is required.

B) Class D airspace; you must receive ATC authorization.

C) Class C airspace; you must receive ATC authorization.

Answer: B - Class D airspace is generally from the surface to 2,500 feet above the airport elevation surrounding airports with an operational control tower, and authorization is required.

Q401. You are flying at 500 feet AGL over a rural field. Above you, the airspace transitions into Class E at 1,200 feet AGL. What is the classification of the airspace you are currently operating in?

A) Class E airspace.

B) Class G airspace.

C) Class C airspace.

Answer: B - Class G airspace is the portion of the airspace that has not been designated as Class A, B, C, D, or E, extending from the surface to the base of the overlying Class E airspace.

Q402. You receive a notification of a 'Warning Area' depicted on your chart. What does this designation indicate regarding the airspace?

A) The area is permanently prohibited to all aircraft.

B) Certain activities must be confined, or limitations may be imposed on aircraft operations.

C) The airspace is controlled by the military and strictly off-limits to civilians.

Answer: B - Special use airspace, such as Warning areas, is designated for activities that must be confined, or where limitations may be imposed on aircraft operations that are not part of those activities.

Q403. You are flying at 2,000 feet MSL over a mountainous region. The chart indicates Class E airspace beginning at 1,200 feet AGL but does not specify an MSL base. What is the lowest altitude you can legally operate in Class E?

A) 1,200 feet AGL.

B) 1,200 feet MSL.

C) 700 feet AGL.

Answer: A - In most areas, the Class E airspace base is 1,200 feet above ground level (AGL).

Q404. You are operating a drone in uncontrolled airspace and you are below the base of any overlying Class E airspace. What is the requirement regarding ATC authorization?

A) You must receive ATC authorization.

B) You do not need ATC authorization.

C) You must contact the airport traffic control tower.

Answer: B - A remote pilot will not need ATC authorization to operate in Class G (uncontrolled) airspace.

Q405. You are flying at 3,500 feet MSL near a Class C airport. You are within the 10 NM radius circle that extends from 1,200 feet to 4,000 feet above the airport elevation. What authorization is required to operate in this airspace?

A) No authorization is required as you are below 4,000 feet.

B) ATC clearance and authorization from the airport operator.

C) ATC authorization before operating in Class C airspace.

Answer: C - According to the provided text, a remote pilot must receive authorization before operating in Class C airspace. This applies regardless of the specific altitude within the 1,200 to 4,000 feet layer.

Q406. You are approaching a towered airport with an operational control tower. You are flying at 2,000 feet MSL, which is below 2,500 feet above the airport elevation. What type of airspace are you in?

A) Class C airspace.

B) Class G airspace.

C) Class D airspace.

Answer: C - Class D airspace generally extends from the surface to 2,500 feet above the airport elevation surrounding airports with an operational control tower.

Q407. You are flying over an area where the sectional chart does not depict a Class E base. At what altitude does Class E airspace begin?

A) 1,200 feet AGL.

B) 14,500 feet MSL.

C) 700 feet AGL.

Answer: B - In areas where charts do not depict a Class E base, the airspace begins at 14,500 feet MSL.

Q408. You are flying over an uncontrolled airport with no tower and no Class E airspace depicted above it. You are at 500 feet AGL. What type of airspace are you in?

A) Class G airspace.

B) Class E airspace.

C) Class C airspace.

Answer: A - Class G airspace is uncontrolled airspace that extends from the surface to the base of the overlying Class E airspace.

Q409. You are following a blue line on a sectional chart. This line represents a Federal Airways. In which class of airspace are these Federal Airways typically located?

A) Class A airspace.

B) Class E airspace.

C) Class G airspace.

Answer: B - Federal Airways, shown as blue lines, are usually found within Class E airspace.

Q410. You are flying at Flight Level 580. What class of airspace are you in?

A) Class A airspace.

B) Class E airspace.

C) Class B airspace.

Answer: B - The text states that all airspace above FL 600 is Class E airspace.

Q411. You are entering a section of airspace designated for military activities that may pose a hazard to non-participating aircraft. This area is not specifically assigned to any country. What type of Special Use Airspace is this?

A) Prohibited Area.

B) Restricted Area.

C) Warning Area.

Answer: C - Warning areas are airspace of international concern that may extend beyond 12 nautical miles. They consist of the same type of activities as Restricted Areas but are not specifically assigned to any country.

Q412. You are flying at 2,000 feet AGL near an airport with an operational control tower. The chart indicates this area is Class D airspace. You are instructed by Air Traffic Control (ATC) to maintain visual separation. Which statement is true regarding your operation?

A) You must receive authorization from the airport management, not ATC.

B) You do not need ATC authorization to operate in Class D airspace.

C) You must receive ATC authorization before operating in Class D airspace.

Answer: C - The provided text states that a remote pilot must receive ATC authorization before operating in Class D airspace.

Chapter 5: Flight Operations

Q413. What rating is granted to a pilot who passes the airman knowledge test for sUAS?

A) Remote Pilot Certificate with an sUAS rating

B) Remote Pilot Certificate

C) Small Unmanned Aircraft Pilot License

Answer: A - The text mentions obtaining the 'Remote Pilot Certificate with an sUAS rating airman knowledge test'.

Q414. Which entity is listed as being located at Washington, DC 20591?

A) The Office of the Secretary

B) National Transportation Safety Board

C) Flight Standards Service

Answer: C - The text lists the address for the 'Flight Standards Service' as Washington, DC 20591.

Q415. Who is the intended audience for the Remote Pilot – sUAS Study Guide?

A) Air traffic controllers

B) Aircraft mechanics

C) Remote pilots preparing to take the test

Answer: C - The text states the guide is for those 'you need to study to prepare to take the Remote Pilot Certificate'.

Q416. You are flying over a large stadium. What is the maximum altitude you may operate at?

A) 500 feet above ground level (AGL)

B) 400 feet above ground level (AGL)

C) 1000 feet above ground level (AGL)

Answer: B - The maximum altitude for a small UAS is 400 feet AGL, unless you are operating within a 400-foot radius of a structure, in which case you may ascend to 400 feet above the structure's top (14 CFR § 107.51).

Q417. You are operating a small UAS at night. What is the minimum equipment requirement?

A) Anti-collision lights

B) Red and green navigation lights only

C) White strobe lights only

Answer: A - When operating at night, the drone must be equipped with anti-collision lighting that is visible from at least 3 statute miles (14 CFR § 107.29).

Q418. While flying at 400 feet AGL over a city, your drone hits a 100-foot tower and bounces off. You immediately switch to an alternate landing site. What is the next altitude limit?

A) 500 feet AGL

B) 400 feet AGL

C) 300 feet AGL

Answer: B - If you are no longer above an obstacle, you must descend to 400 feet AGL immediately. You are not permitted to maintain a higher altitude just because you were previously above an obstacle (14 CFR § 107.51).

Q419. You want to fly over a heliport. What is the general restriction regarding operation within 5 nautical miles of a heliport?

A) You may not fly within 5 nautical miles of a heliport.

B) You may fly anywhere as long as you are under 400 feet.

C) You must obtain authorization from ATC.

Answer: C - Operation within 5 nautical miles of a heliport is prohibited unless you have specific authorization from ATC (14 CFR § 107.41).

Q420. You are flying a drone and realize your visual line of sight is obstructed by a tall building. What should you do?

A) Continue flying because you are using GPS.

B) Land the drone immediately.

C) Switch to night lights to see better.

Answer: B - You must maintain Visual Line of Sight (VLOS) of the small UAS at all times. If the obstruction prevents VLOS, you must land immediately (14 CFR § 107.33).

Q421. You discover a Restricted Area (R-4201) on your sectional chart that you need to cross to get to your survey site. What must you do?

A) Fly through it immediately.

B) Obtain ATC authorization before entering.

C) Fly under the ceiling if it is active.

Answer: B - You may not operate within a Restricted Area unless you have received authorization from the controlling agency, which is typically ATC (14 CFR § 107.41).

Q422. Your drone battery is critically low and you have a good GPS signal. What is the safest action to take?

A) Fly to a location further away to get better signal.

B) Immediately execute a Return-to-Home (RTH) or land.

C) Hover in place until the battery stabilizes.

Answer: B - To ensure safety and prevent an accident, the drone pilot should land immediately when battery levels are critically low (14 CFR § 107.23).

Q423. Which of the following constitutes a violation of the Remote ID broadcast rule?

A) Broadcasting your registration number and location.

B) Broadcasting your location and the aircraft's location.

C) Not broadcasting because you are in an unmanned aircraft free zone.

Answer: C - Remote ID broadcasting is required for all drones registered for recreational or commercial use, except in designated areas or under specific exemptions (14 CFR § 107.200).

Q424. You are flying over a stadium that contains radioactive material. What is the restriction?

A) You may fly if you are under 400 feet.

B) You may fly if you are more than 2,000 feet above the material.

C) You may not fly.

Answer: C - You may not operate a small UAS over a human performer or a hazardous activity (such as radioactive material) unless authorized by a waiver (14 CFR § 107.33).

Q425. You plan to operate your small UAS 45 minutes after sunset at a rural site. There is no ambient lighting. Which lighting requirement applies to your aircraft?

A) An anti-collision light visible for at least 3 statute miles

B) Position lights and a strobe visible for at least 5 statute miles

C) A landing light illuminated whenever the aircraft is above 50 ft AGL

Answer: A - For night operations, the sUAS must be equipped with an anti-collision light that is visible for at least 3 statute miles and has a flash rate sufficient to avoid collision. Position lights and landing-light requirements apply to manned aircraft, not small UAS.

Q426. You are using a visual observer to help maintain situational awareness during a long linear flight. Which statement is correct regarding visual line of sight (VLOS)?

A) You must personally keep the sUAS in sight with unaided vision at all times

B) The visual observer must be able to see the sUAS throughout the flight and communicate with you

C) The visual observer may take over as pilot in command if you lose sight

Answer: B - A visual observer must maintain VLOS of the sUAS and stay in communication with the remote PIC. The PIC remains responsible for the flight and cannot relinquish command to an observer.

Q427. You intend to fly at 300 ft AGL over farmland located 4 nautical miles from a non-towered airport. What is a key operational responsibility during this flight?

A) Scan for and yield to manned aircraft, and avoid creating a collision hazard

B) Obtain explicit permission from the nearest airport owner before operating

C) Remain below 200 ft AGL because you are within 5 miles of the airport

Answer: A - The remote pilot must see and avoid other aircraft and must yield the right of way to all manned aircraft. There is no blanket permission requirement or automatic altitude cap for operating near a non-towered airport.

Q428. While landing your sUAS after a rooftop inspection, it clips a small satellite dish and causes $200 in damage. No one is injured. What are your reporting obligations?

A) File an accident report with the FAA within 10 days

B) File a report with the NTSB because property damage occurred

C) No immediate report is required to the FAA or NTSB

Answer: C - Mandatory reporting is required only for serious injury, fatality, or property damage exceeding $500. In this scenario, the damage is below that threshold and no injury occurred.

Q429. During a bridge inspection flight, you notice winds have increased to the limit of your aircraft's capabilities. Your battery is still at 60%, and you have enough fuel to finish in 15 minutes. What is the best aeronautical decision?

A) Abort the inspection and land in a safe area

B) Reduce your speed and continue so the battery is not wasted

C) Continue the flight but increase altitude to escape turbulence

Answer: A - Good aeronautical decision making prioritizes safety. Once conditions exceed safe limits, continuing increases risk. The safest action is to terminate the operation and land, even if the battery appears sufficient.

Q430. You are flying your sUAS at 200 ft AGL when you notice a manned helicopter approaching from your right at roughly the same altitude. What is your responsibility regarding right of way?

A) You must yield the right of way and ensure that the helicopter does not have to maneuver to avoid you

B) The helicopter must maintain a distance of 500 ft from you because you are a small aircraft

C) You have the right of way because sUAS are smaller and more maneuverable

Answer: A - The sUAS must yield the right of way to all manned aircraft. You are required to operate in a way that does not create a collision hazard and to give way if the manned aircraft is in your path.

Q431. You are operating your drone near an airport when you see a manned aircraft approaching head-on. Under Part 107, what must you do?

A) Climb to avoid a collision

B) Continue on your current course because you have the right of way as a smaller aircraft

C) Yield the right of way to the manned aircraft

Answer: C - Part 107 states that the remote PIC must give way to all manned aircraft, especially when there is a risk of collision. Small UA are not granted any right of way over manned aircraft.

Q432. During your Part 107 flight, your drone collides with a person on the ground, causing a serious injury. What must you do regarding reporting?

A) Report the accident to the FAA within 10 days

B) Report the accident to the nearest law enforcement immediately

C) No report is required if the injury was unintentional

Answer: A - Part 107 requires the remote PIC to report an accident to the FAA within 10 days if it involves serious injury to any person or substantial damage to property. A serious injury is defined as one requiring hospitalization.

Q433. Your cross-country mapping mission will pass through a restricted area (e.g., R-2510) that is currently 'cold' (not active). Can you fly through it?

A) Yes, as long as it is not active

B) No, you must avoid restricted areas unless authorized to operate within them

C) Yes, if you fly below 400 ft AGL

Answer: B - Restricted areas are special use airspace where flight operations are subject to restrictions. Even when not active, sUAS operations are not permitted without the permission of the controlling agency, and Part 107 pilots should avoid such areas.

Q434. A real estate client requests drone photographs of a property 45 minutes after official sunset. Your sUAS is equipped with an anti-collision strobe visible from more than 3 miles. The weather is clear. Are you allowed to conduct the flight under Part 107?

A) Yes, because the anti-collision lighting is visible for at least 3 statute miles.

B) Yes, because 107.29 permits night operations if the aircraft has anti-collision lighting.

C) No, because civil twilight ends 30 minutes after official sunset and this flight would be outside that period.

Answer: C - Part 107.29 permits operations during daylight or civil twilight (30 minutes before official sunrise to 30 minutes after official sunset) provided anti-collision lighting is used. A flight 45 minutes after sunset is outside the civil twilight window.

Q435. While you are operating your sUAS at 250 feet AGL, a manned helicopter enters the area on a direct collision course. What action does Part 107 require you to take?

A) Continue your current course because the helicopter is responsible for avoiding you.

B) Immediately climb above the helicopter because you are below 400 feet.

C) Make every reasonable effort to yield the right-of-way and move out of the helicopter's path.

Answer: C - Part 107.37 states that a small unmanned aircraft must yield the right-of-way to all manned aircraft. You must take action to avoid the helicopter, not assume it will maneuver away.

Q436. You plan to fly a commercial inspection inside the Class C airspace surrounding a medium-hub airport. The flight will stay below 400 feet AGL and remain 3 miles from the airport. What must you do before operating?

A) Obtain prior authorization from ATC before flying in Class C airspace.

B) File a flight plan with the nearest Flight Service Station.

C) No ATC authorization is needed because you are below 400 feet AGL.

Answer: A - Part 107.41 prohibits operation in Class B, Class C, or Class D airspace without prior ATC authorization. This applies regardless of altitude within that airspace.

Q437. You are inspecting the top of a 250-foot-tall communication tower. You keep the sUAS within 100 feet of the tower structure. What is the maximum altitude you may operate the sUAS above ground level?

A) 400 feet AGL

B) 650 feet AGL

C) 250 feet AGL

Answer: B - Under Part 107.51, the maximum allowable altitude is 400 feet above ground level, except that a small unmanned aircraft may operate up to 400 feet above the top of a structure if it is within 400 feet of that structure. 250 + 400 = 650 feet AGL.

Q438. You are hired to film a crowded outdoor festival. You plan to fly directly over the crowd at 25 feet AGL to capture close-up footage. Under Part 107, what is required for this operation?

A) You may operate over the crowd as long as you stay below 50 feet AGL.

B) You may not operate over people who are not directly participating in the operation.

C) You may operate over the crowd if you have a visual observer and an emergency landing plan.

Answer: B - Part 107.39 prohibits operating a small unmanned aircraft over persons who are not directly participating in the operation, and persons under covered structures or inside stationary vehicles. Spectators at a festival are not participants.

Q439. While performing a sUAS delivery test, you want to release a sandbag over a residential backyard. No one appears to be in the yard. At what weight would this release be safe and permitted?

A) Any weight, as long as the aircraft remains below 400 feet AGL.

B) You may drop objects if you first broadcast a warning to people on the ground.

C) It is prohibited if the dropped object creates a hazard to persons or property.

Answer: C - Part 107 prohibits allowing an object to be dropped from a small unmanned aircraft if it creates a hazard to persons or property. Even if nobody appears present, a sandbag over a residential area poses an unacceptable risk.

Q440. You are conducting an aerial survey from a slowly moving pickup truck in a sparsely populated rural area. You are not carrying any cargo and no people are nearby. Is this operation permitted under Part 107?

A) Yes, because operations from a moving land vehicle are allowed over sparsely populated areas when no property is carried.

B) No, because the sUAS must be launched from a stationary location at all times.

C) Yes, but only if you hold a commercial driver's license.

Answer: A - Part 107.25 permits operation from a moving land or water vehicle if the operation is over a sparsely populated area and the small UAS does not transport property for compensation or hire. This survey meets those requirements.

Q441. While flying your sUAS, you notice that the remote control signal is lost, but the aircraft remains airborne and stable. What should you do first?

A) Attempt to re-establish communication and follow the aircraft's preplanned lost-link procedure.

B) Immediately turn off your controller to activate the lost-link fail safe.

C) Continue with the planned mission because the aircraft's GPS is still working.

Answer: A - A remote PIC must always be able to maintain control of the sUAS. If the control link is lost, you should follow the manufacturer's lost-link procedure, which typically involves attempting to re-establish the link while the aircraft executes a safe contingency plan, such as returning to home or landing.

Q442. You are flying your sUAS at an altitude of 150 feet AGL when you see a manned helicopter approaching from your right. Who has the right-of-way?

A) Your sUAS, because it is at a lower altitude.

B) The helicopter, because it is on your right.

C) The helicopter, because all manned aircraft have right-of-way over small unmanned aircraft.

Answer: C - Part 107 requires that the small unmanned aircraft yields the right-of-way to all manned aircraft. The drone must not interfere with the helicopter's operation and should maneuver to avoid a collision.

Q443. A visual observer is assisting you with your sUAS operation. What is the primary responsibility of the visual observer?

A) To maintain visual line of sight with the aircraft and scan for potential collision hazards.

B) To take over the controls if you lose sight of the aircraft.

C) To keep the aircraft within radio range of the remote controller.

Answer: A - The visual observer (VO) is a crewmember who assists the remote PIC by positively identifying the UAS at all times and scanning the airspace for hazards. The VO does not control the aircraft; that is the responsibility of the remote PIC.

Q444. You are a remote pilot in command and plan to operate your sUAS from a moving car in a sparsely populated rural area. No cargo is being carried on the UAS. Is this operation permitted?

A) Yes, as long as the area is sparsely populated and no cargo is transported by the UAS.

B) No, operating from a moving vehicle is always prohibited for small unmanned aircraft.

C) Yes, but only if a visual observer is also in the vehicle.

Answer: A - Part 107 permits operations from a moving land or water vehicle if the operation is conducted in a sparsely populated area and the UAS does not carry cargo. This exception allows sustained flight from a road vehicle under the stated conditions.

Q445. You had a long, stressful workday and feel fatigued. Your scheduled sUAS flight is for a routine mapping project. What is the safest decision?

A) Cancel the flight and fly another day when you are rested.

B) Proceed with the flight as planned because fatigue does not affect UAS operations.

C) Fly with a visual observer who can monitor your performance.

Answer: A - Fatigue can significantly impair decision-making, reaction time, and situational awareness. As a remote pilot in command, you are responsible for ensuring you are fit for duty. If you are fatigued, the safest operation is to cancel or postpone the flight rather than introduce an avoidable human factors risk.

Q446. You are preparing to launch your sUAS in a remote rural area with no other air traffic visible. You notice a manned airplane on a direct collision course with your drone. Who has the right of way and what action must you take?

A) You have the right of way because the aircraft is below 500 feet AGL

B) You must maneuver to the right and yield the right of way to the manned aircraft

C) You must descend immediately because the aircraft is higher in the sky

Answer: B - Small UAS must always yield the right of way to manned aircraft. If a collision risk exists, the remote PIC should maneuver appropriately (typically to the right) and not expect the manned aircraft to alter its course.

Q447. You are flying your sUAS at 200 ft AGL in uncontrolled airspace when you see a manned aircraft approaching from your left at the same altitude. What is the correct action?

A) Climb to 400 ft to ensure vertical separation

B) Maintain your course and let the manned aircraft maneuver around you

C) Give way to the manned aircraft immediately

Answer: C - Small UAS must yield the right of way to all manned aircraft. The remote PIC is expected to maneuver the sUAS to avoid a collision when converging with a manned aircraft.

Q448. While flying a survey mission, your drone passes behind a large building and you briefly lose visual line of sight (VLOS). What should you do?

A) Continue along the planned route since the loss is only temporary

B) Maneuver to regain VLOS or land the aircraft immediately

C) Increase altitude to see over the building and reacquire the aircraft

Answer: B - Part 107 requires the remote PIC and visual observer (if used) to maintain VLOS of the sUAS. If VLOS is lost, the remote PIC must take immediate steps to regain it or land the aircraft.

Q449. You and another remote PIC are flying sUAS that are converging at the same altitude. Your aircraft is to the left of the other aircraft. Who has the right of way?

A) You do, because you are on the left

B) The other aircraft, because it is on your right

C) The aircraft that is flying slower

Answer: B - Part 107 right-of-way rules state that when two small UAS are converging, the aircraft on the right has the right of way. Since the other aircraft is on your right, you must give way.

Q450. You are hired to take aerial photos of a crowded community festival. Over 100 people are in the open area. Can you operate your sUAS directly over the crowd?

A) Yes, if you remain below 400 ft AGL

B) Yes, if you obtain permission from the event organizer

C) No, because operations over human beings who are not directly involved in the operation are prohibited

Answer: C - Unless the operation qualifies under an exception or category (e.g., operations over people rules), Part 107 prohibits flying over human beings not directly participating in the sUAS operation. A crowd at a festival is not directly participating, so the flight cannot be conducted over them.

Q451. You are the remote pilot in command flying a visual line-of-sight operation. Your cell phone rings and you begin a text conversation while flying. What is the best practice in this situation?

A) Quickly finish the text and then return your attention to the UAS

B) Continue flying while texting since your visual observer can watch the aircraft

C) End the call/text and focus exclusively on the sUAS operation

Answer: C - Good crew resource management (CRM) means eliminating distractions and maintaining full awareness of the UAS and airspace. A remote PIC must not let phone use interfere with their primary responsibility to fly the aircraft safely. The best practice is to terminate the distraction.

Q452. While flying your small UAS near a rural airstrip, you notice a manned airplane approaching your right side on a collision course. What is your required action under Part 107?

A) Climb above the airplane so it can pass underneath

B) Give way to the manned aircraft by taking action to avoid it

C) Maintain your course and speed because the UAS has the right-of-way

Answer: B - Part 107.37 requires small UAS operators to yield the right-of-way to all manned aircraft. You must maneuver to avoid a collision; maintaining course and speed would violate this requirement.

Q453. You are conducting a nighttime Part 107 operation. To meet the FAA's safety lighting requirement, what must your anti-collision lighting be capable of?

A) Being visible for at least 3 statute miles

B) Being visible for at least 1 statute mile

C) Being visible for at least 5 nautical miles

Answer: A - Under §107.29, a small UAS operated at night must have anti-collision lighting that is visible for at least 3 statute miles. This requirement helps other aircraft see the drone during nighttime operations.

Q454. While on a survey flight at 400 feet AGL, you need to inspect a 250-foot-tall water tower that is isolated in a field. Under Part 107, what is the maximum altitude you may fly to inspect the top of the tower without a waiver?

A) 400 feet AGL

B) 650 feet AGL

C) 500 feet AGL

Answer: B - Part 107.51 limits altitude to 400 feet AGL, but allows flying above this limit if the aircraft is within 400 feet of a structure. You may go up to 400 feet above the structure's uppermost limit (250 ft + 400 ft = 650 ft AGL).

Q455. You are operating your sUAS when you see a manned airplane approaching head-on. Both aircraft are at the same altitude. What action must the remote pilot in command take?

A) Climb immediately to pass above the airplane

B) Descend immediately to pass below the airplane

C) Yield the right-of-way to the airplane

D) D) Maintain course and speed

Answer: C - 14 CFR §107.37 requires that small unmanned aircraft yield the right-of-way to all manned aircraft. The remote pilot must maneuver away to avoid a collision, but the rule does not specify a particular direction.

Q456. During a preflight inspection, which of the following must the remote pilot in command verify?

A) That the small UAS control link is functioning properly

B) That the remote pilot's certificate is physically displayed on the aircraft

C) That a visual observer is assigned when flying in Class G airspace

Answer: A - 14 CFR §107.49 requires a preflight inspection to ensure the small UAS is in a condition for safe operation. This includes checking that the control link is functional and that the aircraft's systems are working properly.

Q457. You are flying over a stadium with a roof height of 200 feet. Your drone is at 600 feet above ground level (AGL). Is this operation compliant with Part 107 altitude regulations?

A) No, because you must always be within 400 feet of an object.

B) No, because the drone is above 400 feet.

C) Yes, provided it remains higher than the nearby structures.

Answer: C - Part 107 allows a sUAS to operate at more than 400 feet AGL only if it remains at or below 400 feet above the ground level of the structure upon which it is operating.

Q458. You are conducting a visual flight over a forest canopy using First Person View (FPV) goggles. You have a visual observer assisting you. Are you in compliance with the Visual Line of Sight (VLOS) requirement?

A) Yes, as long as the visual observer is watching the drone.

B) Yes, FPV goggles are allowed as long as you have an observer.

C) No, the pilot must maintain direct visual sight of the sUAS at all times.

Answer: C - Under Part 107, the remote pilot in command must maintain visual sight of the sUAS at all times during the flight. Using FPV goggles negates the pilot's direct visual sight, violating the VLOS rule.

Q459. The current weather conditions show a ceiling of 450 feet and visibility of 2 miles. Can you legally conduct a Part 107 flight under these conditions?

A) Yes, as long as you are more than 5 miles from an airport.

B) No, you must have a ceiling of at least 500 feet and visibility of 3 miles.

C) Yes, provided you are flying during the day.

Answer: B - Part 107 requires Visual Meteorological Conditions (VMC), which mandate a ceiling of at least 500 feet and visibility of at least 3 miles.

Q460. You are flying in a residential area at a ground speed of 110 mph. Is this speed compliant with Part 107 regulations?

A) No, the maximum speed is 100 mph (87 knots) in any airspace.

B) Yes, speed limits only apply in Class B airspace.

C) Yes, provided you are flying over water.

Answer: A - Part 107 restricts the speed of a small UAS to no more than 100 mph (87 knots) in any airspace or terrain. Exceeding this limit in a populated area is a violation.

Q461. It is night, and you are operating a small UAS. Your drone has red navigation lights on the wings and a red anti-collision light on the tail. Is your lighting setup compliant?

A) No, the anti-collision light must be white and visible to 3 miles.

B) Yes, red lights are sufficient for night operations.

C) No, you cannot fly at night unless you have a waiver.

Answer: A - For night operations, the sUAS must have anti-collision lighting that is visible for at least 3 statute miles. This light must be white.

Q462. You are flying over a city and a small manned aircraft approaches your drone from the opposite direction (head-on). What is the required action to avoid a collision?

A) You must yield right of way to the manned aircraft.

B) You should increase speed to overtake the manned aircraft.

C) The manned aircraft has the right of way; you must remain steady.

Answer: A - Part 107 states that the pilot of a manned aircraft has the right of way over a sUAS. The remote pilot in command must take action to avoid a collision.

Q463. You are flying near a tall communications tower. You set your altitude to 650 feet AGL. Is this compliant with the 'higher than a structure' rule?

A) Yes, as long as you are more than 500 feet AGL.

B) No, you must always remain below 500 feet AGL.

C) Yes, provided the tower is shorter than the drone.

Answer: C - Part 107 allows a sUAS to operate higher than 400 feet if it remains higher than the structure upon which it is operating (or nearby structures). 650 feet is higher than a typical tower.

Q464. You are flying in a VLOS manner but the drone enters a small gap between two trees that obscures your view for a few seconds. You have a visual observer watching the drone's location. Are you still compliant?

A) Yes, as long as the observer is providing feedback.

B) Yes, this is acceptable as long as you do not enter the trees.

C) No, the pilot must maintain VLOS at all times.

Answer: C - The remote pilot in command is solely responsible for maintaining visual contact with the sUAS at all times. An observer can assist, but if the pilot loses sight, they are in violation of the VLOS rule.

Q465. You are flying a drone through a dense forest where the canopy obscures your view of the aircraft. You rely entirely on the live camera feed to navigate and maintain orientation. What is the result of this flight?

A) Visual Line of Sight (VLOS) is maintained because you can see the drone through the camera.

B) Visual Line of Sight (VLOS) is broken because you are not seeing the drone with your own eyes.

C) Visual Line of Sight (VLOS) is maintained because you are following the path shown on the controller screen.

Answer: B - The Remote Pilot must maintain direct unaided visual sight of the small UAS to control it safely. A camera feed does not satisfy the VLOS requirement.

Q466. A drone operator is taking photos of a large stadium from the parking lot. What is the maximum altitude they can legally fly without an airspace authorization or waiver?

A) 500 feet above ground level

B) 400 feet above ground level

C) 1,000 feet above ground level

Answer: B - Under Part 107, the maximum altitude for a small UAS is generally 400 feet above ground level, unless higher altitudes are necessary for operations that comply with other parts of the regulation.

Q467. While flying over a public park, your drone accidentally drifts closer than 50 feet to a group of people standing on the ground. What immediate action must you take?

A) Nothing, as long as you do not physically strike anyone.

B) Immediately land the drone to avoid violating the minimum distance requirements.

C) Increase your speed to clear the area faster.

Answer: B - The Remote Pilot must maintain a minimum distance of 50 feet from people or structures not directly involved in the operation. A breach of this distance requires immediate action to land.

Q468. You are flying near a school that has a designated Helicopter Landing Area (HLS). What are the minimum horizontal distance requirements for your flight?

A) 25 feet from the edge of the HLS

B) 500 feet from the edge of the HLS

C) 1,000 feet from the edge of the HLS

Answer: A - Part 107 requires a small UAS to stay at least 25 feet horizontally and 500 feet vertically from an HLS.

Q469. You plan to fly near a small private airport. You check the FAA website and see a Notice to Airmen (NOTAM) restricting operations in that airspace. What is your responsibility regarding this NOTAM?

A) You are exempt from NOTAMs if you are flying under Part 107.

B) You must comply with all applicable NOTAMs.

C) You must contact the airport manager to request permission to ignore the NOTAM.

Answer: B - Remote pilots are required to comply with all Notices to Airmen (NOTAMs) affecting the area of operation.

Q470. What equipment is required to legally operate a drone at night under Part 107 regulations?

A) Anti-collision lights only

B) Landing lights only

C) No additional equipment is required for night operations.

Answer: A - Anti-collision lights are required to be on and visible from at least 3 statute miles during night operations to ensure the aircraft is seen by other pilots.

Q471. You are flying over an open field at a constant speed of 110 mph. Is this flight legal under standard Part 107 rules?

A) Yes, as long as you stay below 400 feet.

B) Yes, speed is unlimited in uncontrolled airspace.

C) No, the speed must be no more than 100 mph or 87 knots.

Answer: C - The maximum speed of a small UAS in Class G airspace is 100 mph (87 knots) or the manufacturer's maximum speed, whichever is lower.

Q472. You are piloting a drone through a dense urban area where you cannot see the drone with your own eyes. A friend stands nearby and tells you where the drone is relative to buildings. Does this satisfy the visual line of sight requirement?

A) No, the pilot must maintain visual line of sight with their own eyes.

B) Yes, a Visual Observer is sufficient to satisfy VLOS.

C) Yes, as long as you are using a camera feed to verify their position.

Answer: A - A Visual Observer (VO) is an observer who assists the pilot but does not count as the pilot for VLOS requirements. The pilot must maintain their own direct unaided visual sight of the sUAS.

Q473. You are operating a small UAS under Part 107. You look down at your GPS and notice you are at 500 feet above ground level (AGL). Is this operation compliant with the maximum altitude regulation?

A) No, the maximum altitude is 400 feet AGL.

B) Yes, because it is under 600 feet.

C) Yes, provided you have a waiver.

Answer: A - Under 14 CFR §107.51, the maximum altitude for a small UAS is 400 feet above ground level, unless you have an authorized waiver from the FAA.

Q474. While flying your drone, you encounter another small UAS pilot. According to right-of-way rules, which aircraft has the right of way?

A) The aircraft that is flying lower.

B) The aircraft that is flying faster.

C) The larger aircraft.

Answer: C - Under 14 CFR §107.41, the larger aircraft has the right of way, and the operator of the smaller aircraft must give way to avoid collision.

Q475. You are flying a drone weighing more than 0.55 lbs and want to fly over an open-air assembly of people (like a sporting event). What is required to operate in this scenario?

A) No special permission is needed if the people are not looking at the drone.

B) You must notify the FAA at least 24 hours in advance.

C) You must obtain prior written authorization from the person organizing the event.

Answer: C - Under 14 CFR §107.51, if you are operating a UAS weighing more than 0.55 lbs over an open-air assembly of people, you must obtain prior written authorization from the person organizing the event.

Q476. You want to conduct a night flight operation. What equipment is required to be installed on your drone?

A) A strobe light.

B) Anti-collision lights visible from 3 miles away.

C) A red and green navigation light system.

Answer: B - Under 14 CFR §107.29, anti-collision lights must be installed and turned on during night operations if the aircraft is capable of flight above 550 feet AGL.

Q477. You are flying a drone over a highway where cars are moving. Can you fly directly over the moving vehicles?

A) Yes, as long as you maintain visual line of sight.

B) No, you cannot operate over moving vehicles.

C) Yes, provided the vehicles are moving below 30 mph.

Answer: B - Under 14 CFR §107.33(c), a small UAS may not be operated over a person or vehicle not directly involved in the operation unless the operator has a waiver or authorization from the Administrator.

Q478. You are conducting a commercial flight under Visual Flight Rules (VFR). What are the minimum weather conditions required?

A) 3 miles of visibility and 1,000 feet above ground level (AGL) cloud clearance.

B) 1 mile of visibility and 500 feet AGL cloud clearance.

C) 5 miles of visibility and 3,000 feet AGL cloud clearance.

Answer: A - Under 14 CFR §107.15, when operating under VFR, you must maintain at least 3 miles of visibility and be at least 1,000 feet above ground level.

Q479. You are flying over a stadium during a football game. The drone weighs 2 pounds. What must you do?

A) You must obtain a waiver from the FAA.

B) You must obtain prior written permission from the event organizer.

C) You must land immediately.

Answer: B - Since the drone weighs more than 0.55 lbs, operating over an open-air assembly of people requires prior written authorization from the person organizing the event (14 CFR §107.51).

Q480. What is the maximum speed limit for a small UAS under Part 107?

A) 100 mph.

B) 87 knots.

C) 65 mph.

Answer: A - Under 14 CFR §107.33, the maximum speed of a small UAS is 100 mph (or 87 knots).

Q481. You are flying your drone over a large outdoor stadium during a concert to capture footage for a client. You are maintaining visual line of sight. What additional requirement applies to your operation?

A) You must maintain a distance of at least 500 feet from the stadium walls.

B) You must have an observer assist you in maintaining visual line of sight.

C) You must submit a waiver request to the FAA prior to takeoff.

Answer: B - When operating a small UAS for compensation or hire over a densely populated area, you must have an observer to help maintain visual line of sight, in addition to your own visual line of sight. This ensures safety and situational awareness.

Q482. While flying at an altitude of 400 feet above the ground, you notice you are flying directly over a schoolyard. What is the most appropriate action to take regarding your altitude?

A) You may continue to fly at 400 feet as it is below the altitude limit.

B) You must immediately increase your altitude to 500 feet to clear the area.

C) You must immediately descend to an altitude below 400 feet.

Answer: C - Part 107 requires operators to remain at least 400 feet above any person, vessel, vehicle, or structure unless you are within a 500-foot radius of a takeoff or landing site. Flying over a schoolyard populated by people requires descending below 400 feet.

Q483. You are flying in the vicinity of a Class C airport with an operational control tower. You are more than 4 nautical miles from the airport's primary airport, but you are within the lateral boundaries of the controlled airspace. What must you do?

A) You may fly under visual flight rules without communicating with ATC.

B) You must establish two-way radio communication with ATC and obtain permission to enter.

C) You must remain below 500 feet AGL to avoid ATC interference.

Answer: B - To operate a small UAS within Class C airspace (or any controlled airspace with a control tower), you must establish two-way radio communication with the Air Traffic Control (ATC) facility and receive authorization or clearance before entering.

Q484. You are performing a visual inspection of your drone's propellers before a flight. The drone's battery is installed and the motors are connected. What is the correct safety procedure?

A) You must power off the drone and disconnect the battery before performing maintenance.

B) You can perform the inspection while the drone is powered on.

C) You must hold the drone in the air while inspecting the propellers.

Answer: A - You must ensure that the small UAS is powered off and the battery is disconnected before performing any maintenance or inspections. This prevents accidental activation or injury from spinning propellers.

Q485. You are operating a small UAS at night. What lighting requirement must you meet to comply with regulations?

A) You must use a strobe light visible for 3 miles.

B) You must have anti-collision lighting visible from at least 3 statute miles.

C) You are not required to use any lights as long as you have a visual observer.

Answer: B - When operating a small UAS during civil twilight (night), you must have anti-collision lighting that is capable of being seen by other aircraft for at least 3 statute miles.

Q486. You are flying your drone and a manned aircraft approaches your position. What is the correct action regarding right of way?

A) You must give way to the manned aircraft and alter your course to avoid a collision.

B) You must remain at your current altitude and heading.

C) You should alert the pilot of the manned aircraft to land immediately.

Answer: A - Small UAS operators must give way to any manned aircraft. You must take appropriate action to avoid a collision, which typically involves descending or changing course.

Q487. You are operating your drone over a crowd of people at a fairground. The crowd is stationary, and you are maintaining a distance of 100 feet from the nearest person. Is this operation compliant?

A) Yes, this is compliant because you are staying within 100 feet of the people.

B) No, unless you are operating under a specific waiver from the FAA.

C) Yes, this is compliant because you are operating for recreation, not compensation.

Answer: B - To operate at an altitude less than 500 feet above a person, vessel, vehicle, or structure, you must be operating under a specific Part 107 waiver or be a recreational flyer following specific hobbyist rules (like flying over a crowd being generally prohibited). However, under standard Part 107 operations, staying under 500 feet over people is prohibited without a waiver.

Q488. What is the maximum takeoff weight limit for a small unmanned aircraft that can be operated under Part 107?

A) 55 pounds.

B) 55 kilograms.

C) 100 pounds.

Answer: A - A small unmanned aircraft is defined as an unmanned aircraft weighing less than 55 pounds (including its fuel and any person on board).

Chapter 6: Regulations & Safety

Q489. Which chapter of the study guide covers Airspace Classification, Operating Requirements, and Flight Restrictions?

A) Chapter 2

B) Chapter 3a

C) Chapter 1

Answer: A - The Table of Contents lists Chapter 2 as covering 'Airspace Classification, Operating Requirements, and Flight Restrictions'.

Q490. Which chapter of the provided study guide focuses on Aviation Weather Sources?

A) Chapter 3a

B) Chapter 1

C) Chapter 2

Answer: A - The Table of Contents explicitly lists Chapter 3a as 'Aviation Weather Sources'.

Q491. What rating does the Remote Pilot – Small Unmanned Aircraft Systems Study Guide prepare pilots for?

A) Remote Pilot Certificate with an sUAS rating

B) Private Pilot Certificate

C) Commercial Pilot Certificate

Answer: A - The Preface states the study guide communicates knowledge areas for the 'Remote Pilot Certificate with an sUAS rating airman knowledge test'.

Q492. In the context of the provided document, what is the acronym for the Federal Aviation Administration?

A) FAA

B) FAA-G

C) FSD

Answer: A - The acronym 'FAA' appears multiple times on the cover page and within the Preface.

Q493. What is the document identifier for the provided study guide?

A) FAA-G-8082-22

B) FAA-S-8082-22

C) FAA-DOC-8082

Answer: A - The document identifier 'FAA-G-8082-22' is printed at the top of the provided text.

Q494. Which organization is listed as the publisher of the study guide?

A) Flight Standards Service

B) National Air Traffic Controllers Association

C) Department of Transportation

Answer: A - The Preface lists the 'Flight Standards Service' as the publisher under the 'Washington, DC 20591' address.

Q495. While flying in a rural area, your small UA approaches a tree line and you begin losing sight of it. In accordance with Part 107, what must you ensure?

A) You may continue as long as the aircraft remains within 400 feet horizontal distance

B) You must maintain visual line of sight with the aircraft at all times, so you must not let it go behind the tree line

C) You must switch to a visual observer if you cannot see the aircraft yourself

Answer: B - Part 107 requires the remote pilot in command to maintain visual line of sight (VLOS) with the small UA at all times. A visual observer can be used to supplement, but the aircraft must remain within VLOS of the pilot or observer, not simply within 400 feet.

Q496. You are flying your Part 107 drone near a tall communications tower that is 600 feet AGL. Within what horizontal distance of the tower may you legally fly above 400 feet AGL?

A) Within 400 feet of the tower

B) Within 1,000 feet of the tower

C) You may never fly above 400 feet AGL, even near a structure

Answer: A - Part 107 allows operations above 400 feet AGL if the small UA remains within 400 feet of a structure. This exception permits flight over the top of the tower, but only in close proximity to it.

Q497. Your remote pilot certificate was issued 24 months ago. You wish to continue operating under Part 107. What must you do before your current certificate expires?

A) Complete a recurrent aeronautical knowledge test and obtain a new certificate

B) Pass a practical flight test administered by an FAA examiner

C) Submit a renewal application to the FAA and pay a fee

Answer: A - Under Part 107, remote pilot certificates must be renewed within the previous 24 calendar months by completing a recurrent knowledge test. No practical test or renewal application is required; passing the recurrent exam updates the certificate.

Q498. You are a Part 107 pilot operating in a sparsely populated rural area. Is it legal to operate your small UA from a moving vehicle?

A) Yes, as long as the vehicle is moving slowly and the operation is not over people

B) No, operations from a moving vehicle are always prohibited under Part 107

C) Yes, because you are in a rural area and you are not carrying any property for compensation

Answer: A - Part 107 permits operations from a moving vehicle only if the operation is over a sparsely populated area and does not involve the carriage of property over people for compensation or hire. The vehicle may be moving, but the pilot must be able to maintain control and safety.

Q499. You plan to fly your Part 107 drone during civil twilight. What equipment or lighting must be visible on your small UA?

A) A green anti-collision light visible for at least 3 statute miles

B) Any lighting that makes the aircraft easily visible to other air traffic

C) A red or white anti-collision light visible for at least 3 statute miles

Answer: C - Part 107 allows operations during civil twilight if the small UA has an anti-collision lighting system that is visible for at least 3 statute miles. The lighting must be a red or white strobe or other anti-collision light, not a green navigational light.

Q500. You are planning a bridge inspection flight in controlled airspace. The latest METAR reports visibility as '1 1/2 SM' and shows 'BR' (mist). Can you legally operate your drone under Part 107?

A) Yes, because mist does not affect unmanned aircraft operations

B) Yes, if you maintain visual line of sight and stay below 400 feet

C) No, because Part 107 requires at least 3 statute miles visibility

Answer: C - Part 107.51 requires minimum flight visibility of 3 statute miles from the control station. A reported visibility of 1 1/2 SM is below the legal minimum, so flight is not permitted.

Q501. You are planning a sUAS flight over an open field. The airspace is Class G from the surface to 700 feet AGL. You are not flying near any structure. What is the maximum altitude AGL you may operate your drone under Part 107 without additional authorization?

A) 700 feet above ground level

B) 100 feet above ground level

C) 400 feet above ground level

Answer: C - Part 107.51 limits small UAS operations to 400 feet above ground level (AGL) unless the drone is flown within a 400-foot radius of a structure and does not exceed 400 feet above the structure's uppermost limit. Since you are not near a structure, 400 feet AGL is the maximum.

Q502. You need to conduct a roof inspection at a property that lies inside the lateral boundary of a Class D airport's airspace. What must you do before flying your drone in this controlled airspace?

A) Call the airport's tower operator by phone and state your name

B) File a flight plan with the nearest Flight Service Station

C) Obtain ATC authorization through an approved means such as LAANC

Answer: C - Class D airspace is controlled airspace. Under Part 107.41, a person may not operate a small UAS in Class B, C, D, or E airspace unless the operation is authorized by Air Traffic Control (ATC). The FAA provides LAANC (Low Altitude Authorization and Notification Capability) and other approved methods to obtain this authorization.

Q503. You are flying your drone during the day and observe a manned aircraft approaching head-on at the same altitude. What action must you take to comply with right-of-way rules?

A) Maneuver to the right immediately

B) Continue your course because you are smaller and reposition at the last moment

C) Yield the right of way and maneuver out of its path

Answer: C - Under Part 107.37, the remote pilot must yield the right of way to all other manned aircraft. When an aircraft is approaching head-on, both are expected to turn right, but the sUAS must ensure it cannot interfere. Simply turning right is only part of the requirement; the fundamental duty is to yield the right of way.

Q504. You plan to fly a commercial real estate mission at dusk after sunset. Your drone is equipped with a strobe light. Which statement about night operations under Part 107 is true?

A) Night operations are prohibited regardless of equipment

B) Night operations are permitted if the anti-collision light is visible for at least 3 statute miles

C) Night operations are only permitted if you first obtain a blanket waiver from the FAA

Answer: B - Under Part 107.29, night operations are allowed if the small UAS is equipped with an anti-collision light that is visible for at least 3 statute miles. The remote pilot must also have completed the appropriate night-safety training if their knowledge test was passed before the effective date of the night-ops rule.

Q505. You are flying your 2.6-pound drone over a park where a group of people is standing. You are not operating under a waiver, and your drone does not meet any of the FAA's Category 1-3 safety standards. Is it legal to fly directly over the group?

A) Yes, as long as you remain at least 50 feet above the people

B) Yes, if you maintain visual line of sight and avoid the people to the extent possible

C) No, operations over human beings are prohibited unless the drone meets the applicable open-category requirements

Answer: C - Part 107.39 generally prohibits operating a small UAS directly over a human being unless the drone meets certain category requirements (e.g., a Category 1 aircraft under 0.55 pounds may be flown over people if no injury can be caused). A 2.6-pound drone without meeting those standards may not be flown over people without a waiver.

Q506. You are inspecting a 500-foot-tall communications tower. You keep your drone within 300 feet of the tower as you climb its side. What is the maximum altitude above ground level you may fly your drone without a specific altitude waiver?

A) 400 feet AGL

B) 500 feet AGL

C) 900 feet AGL

Answer: C - Part 107.51 allows the drone to be flown above 400 feet AGL if it is within a 400-foot radius of a structure and does not fly higher than 400 feet above the structure's uppermost limit. For a 500-foot tower, the maximum is 500 ft + 400 ft = 900 feet AGL.

Q507. Before taking off for a commercial job, you find that your drone has not been flown in several weeks and the propeller blades show small paint chips. What must you do to comply with Part 107 before the flight?

A) Do nothing because paint chips are not damage

B) Conduct a preflight inspection to ensure the drone is in a condition for safe operation

C) Contact the drone manufacturer and request a 100-hour inspection

Answer: B - Part 107.49 requires a remote pilot in command to conduct a preflight inspection of the small UAS and ensure it is in a condition for safe operation. The pilot must assess any damage, such as paint chips on propellers, and determine whether the aircraft is still airworthy. If not, the flight cannot proceed.

Q508. While flying in a residential area, your drone strikes a parked car and causes $1,200 of damage to the vehicle. No one is injured. When must you report this accident to the FAA?

A) Within 10 days

B) Immediately, by telephone, before landing

C) No report is required because no one was injured

Answer: A - Part 107.9 requires a remote pilot to report any accident that results in serious injury to a person, death of a person, or property damage to non-UAS property if the cost exceeds $500. Since the damage was $1,200, you must report the accident to the FAA within 10 days.

Q509. While inspecting a 300-foot-tall cellular tower, your remote pilot operations must stay within 400 feet of the structure to follow the structure exemption. What is the maximum altitude above ground level (AGL) you may fly over the tower top?

A) 400 feet AGL

B) 700 feet AGL

C) 500 feet AGL

Answer: B - Part 107 allows a small UAS to fly above the 400-foot limit if it is flown within 400 feet of a structure. The maximum altitude is 400 feet above the top of the structure. Since the tower is 300 feet tall, you may fly up to 300 + 400 = 700 *feet AGL, provided you remain within the 400-foot radius.*

Q510. You are flying your drone during a late afternoon job. As the sun sets, you want to continue shooting until the light fades. Under Part 107, what is allowed for civil twilight operations?

A) You must operate with anti-collision lighting visible for at least 3 statute miles

B) You may continue flying without lighting as long as you are within visual line of sight

C) You must obtain a special waiver before any flight after sunset

Answer: A - Part 107 permits operations during civil twilight (30 minutes before official sunrise to 30 minutes after official sunset) if the small UAS has its anti-collision lights turned on and the lighting is visible for at least 3 statute miles. A waiver is not required for civil twilight.

Q511. You have arrived at a job site and a law enforcement officer asks to see your remote pilot certificate and government-issued photo ID. You left your physical certificate at home but you have a clear photo of it on your phone. What should you do?

A) Show the photo on your phone because it proves you are certified

B) Inform the officer that you are not required to carry the certificate while flying

C) Tell the officer you must produce the certificate and ID when requested, and a photo is not a valid substitute

Answer: C - A remote PIC must comply with a law enforcement officer's request to show the remote pilot certificate and government-issued photo ID. A photograph of the certificate is not a valid document; you must have the physical certificate in your possession.

Q512. While flying your sUAS over a parking lot, you see a large crowd of people gathered for a sidewalk event directly under your flight path. Your UAS weighs 2.3 kg and is not category-rated for flight over people. What action should you take?

A) Request a verbal waiver from the event organizer and proceed

B) Continue flying, but maintain an altitude above 200 feet so the crowd is safe

C) Manually hover away from the crowd and avoid flying directly over people

Answer: C - Part 107 prohibits operations over a person who is not directly participating in the operation unless the UAS meets certain category requirements. You must avoid the crowd. There is no blanket altitude exemption or authority granted by a non-FAA official.

Q513. You are flying a small UAS which weighs 1.8 kg. A fellow pilot asks you to fly his drone that also weighs 1.8 kg, but the drone lacks a registration number. What should you do?

A) Accept the job, then register the unmanned aircraft before flying

B) Refuse to fly the drone because it is not registered as required

C) Fly it only if an amateur radio license is displayed

Answer: B - All small UAS used for Part 107 operations must be registered with the FAA. A valid registration number must be clearly visible on the aircraft. You cannot legally operate an unregistered sUAS even if the aircraft weight is below the registration threshold.

Q514. You are a certificated remote pilot flying a small UAS for commercial real estate photography. A security official at a local park asks to see your documentation during an inspection. Under Part 107, which items must you make available?

A) Your original aeronautical knowledge test score report

B) A printed copy of your FAA flight plan filed for the flight

C) Your remote pilot certificate and a government-issued photo ID

Answer: C - Under 14 CFR §107.7, the remote pilot in command must present their remote pilot certificate and a government-issued photo ID upon request. A test score report is not proof of certification, and Part 107 does not require filing a flight plan.

Q515. You are using first-person view (FPV) goggles to fly a small UAS during a commercial inspection. Another pilot is acting as your visual observer. What must be true for the flight to comply with Part 107 visual line of sight rules?

A) The visual observer must be able to see the UAS at all times without the use of FPV goggles

B) The FPV goggles may serve as your sustained visual line of sight if they have a 30-degree field of view

C) The UAS must be equipped with ADS-B Out and a working strobe to use FPV goggles

Answer: A - Part 107.31 requires the remote pilot in command or a designated visual observer to maintain visual line of sight with the UAS using unaided vision. FPV goggles by themselves do not satisfy the VLOS requirement.

Q516. You want to operate a 2.5 lb drone over a crowd gathered for an outdoor festival. Under Part 107, what is the primary condition that permits this operation?

A) The drone is under 0.55 lb and you hold a current remote pilot certificate

B) The drone meets the applicable category requirements for operations over people and is operated accordingly

C) The drone is equipped with a visible anti-collision light and you have a visual observer

Answer: B - Part 107.36 allows operations over people only when the small UAS meets the applicable category requirements (Category 1, 2, 3, or 4). A remote pilot certificate, lighting, or visual observer alone does not satisfy the over-people rule.

Q517. You are hired to take aerial photos of a residential property for a real estate listing. You do not have any waivers. Which of the following operations would be prohibited under Part 107?

A) Flying at 300 feet AGL over a house on the property

B) Flying at 200 feet AGL over the backyard with no people present

C) Flying over a car moving on the driveway to capture a shot

Answer: C - Part 107.39 prohibits operating a small UAS over moving vehicles unless the vehicle is covered or the operation is specifically designed to avoid the vehicle (e.g., with a crashworthy UAS). Capturing a photo over a moving car does not meet those conditions, so it is prohibited.

Q518. You are flying a small UAS in a rural area under Part 107. What is the maximum groundspeed you may fly without a waiver?

A) 50 miles per hour

B) 100 miles per hour

C) 150 miles per hour

Answer: B - Part 107.51(b) limits the small UAS groundspeed to no more than 100 mph (87 knots) unless authorized by a waiver.

Q519. You plan to operate your small UAS during civil twilight so you can capture sunset photos. What is required before you begin flying?

A) Use anti-collision lighting that is visible for at least 3 statute miles

B) Obtain a night operations waiver from the FAA

C) Ensure the aircraft is no heavier than 0.55 pounds

D) D) File a flight plan with the nearest air traffic control facility

Answer: A - Under 14 CFR §107.29, during civil twilight (defined as the period from sunrise to sunset and sunset to sunrise, plus such time as the center of the sun is up to 6 degrees below the horizon) and at night, the small UAS must have anti-collision lighting visible for at least 3 statute miles. A waiver is not required for civil twilight operations if lighting is used.

Q520. You are operating a small UAS that weighs 0.55 pounds and has no exposed rotating parts. Without a waiver, are you allowed to hover over a crowd at a public park?

A) Yes, because it is a Category 1 small UAS

B) No, because operations over people are always prohibited without a waiver

C) Yes, but only if you have a visual observer and a protective propeller guard

Answer: A - A small UAS weighing 0.55 pounds or less with no exposed rotating parts qualifies as a Category 1 aircraft under 14 CFR §107.4. Category 1 operations are permitted over people and over open-air assemblies without a waiver, unless other Part 107 limitations apply.

Q521. You are hired to photograph a residential property from the air. The homeowner gives you permission to fly over the property, but pedestrians are walking in the neighbor's driveway. What should you do?

A) Fly quickly above 100 feet to minimize risk to the pedestrians.

B) Fly over the neighbor's driveway because the homeowner gave permission for the property.

C) Avoid flying over the pedestrians and their driveway, as they are not directly participating in the operation.

Answer: C - Part 107.39 prohibits operating a small UAS over human beings unless they are directly participating in the operation or under a covered structure. The pedestrians in the neighbor's driveway are not participating, so you must avoid flying over them.

Q522. While flying in Class G airspace at 300 feet AGL, you notice a manned aircraft approaching at the same altitude from your left. According to Part 107, what is your responsibility?

A) You have the right of way because you are smaller and more maneuverable.

B) You should climb to increase vertical separation.

C) You must yield the right of way to the manned aircraft.

Answer: C - Part 107.37 requires that the remote pilot yield the right of way to all other aircraft. A small UAS never has the right of way over a manned aircraft.

Q523. You are inspecting a 500-foot-tall communications tower. Your drone remains within 100 feet horizontally of the tower. What is the maximum altitude above ground you may fly without a waiver?

A) 400 feet AGL

B) 500 feet AGL

C) 900 feet AGL

Answer: C - Part 107.51 permits flying higher than 400 feet AGL if the drone is within 400 feet of a structure; you may fly no higher than 400 feet above the structure's uppermost limit. Here, 500 feet + 400 feet = 900 feet AGL.

Q524. While flying a mission, your small UAS passes behind a large tree, and you momentarily lose visual line of sight. Under Part 107, what action must you take?

A) Continue flying using the FPV monitor until the drone re-emerges.

B) Immediately maneuver to regain visual line of sight or land.

C) Ask a visual observer to watch the FPV monitor for you.

Answer: B - Part 107.31 requires that the remote pilot and any visual observer maintain continuous visual line of sight with the small UAS. If VLOS is lost, you must take immediate steps to regain it or land.

Q525. You want to fly your Part 107 drone at night to capture a fireworks display. The drone is not equipped with any lighting. Can you legally operate?

A) Yes, if you stay below 400 feet and have the event organizer's permission.

B) No, you must have anti-collision lights visible for at least 3 statute miles.

C) Yes, but only in uncontrolled (Class G) airspace.

Answer: B - Under Part 107.29, night operations require the small UAS to have anti-collision lights that can be seen for at least 3 statute miles. Without such lighting, the operation is not authorized.

Q526. You plan to fly a mapping mission within the lateral boundaries of a Class C airport's surface area. What must you obtain before takeoff?

A) A NOTAM filed with the FAA to alert other pilots.

B) Verbal permission from the airport manager.

C) Authorization from air traffic control, typically via LAANC or a waiver.

Answer: C - Part 107.41 prohibits operations in controlled airspace (including Class C) without prior ATC authorization. LAANC or a FAA waiver is the standard method for obtaining that authorization.

Q527. A client asks you to inspect a large farm in a remote area. The airspace above the farm is Class G. What is the maximum altitude you may fly the drone above ground level?

A) 400 feet AGL

B) 1,200 feet AGL

C) 500 feet AGL

Answer: A - Part 107.51(a) states that in Class G airspace, a small UAS may not be flown higher than 400 feet above ground level, unless within 400 feet of a structure.

Q528. After a rough landing, you notice a crack in the drone's motor mount. You have another job scheduled in one hour. What is the proper course of action?

A) Ground the drone and have it inspected or repaired before further operation.

B) Use a lower-capacity battery to reduce rotational stress on the motor mount.

C) Proceed with the job but reduce the drone's flying speed.

Answer: A - Part 107.49 requires a preflight inspection to ensure the small UAS is in a condition for safe operation. A cracked motor mount is a safety hazard, so the drone must be repaired or inspected before any further flight.

Q529. You are flying a drone at 400 feet above a public park. Suddenly, you notice a manned aircraft approaching your drone. Who has the right of way?

A) The manned aircraft, because it is larger and more maneuverable.

B) The drone pilot, because they are operating under Part 107 regulations.

C) The drone pilot, because the drone is smaller and less maneuverable.

Answer: A - Under 14 CFR 107.37, small unmanned aircraft must give way to manned aircraft. The manned aircraft has the right of way because it is larger and more maneuverable.

Q530. You are operating a small unmanned aircraft and you notice that your remote pilot certificate or aircraft registration has been lost or destroyed. What must you do?

A) Continue flying as long as you have a digital copy on your phone.

B) Land immediately and do not fly again until you have replaced the documentation.

C) Land immediately and report the loss to the nearest airport tower.

Answer: B - Under 14 CFR 107.41, if a remote pilot certificate or aircraft registration is lost or destroyed, the aircraft must not be operated until the required documentation is replaced.

Q531. You plan to fly over a highway with heavy traffic. Is this permitted under Part 107 regulations?

A) Yes, as long as you are flying below 400 feet.

B) No, operating over a highway obstructs the free flow of traffic.

C) Yes, provided you have a visual observer watching the road.

Answer: B - According to 14 CFR 107.39, a small UA may not be operated over a highway or a road where there is substantial traffic flow, even at low altitudes, because it obstructs the free flow of traffic.

Q532. You are flying a drone that is larger than 55 pounds. You must keep the drone within visual line of sight (VLOS). Who can assist you in maintaining VLOS?

A) No one, the pilot must do it alone.

B) A visual observer (VO).

C) The operator of any nearby manned aircraft.

Answer: B - Under 14 CFR 107.25, a visual observer is allowed to assist the pilot in maintaining visual contact with the small unmanned aircraft.

Q533. You are flying a drone at 500 feet AGL over a large, open field. Is this within the altitude limits?

A) Yes, you can fly as high as you want as long as you stay in VLOS.

B) No, the maximum altitude is 400 feet above the ground level.

C) Yes, provided you are not near an airport.

Answer: B - Under 14 CFR 107.51, a small UA may not operate at more than 400 feet above the ground level, unless it is closer than 400 feet to a structure.

Q534. You are flying near a Class C airport and your altitude is below 1,000 feet. Do you need to obtain ATC clearance?

A) No, as long as you are under 1,000 feet.

B) Yes, you must obtain ATC clearance at least 30 minutes before takeoff.

C) Yes, you must obtain ATC clearance before entering the airspace.

Answer: C - According to 14 CFR 107.39, you must obtain ATC clearance before operating a small UA in Class B, C, or D airspace or at an airport with an operating control tower.

Q535. While flying, your drone experiences a battery failure warning light. What is the appropriate action?

A) Land the drone immediately and do not attempt to fly again.

B) Fly back to the home point to save battery for the return trip.

C) Increase the drone's speed to reach the ground faster.

Answer: A - Under 14 CFR 107.39, if a condition develops that affects the safety of the flight, the remote pilot in command must discontinue the flight immediately.

Q536. You are flying your drone at 400 feet AGL in rural farmland when a sudden storm front approaches, causing turbulence. You need to ascend to avoid the storm. What is the maximum altitude you are allowed to operate at?

A) 500 feet AGL

B) 1000 feet AGL

C) 400 feet AGL

Answer: C - The maximum altitude for a small UA is 400 feet AGL, unless you are within a controlled airspace and receive authorization from ATC. Weather conditions do not allow for exceeding this limit.

Q537. While flying over a residential neighborhood for a photography project, a resident approaches you and demands you stop taking pictures, claiming you are violating their privacy. You are maintaining a safe altitude and distance. What should you do?

A) Land immediately and apologize to avoid a conflict.

B) Fly away immediately to the nearest open field.

C) Maintain altitude and continue, respecting the resident's privacy but not intentionally avoiding the area.

Answer: C - Remote pilots must respect the privacy of others when photographing or recording individuals in areas where they have a reasonable expectation of privacy. However, this does not prohibit flight over private property entirely.

Q538. You are flying near a large sports stadium during a game and spot a weather balloon drifting near the stadium. You decide to fly over the stadium to get a closer look at the balloon. Is this allowed?

A) Yes, you can fly anywhere to observe hazards as long as you remain VLOS.

B) No, you must not operate in a manner that creates a hazard to other aircraft or persons.

C) Yes, provided you are more than 1 mile away from the stadium.

Answer: B - Remote pilots must not operate in a manner that creates a hazard to other aircraft or persons. Flying dangerously close to a populated stadium or potential hazards is prohibited.

Q539. Your drone's battery is depleting rapidly, and you need to land immediately. You are flying near a Class B airport with low traffic. What is the safest course of action?

A) Land at the nearest airport terminal to refuel.

B) Land immediately in the nearest taxiway or runway.

C) Land at the nearest safe area away from the airport.

Answer: C - In an emergency, the pilot should land at the nearest safe area. Operating on or near an active airport surface without authorization is prohibited and dangerous.

Q540. You are flying at night for the first time using a small drone. Does your drone require anti-collision lights?

A) No, anti-collision lights are only required for drones larger than 2 pounds.

B) Yes, anti-collision lights are required for night operations.

C) No, small drones do not require lights at night.

Answer: B - When operating a small UA at night, the pilot must have anti-collision lighting visible for at least 3 statute miles.

Q541. You and a friend each purchased a new drone. Your drone weighs 1.2 pounds, and your friend's drone weighs 1.0 pound. Who is responsible for registering the drones with the FAA?

A) Only the drone weighing 1.2 pounds, as it is over the 0.55-pound threshold.

B) Only the drone weighing 1.0 pound, as it is over the 0.55-pound threshold.

C) Both drones, as both weigh more than 0.55 pounds.

Answer: C - Any person who operates a UAS that weighs more than 0.55 pounds (250 grams) but not more than 55 pounds (including accessories and equipment attached for flight) must register with the FAA before outdoor flight.

Q542. You are flying in a canyon with tall trees on both sides. You can see the drone clearly, but you cannot see the spot where you intend to land. Is this operation compliant with VLOS rules?

A) Yes, as long as you can see the drone itself.

B) No, you must see the spot where the drone will land.

C) Yes, provided you are within 50 feet of the ground.

Answer: B - Visual Line of Sight requires the remote pilot to see the small UA and the spot where it will land, ensuring you have unobstructed lines of sight to the landing area.

Q543. You are driving down a highway and spot a minor traffic accident. You want to fly your drone to take photos for documentation. Is this allowed?

A) Yes, you can operate from a moving vehicle if you are a professional news reporter.

B) No, you cannot operate from a moving vehicle.

C) Yes, you can operate from a moving vehicle as long as you are not careless or reckless.

Answer: B - Remote pilots may not operate a small UA from a moving vehicle unless the vehicle is stationary and the operator is in continuous visual contact with the UA.

Q544. You are flying your drone at a local park and notice a manned aircraft is approaching your position. You have time to maneuver. What is your immediate responsibility?

A) Immediately land the drone and leave the airspace to avoid the aircraft.

B) Increase your speed to get out of the way faster.

C) Immediately maneuver the drone out of the path of the aircraft.

Answer: C - According to 14 CFR 107.29, if a manned aircraft is approaching, the remote pilot must immediately take action to avoid the aircraft.

Q545. You are photographing a stadium from a nearby hill. The top of the stadium is exactly 400 feet tall. Can you fly your drone above the stadium to get a wider angle of the field?

A) Yes, but you must fly at least 1,000 feet AGL to stay clear of the stadium structure.

B) Yes, you can fly up to 500 feet AGL as long as you see the drone.

C) No, you must stay below 400 feet above ground level (AGL).

Answer: C - Part 107.51 prohibits operating a small unmanned aircraft at an altitude that will interfere with operations authorized under Part 77 or at more than 400 feet AGL.

Q546. You are flying at a crowded festival where many people are gathered. You are not within a closed structure. What is the maximum number of people you are allowed to have present in the area where you are flying?

A) No more than 10 people.

B) No more than 1,000 people.

C) No more than 1 person.

Answer: C - For standard operations over people not in a closed structure, the remote pilot must not operate the aircraft over more than one person at a time.

Q547. You are flying in an area with tall trees and limited visibility. You must maintain visual line of sight with your drone. What does this requirement specifically mean?

A) You can use a telephoto lens to see the drone from a distance.

B) You only need to see the drone if there are other aircraft nearby.

C) You must see the drone with your own eyes and ensure no obstruction blocks the line of sight.

Answer: C - Visual line of sight requires the remote pilot to see the unmanned aircraft with their own eyes (not through a screen or binoculars) and ensure there is no obstruction blocking the view.

Q548. You are preparing to fly, but you check the wind speed and see it is 18 knots with gusts to 25 knots. The drone manufacturer lists a maximum wind speed of 15 knots. What should you do?

A) Launch immediately because the drone can handle the wind.

B) Check the B4UFLY app to see if it is a no-fly zone rather than checking wind.

C) Not launch the drone because the wind exceeds the manufacturer's specifications.

Answer: C - Part 107.21 requires the remote pilot to not operate the aircraft in a manner that is beyond the aircraft's limitations, including weather conditions like wind speed.

Q549. You receive a notification that your drone's Remote ID signal is being jammed or blocked while flying in an open area. What is the correct course of action?

A) Continue flying and ignore the notification.

B) Increase the drone's altitude to try to clear the interference.

C) Land the drone immediately as the Remote ID requirement is being violated.

Answer: C - Part 89 requires drones to broadcast Remote ID. If the signal is blocked or jammed, the pilot is failing to comply with the requirement, necessitating an immediate landing.

Q550. You purchased a drone as a gift for your friend, but they haven't received it yet. Who is responsible for registering the drone with the FAA?

A) The person who bought the drone.

B) The person who builds the drone.

C) The person who received the drone.

Answer: C - Registration is the responsibility of the person who uses the aircraft for hobby or recreation. If it is a gift, the recipient is the user and must register it.

Q551. During a flight, your drone experiences a critical failure, such as a loss of control, and is drifting toward a group of people. What is your immediate responsibility?

A) Fly higher to get away from people before landing.

B) Try to reset the drone's compass to regain control.

C) Land the drone as soon as possible to prevent damage to property or injury.

Answer: C - Part 107.65 requires the remote pilot to land immediately if the aircraft malfunctions, to prevent damage to property or injury to people.

Q552. You are operating a drone that weighs 0.6 pounds. Do you need to register it with the FAA?

A) No, because it weighs less than 0.55 pounds.

B) Yes, because it exceeds the 0.55-pound weight limit.

C) No, because it weighs less than 55 pounds.

Answer: B - Under Part 107 regulations, a person may not operate a UAS that weighs more than 0.55 pounds (250 grams) unless it is registered with the FAA.

Q553. You are planning to fly your drone at night in a populated city. What is required to legally conduct this flight?

A) You do not need anything special, as night operations are always permitted.

B) You must obtain a waiver from the FAA.

C) You must have an anti-collision light installed and functioning.

Answer: C - To fly at night, the drone must be equipped with an anti-collision light that is capable of being seen for at least 3 miles.

Q554. You are applying for a waiver to fly in Class B airspace where normally no drones are allowed. When should you submit your application?

A) At least 60 days before the desired flight date.

B) At least 10 days before the desired flight date.

C) Immediately before the desired flight date.

Answer: A - Applications for waivers must be submitted at least 60 days before the desired date of operation to allow sufficient time for processing.

Q555. You have just passed the FAA Part 107 aeronautical knowledge test. What additional requirement must you fulfill to become a certified remote pilot?

A) You must complete a flight review with an instructor.

B) You must pass the TRUST (Transportation Security Administration) safety exam.

C) You must submit your registration to the local airport control tower.

Answer: B - To become a certified remote pilot, an individual must successfully complete the Part 107 exam and pass the TSA Transportation Security Administration (TSA) screening.

Q556. You are using a pair of binoculars to track the position of your drone. Is this considered Visual Line of Sight (VLOS)?

A) Yes, binoculars help you see further.

B) No, VLOS requires you to see the drone with your own eyes, unaided.

C) Yes, as long as you can see the drone, the method doesn't matter.

Answer: B - Visual Line of Sight means you must see the drone with your own eyes (unaided by binoculars, cameras, or other devices) at all times during flight.

Q557. You are flying near a 700-foot tall communications tower. What is the maximum altitude you can fly at that location?

A) 400 feet above the ground.

B) 700 feet above the ground.

C) 500 feet above the ground.

Answer: A - The maximum altitude for a small UAS is 400 feet above the ground, unless you are within 500 feet of a structure, in which case you must remain below that structure's top.

Q558. You wish to fly your drone over a football stadium during a game. What is the most likely regulatory hurdle for this flight?

A) You must complete a flight review within the last 24 months.

B) You must notify the local air traffic control tower.

C) You must obtain a waiver from the FAA to operate over people.

Answer: C - Operating over people without their prior consent generally requires a waiver from the FAA, unless the drone is being operated under a Certificate of Waiver or Authorization (COA).

Q559. You are flying over a restricted airspace area designated by the military. What is your responsibility regarding this area?

A) You must not enter the area without prior authorization.

B) You must file a flight plan with ATC 24 hours in advance.

C) You must request permission from the nearest airport manager.

Answer: A - It is illegal to operate a UAS in prohibited or restricted areas unless authorized by the ATC or using an authorized airspace provider.

Q560. You are planning to fly your small UAS near a Class C airport. What is required to legally operate in that airspace?

A) You must obtain a visitor agreement from the airport operator.

B) You must stay at least 5 miles away from the airport.

C) You must file a flight plan with ATC before every flight.

Answer: A - Under Part 107.31, remote pilots must obtain permission from the airport operator before operating within 5 nautical miles of a Class C airport. This is typically done via a Visitor Agreement.

Q561. What is the maximum altitude you may operate a small UAS under Part 107 regulations?

A) 500 feet above ground level.

B) 1,000 feet above ground level.

C) 400 feet above ground level.

Answer: C - Section 107.51 states that a small UAS may not operate more than 400 feet above the ground level, unless within a structure or authorized by ATC with a waiver.

Q562. You are flying at 500 feet and need to drop a small package to a person on the ground. What is required to do this legally?

A) You can drop the package as long as you are directly above the person.

B) You must ensure that the object will not cause injury or damage to persons or property.

C) Dropping objects from a UAS is strictly prohibited under all circumstances.

Answer: B - Section 107.51 prohibits dropping an object from a UAS if that object creates a hazard to persons or property. The operator must take precautions to ensure safety.

Q563. You wish to fly your drone at night. What specific equipment is required?

A) Anti-collision lights visible for at least 3 statute miles.

B) A spotlight to illuminate the area below the drone.

C) A waiver from the FAA to operate at night.

Answer: A - Section 107.29 requires that small UAS operated at night must have anti-collision lighting that is capable of being seen for at least 3 statute miles and has an intensity sufficient to be seen by other crew members and in adverse weather conditions.

Q564. You are flying near a Class B airspace. What is the standard procedure for entering this airspace?

A) Enter the Bravo shelf to get a better view of the airport.

B) Contact the nearest Flight Service Station (FSS) to receive an authorization.

C) Fly directly over the airport runway to ensure you have clearance.

Answer: B - For Class B airspace, pilots must contact ATC and receive an authorization or clearance before entering. The

Bravo shelf is not a public airspace for general drone use.

Q565. While flying, you capture a photo of a neighbor's backyard. Under privacy regulations, what should you do?

A) You can share the photo as long as you do not show the neighbor's face.

B) You must obtain the neighbor's consent before taking or sharing the photo.

C) There are no privacy regulations regarding photography taken by drones.

Answer: B - FAA Part 107.40 states that remote pilots should not photograph or record identifiable private property or individuals without consent, as privacy laws may still apply even if you are flying legally.

Q566. You are maintaining visual line-of-sight (VLOS) with your drone, but you are using a pair of binoculars to see it. Are you complying with the regulation?

A) Yes, binoculars are an acceptable form of VLOS.

B) No, you must maintain direct visual contact without optical aids.

C) Yes, as long as the drone is small enough to see with the naked eye.

Answer: B - Section 107.33 defines VLOS as the ability to see the aircraft without using binoculars or any other optical aid. Using binoculars technically breaks VLOS.

Q567. You are operating a drone that weighs 1.5 pounds. Do you need to register it with the FAA?

A) No, drones under 0.55 pounds do not need registration.

B) Yes, all drones must be registered regardless of weight.

C) No, only drones over 55 pounds need registration.

Answer: A - Section 107.21 requires registration only for drones weighing more than 0.55 pounds but less than 55 pounds that are used for hobby or recreational purposes, or for commercial operations.